

Lab 3

Lab 3: Vector Network Analyzer and L Matching Networks

In this lab you are going to calibrate and use a vector network analyzer (VNA), an instrument used to measure reflection coefficients. (You'll also simulate its behavior, a particularly instructive exercise.) We will use the VNA to characterize a circuit that you will solder onto a printed circuit board: an L matching network.

After this lab, you will be able to:

1. Interpret Smith Charts for common loads and parasitic effects that you will see in labs.
2. Simulate the reflection coefficient of a load using S-Parameter simulation and plot Smith Charts. (Optionally using RF packages like Scikit-RF and PyPlot.)
3. Calibrate a VNA
4. Design an L matching network
5. Solder surface mount passive and connector components
6. Characterize components using a VNA

Theory Questions: Smith Chart practice

1. For each of the following circuits, calculate their impedance, calculate their reflection coefficient, and plot them as a single

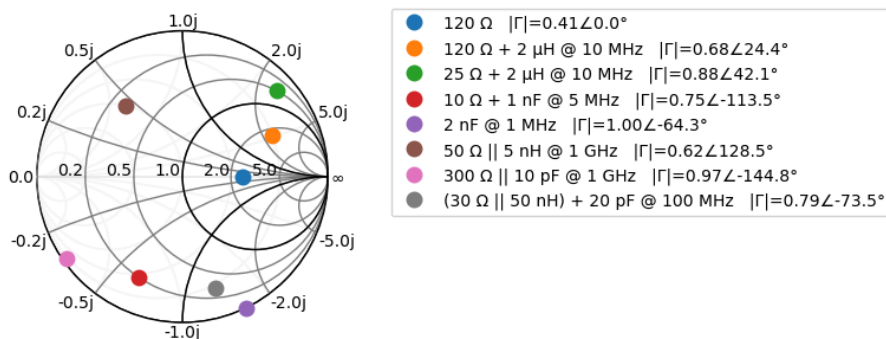
point on a Smith chart. Assume a Smith chart normalization of $Z_0 = 50 \Omega$. Give the impedance in terms of real and imaginary parts (i.e., $Z = R + jX$), and give the reflection coefficient in terms of magnitude and phase (i.e., $\Gamma = |\Gamma| \angle \Gamma^\circ$).

You must do all of these without a simulator (using MATLAB or any other calculator of your choice is fine as long as you do the algebra yourself). However, AFTER you have solved each one, use a digital Smith chart tool to check your work (e.g., <https://onlinesmithchart.com/>).

As you go, check your work and your intuition by considering whether the point you plotted on the Smith chart matches the reflection coefficient you calculated. For each of the “consider” questions, you do not need to report your answers or responses. Just ask yourself the questions and discuss them with your lab partners.

To be graded 0

Claude plot using my calculated Gammas



end of grade as 0

The naming convention for this is

xxxx resistance, yyyy unit (i.e. micro μ or nano n) and then type (l for inductor, c for capacitor)

1. A 120-ohm resistor

$$Z = 120 + 0j$$

The reflection if it comes after a 50Ω transmission line is

$$\Gamma = \frac{120 - 50}{120 + 50}$$

$$\Gamma = 0.41$$

with no phase.

Smith Chart: $\Gamma = 0.412$ so it agrees.

2. A 120-ohm resistor in series with a 2-uH inductor at 10 Mhz

$$120 + Ls$$

$$= 120 + 2\pi * 2 * 10^{-6} * 10^7$$

$$= 120 + 125.6j$$

We have to normalize to get to reflection coefficient. this gets

$$2.4 + 2.5j$$

The reflection coefficient is

$$\frac{(120 - 50 + 125j)}{(120 + 50 + 125j)} = 0.68(24.4^\circ)$$

3. A 25-ohm resistor in series with a 2-uH inductor at 10 Mhz

We have

$$Z = 25 + 125.6j$$

(same L as before)

This gives us

$$\Gamma = \frac{(25 - 50 + 125j)}{(25 + 50 + 125j)}$$

Which is

$$0.88\angle 42.29$$

1. Consider: how much of an effect does the 2-uH inductor have on the reflection coefficient when combined with a 120-ohm resistor vs. a 25-ohm resistor? At 10 MHz, how does the impedance of the 2-uH inductor compare to the impedance of a 120-ohm vs. a 25-ohm resistor?
2. A 10-ohm resistor in series with a 1-nF capacitor at 5 MHz

$$z = 10 + 10^{-9} * 5 * 10^6$$

this gives us

$$z = 10 + \frac{1}{j2\pi fc}$$

$$= 10 - \frac{1}{2\pi * 5 * 10^6 * 10^{-9}}j$$

$$= 10 - 31.8j$$

This gives a reflection coefficient

$$\frac{10 - 31.8j - 50}{10 - 31.8j + 50}$$

Which gives

$$\Gamma = 0.75 \angle -113.592$$

3. A 2-nF capacitor at 1 MHz

This has

$$\begin{aligned} z &= -\frac{1}{2\pi f_c} j \\ &= -79j \end{aligned}$$

This gives

$$\Gamma = \frac{-50 - 79j}{50 - 79j}$$

The magnitude is just -1 , with phase shift $\angle = -64^\circ$.

4. A 50-ohm resistor in parallel with a 5-nH inductor at 1 GHz

The parallel impedance is

$$\begin{aligned} Z &= \frac{1}{\frac{1}{50} + \frac{1}{j2\pi \cdot 10^9 \cdot 5 \cdot 10^{-9}}} \\ &= 14.15 + 22.52j \end{aligned}$$

This has

$$\Gamma = 0.62 \angle 128.51$$

5. For the parallel loads, I recommend you start by calculating the admittance, $Y = G + jB$. To get the impedance, you can (1) plot the admittance on a Smith chart with admittance circles, then read the impedance off of the chart, (2) do a series-to-parallel transformation to find an equivalent series circuit, or (3) do the algebra to get impedance

from $Z = 1/Y$.

6. A 300-ohm resistor in parallel with a 10-pF capacitor at 1 GHz

$$Z = \frac{1}{\frac{1}{300} + 2\pi(10)j}$$

$$z = 8.4 * 10^{-7} - 0.01j$$

$$\Gamma = 0.96 \angle -144$$

7. Consider: What does a capacitor in parallel with a resistor look like at low vs high frequencies? How do you model a capacitor at DC vs. really high frequencies? Does your calculation here for a parallel RC circuit make sense based on your conceptual understanding of how shunt capacitors behave?
8. A 30-ohm resistor in parallel with a 50-nH inductor, in series with a 20-pF capacitor, at 100 MHz. To clarify, you are finding the impedance looking into one end of a 20-pF capacitor, while the other end of the capacitor is attached to a 50-nH inductor and 30-ohm resistor in parallel with each other.

$$Z = \frac{1}{\frac{1}{30} + \frac{1}{2\pi \cdot 100 \cdot 10^6 \cdot 50 \cdot 10^{-9}j}} + \frac{1}{2\pi \cdot 100 \cdot 10^6 \cdot 20 \cdot 10^{-12} * j}$$

This gives us

$$Z = 15.6 - 64.5j$$

so

$$\Gamma = 0.79 \angle -73.45$$

Theory and Simulation: VNA

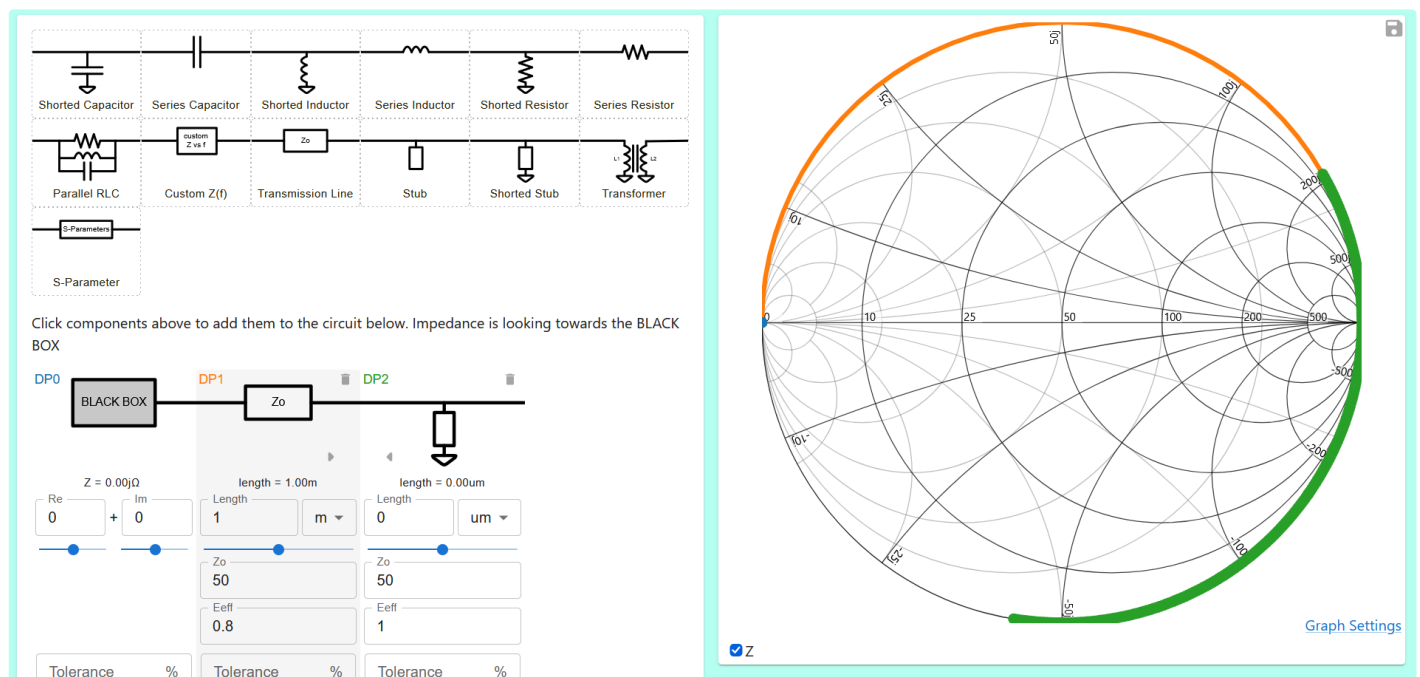
These questions center on using a vector network analyzer (VNA) to take a measurement of the reflection coefficient of a device under test. For now, it's OK to think of the vector network analyzer as generating sinusoids of different frequencies while measuring magnitude and phase of reflected sinusoids, not the sum of sinusoids that form the standing wave pattern. In Lab 2, your oscilloscope could only show you the total voltage wherever you connected a channel to. The magic of a VNA is that it can give you measurements directly related to forward and reverse traveling waves, not just the standing wave. This means your

VNA will produce two points of data at each measurement frequency point: $|V_{reflected}(f, -S)|$ and $\angle V_{reflected}(f, -S)$, where f is the driving frequency (swept during operation) and S is the length of the attached cable. These mathematical expressions presume that we define $x = 0$ at the load and $x = -S$ at the source. These reflections can be compared with the forward wave, which is a known quantity $V_{forward}(f, -S)$ because we transmit onto the line.

For these questions, note that SMA cables have a velocity factor of 0.8 and a characteristic impedance of 50 ohms. Note also that vector network analyzers all have 50 ohms of source impedance on each port.

A 1-m long SMA cable is terminated in a short circuit and driven by a VNA. The VNA is set to measure the frequency range from 70 MHz to 130 MHz.

Draw a Smith Chart indicating the locus of reflection coefficients the VNA will perceive when it compares the magnitude and phase of reflected waves to the incident wave.



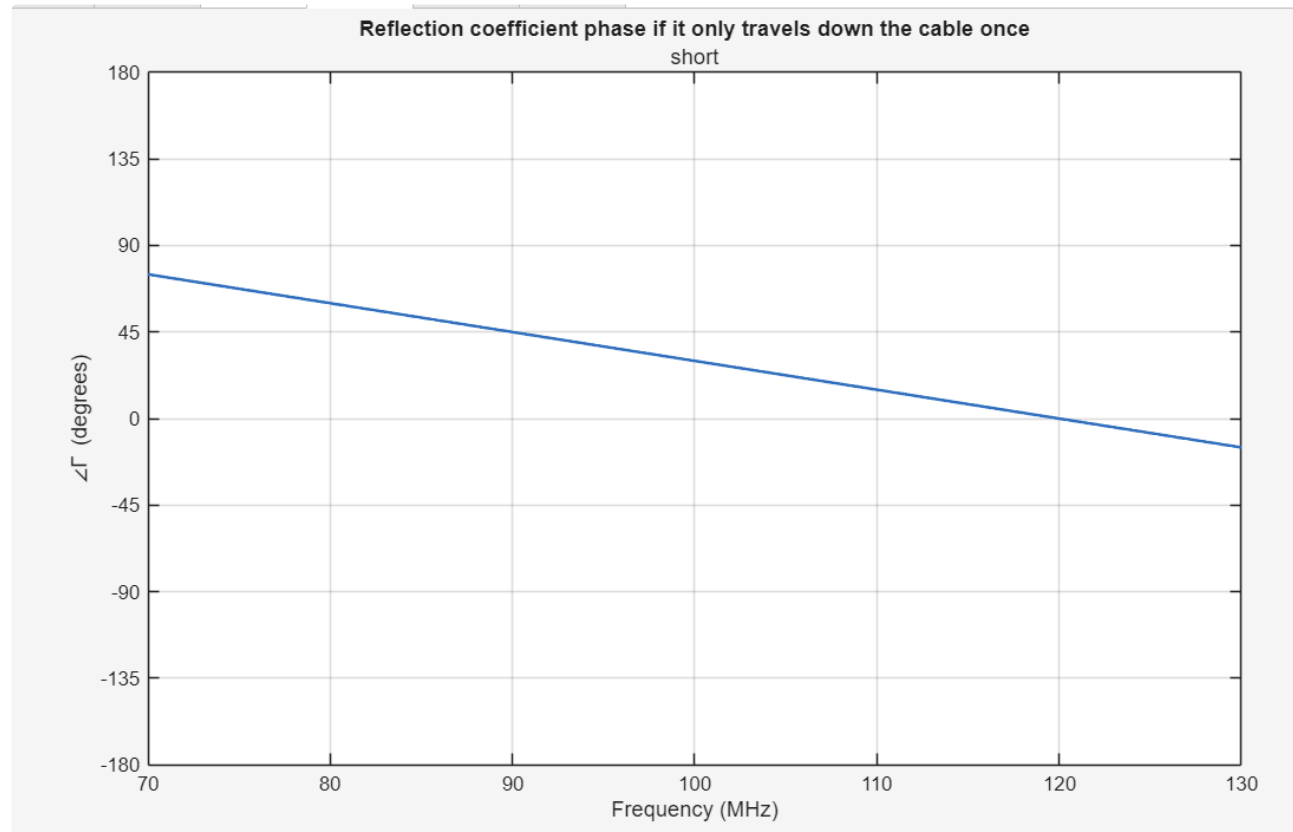
We can work this out mathematically.

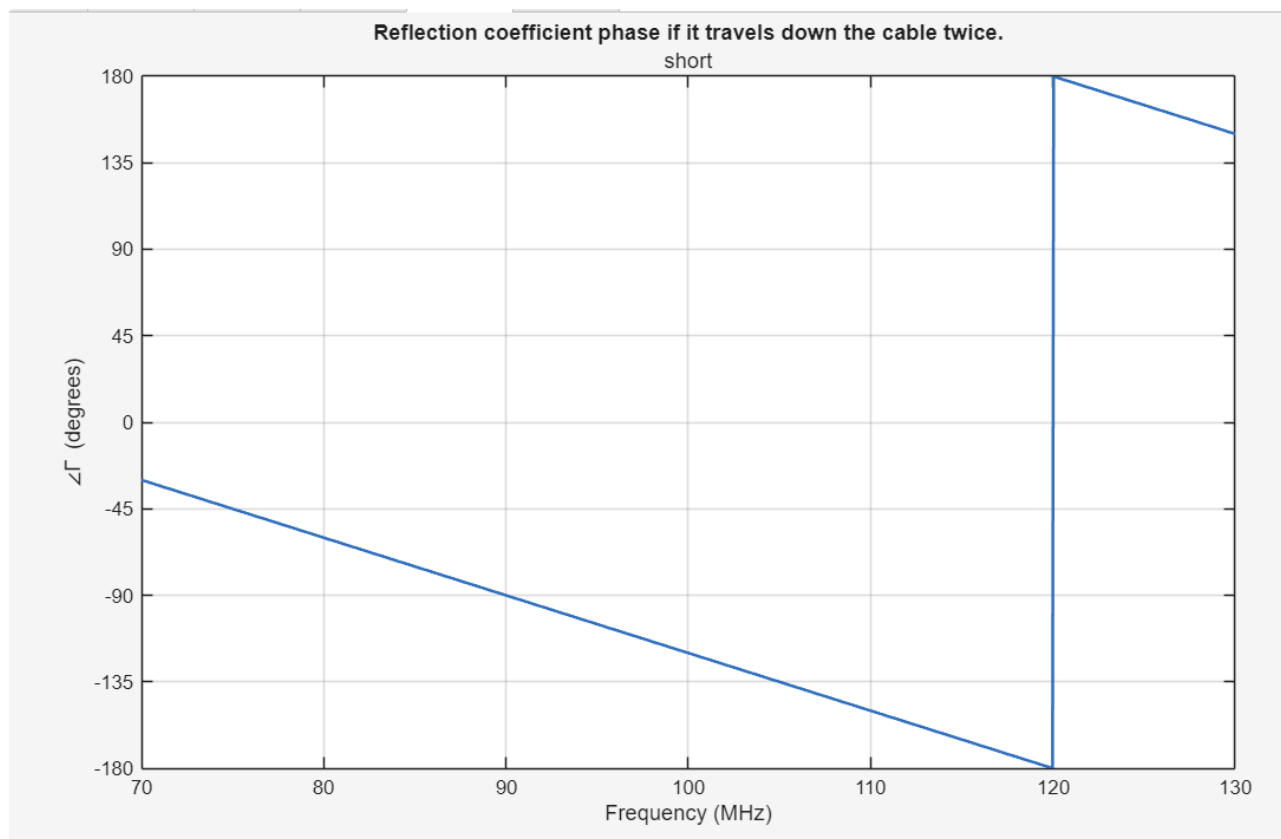
There are $\Delta\phi = (2L\% \lambda) \frac{2\pi}{\lambda}$ (the modulo, aka remainder) as a fraction of wavelengths phase accumulated going to the end of the wire (two traversals). $\lambda = \frac{0.8c}{f}$

This gives us the angle of our reflection coefficient, while the magnitude is always 1. We set it to start at -1 , since the reflected wave should cancel out the forward wave. The phase/ angle of our reflection coefficient is just the accumulated phase across the length of the cable (traveling one way, not two). This is $e^{i2\pi \frac{\Delta}{L}}$, where L is the length of the cable.

This feels super duper weird. If the wire is grounded, we need $\Gamma = -1$. If we allow the angle of Γ to vary with electrical length (i.e. what phase of the wave the reflection is coming from), then we could measure a voltage standing wave depending on the electrical length of the wire!

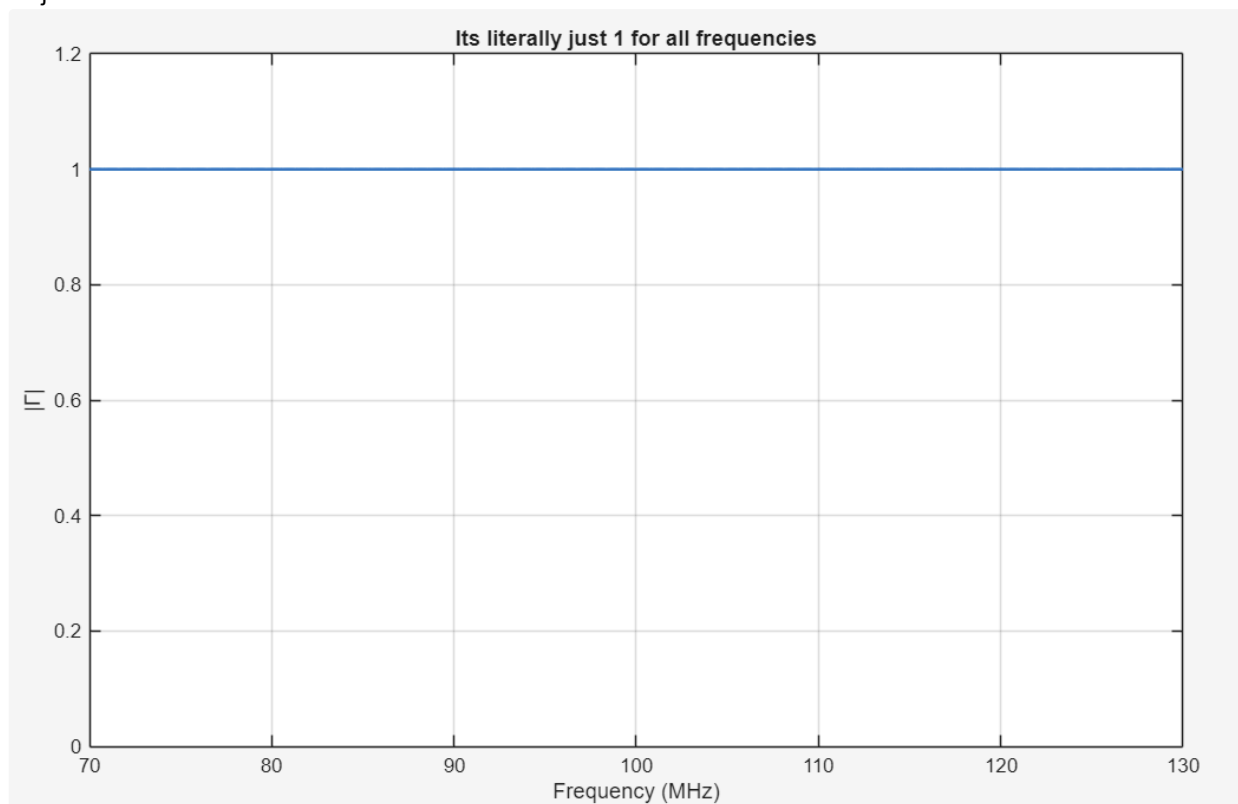
Draw a plot of reflection coefficient phase vs. frequency for this combination of cable and load. Please provide a wrapped phase plot (search online/ask an AI chatbot what wrapped vs. unwrapped phase is, if necessary).





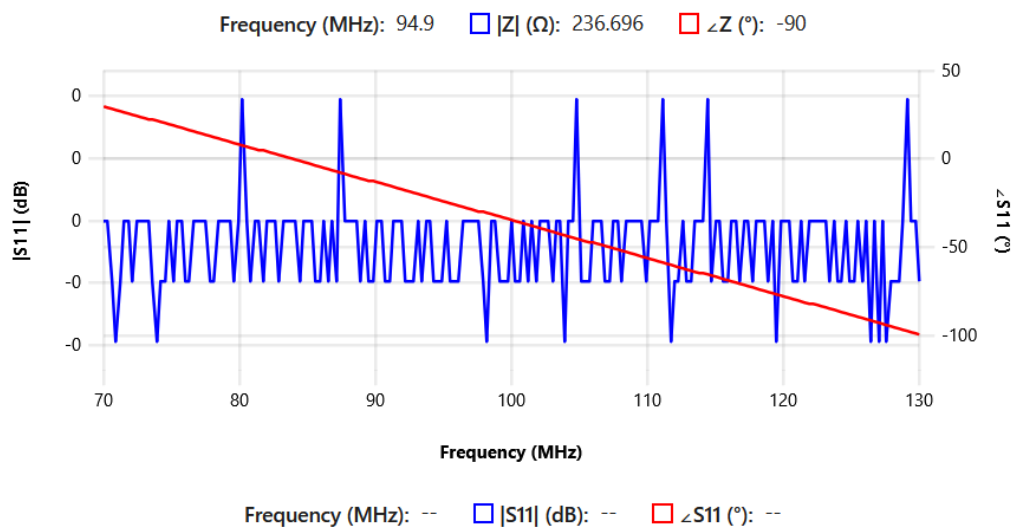
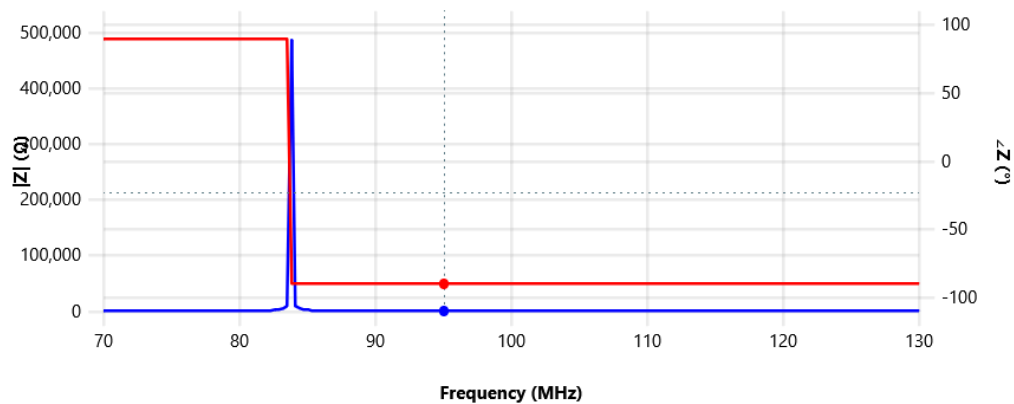
Draw a plot of the magnitude of the reflection coefficient vs. frequency for this combination of cable and load.

Its just one.



Checking this with the online smith chart plots:

Note that the left Y axis on the lower plot is in dB. 0 dB means that the ratio is 1, so $\Gamma \approx 1$ for the whole plot. The phase, which goes from 50 to -100 degrees across this span is what we care about.



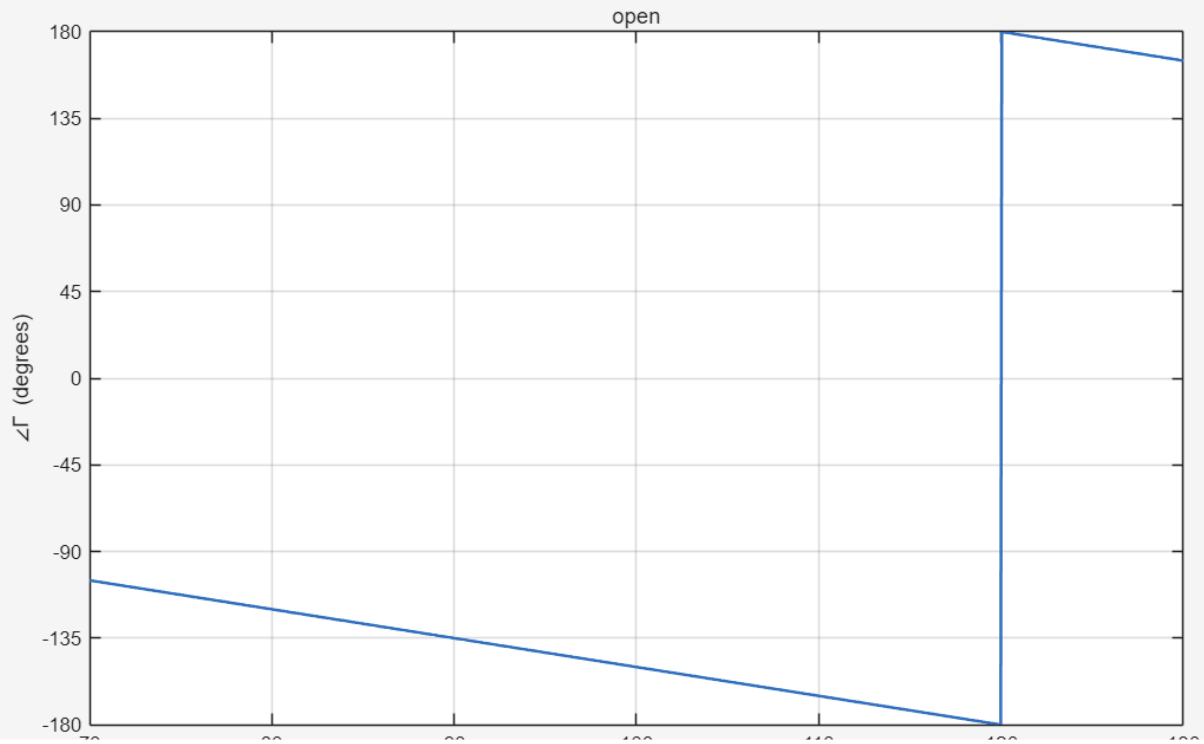
I'm confused by the magnitude, but phase looks right ish? (just starting at +45 instead of -45).

Repeat parts a-c assuming the following conditions: 1-m cable with open termination, 1-m cable with 50-Ohm termination, 10-m cable with short termination.

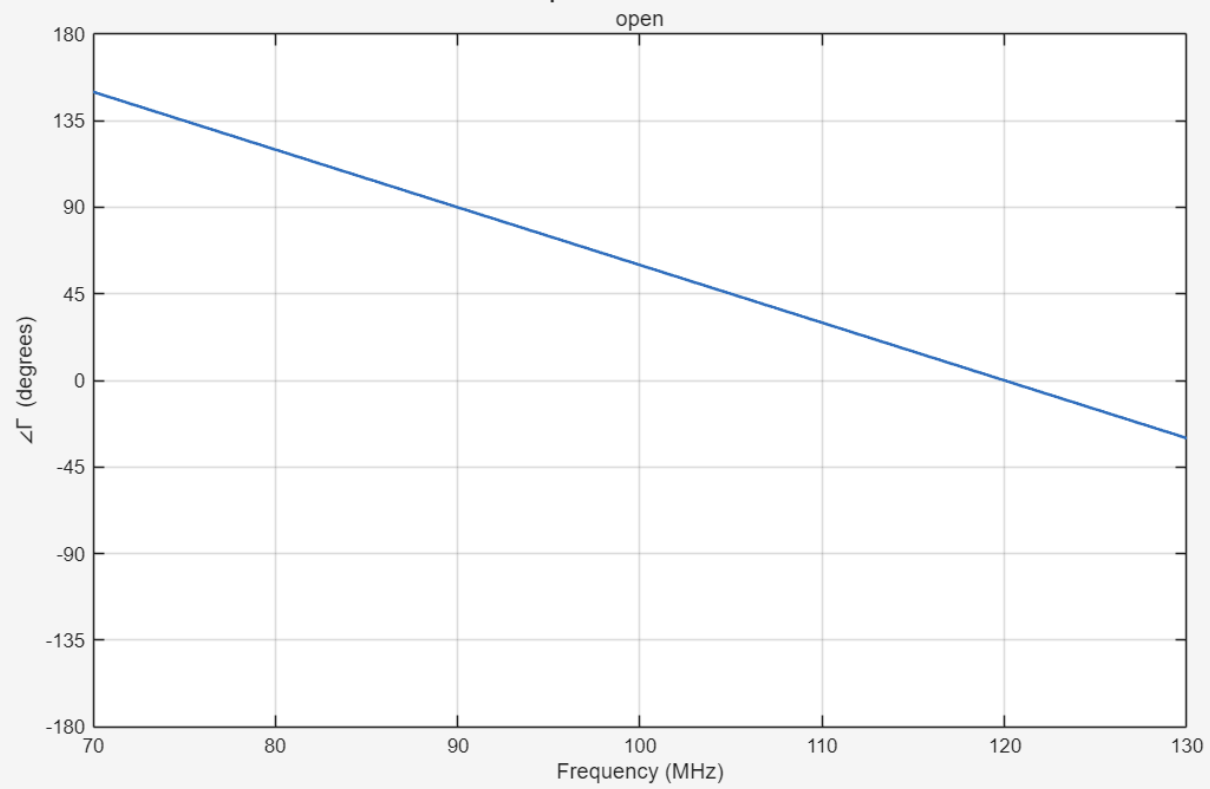
Open Termination 1 m cable

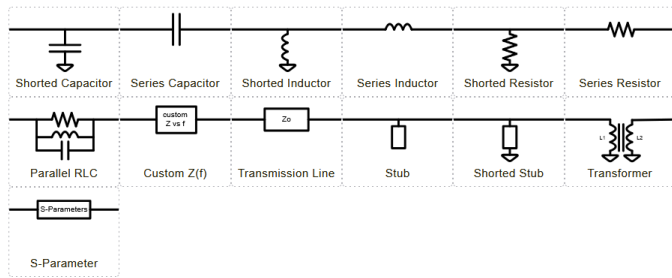
For the open termination, I just changed $\angle \Gamma$ from starting at $-\pi$ to 0 before applying the phase from traveling across the wire.

Reflection coefficient phase if it only travels down the cable once

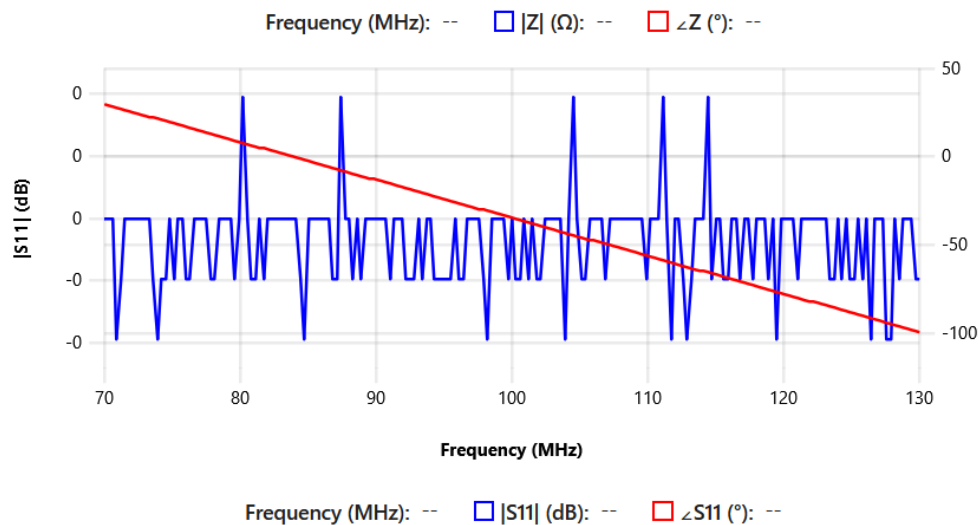
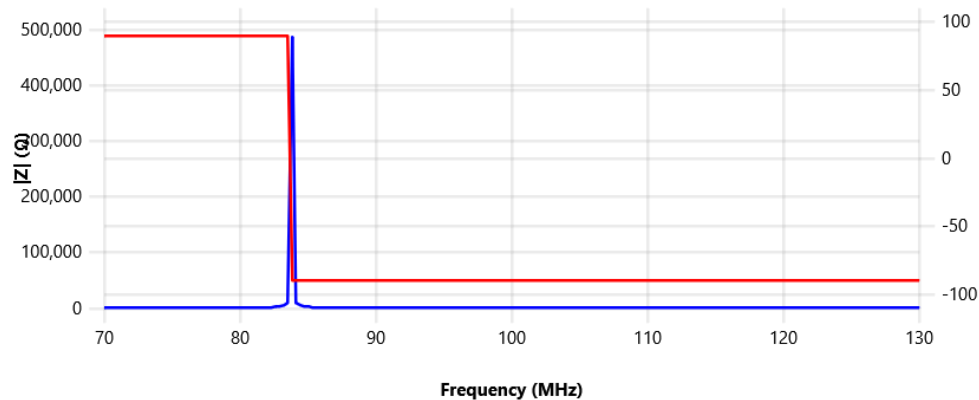
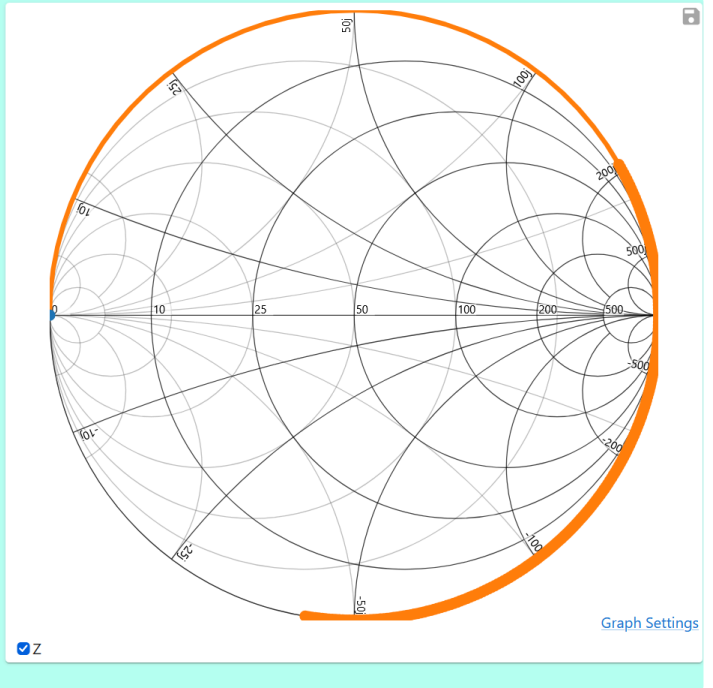
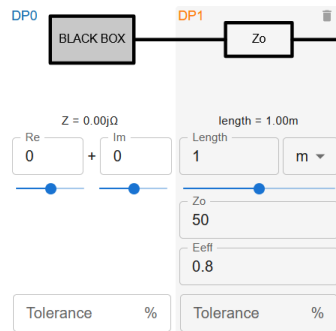


Reflection coefficient phase if it travels down the cable twice.



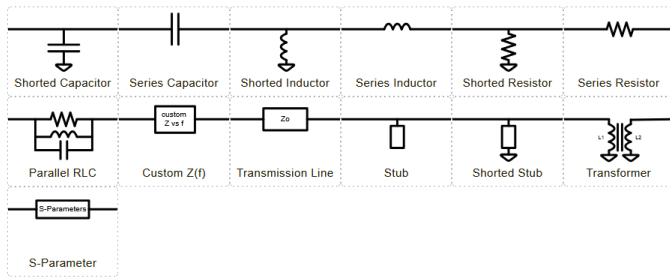


Click components above to add them to the circuit below. Impedance is looking towards the BLACK BOX

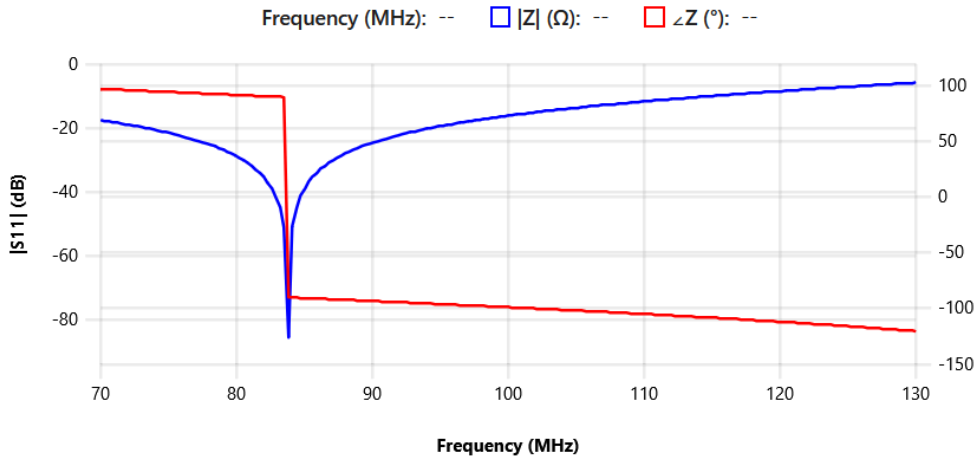
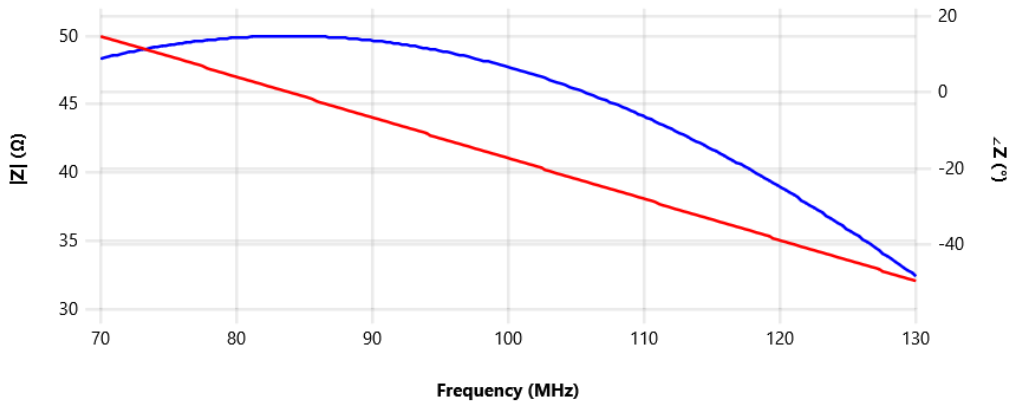
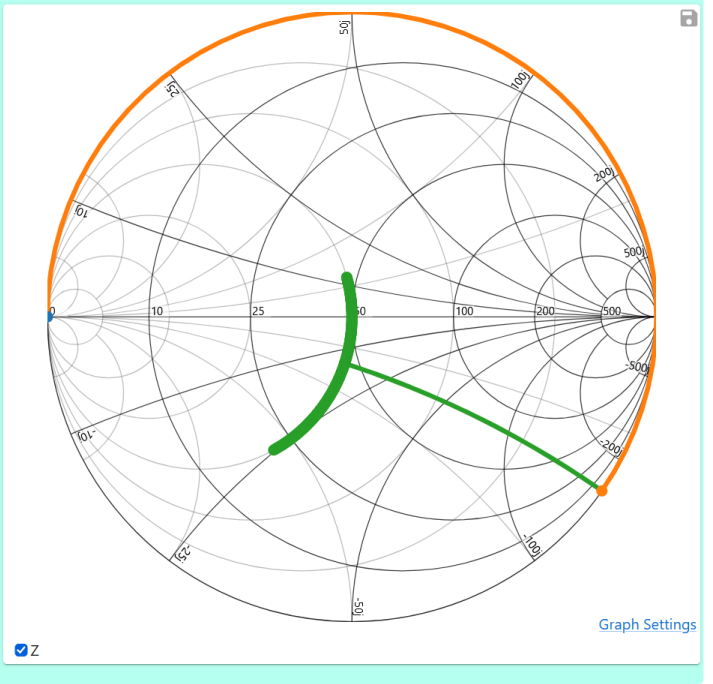
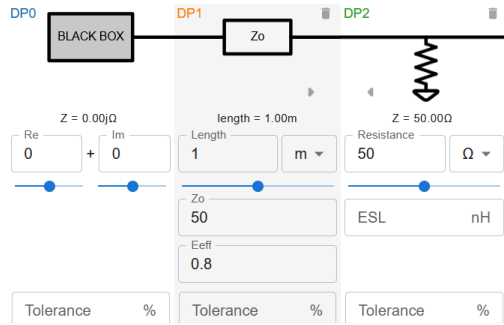


50 ohm termination

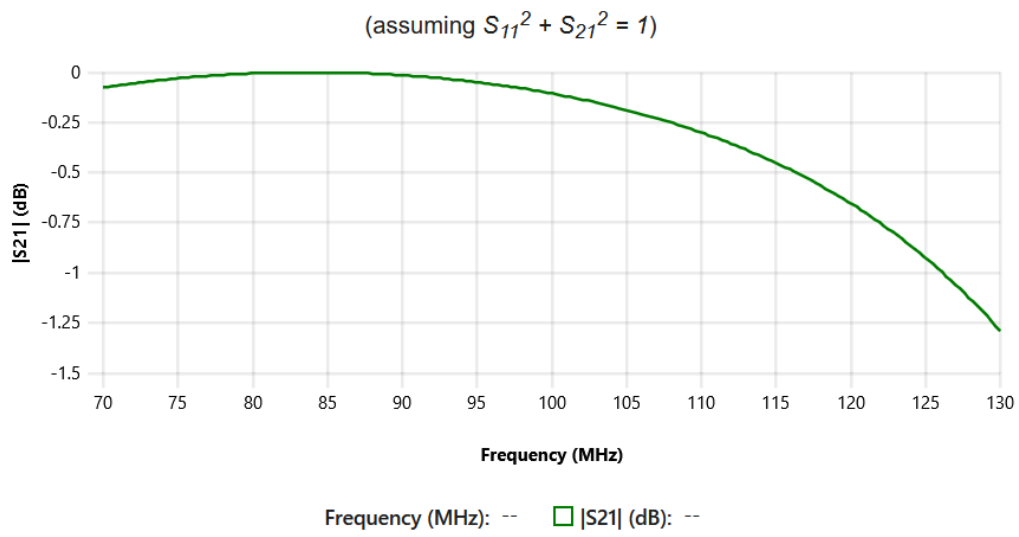
In my head, I was imagining zero reflections, so angle doesn't matter (and is not defined for polar coordinates). Therefore, I only did the online smith chart, because i assumed it would just be $|\Gamma| = 0, \angle \Gamma = NaN$.



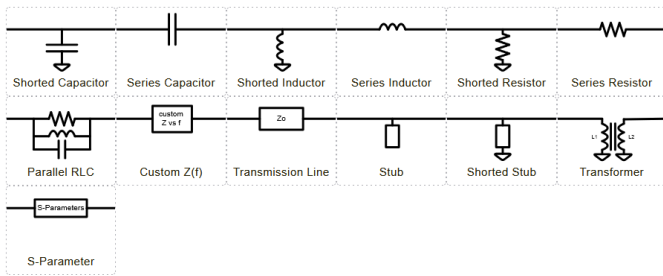
Click components above to add them to the circuit below. Impedance is looking towards the BLACK BOX



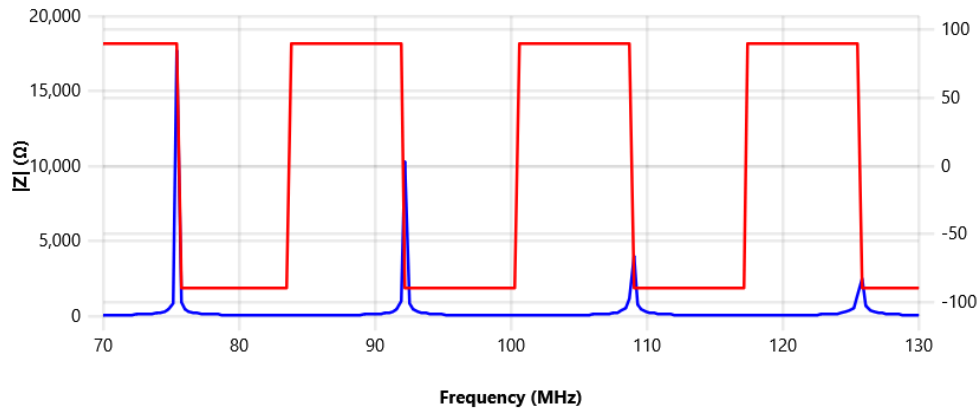
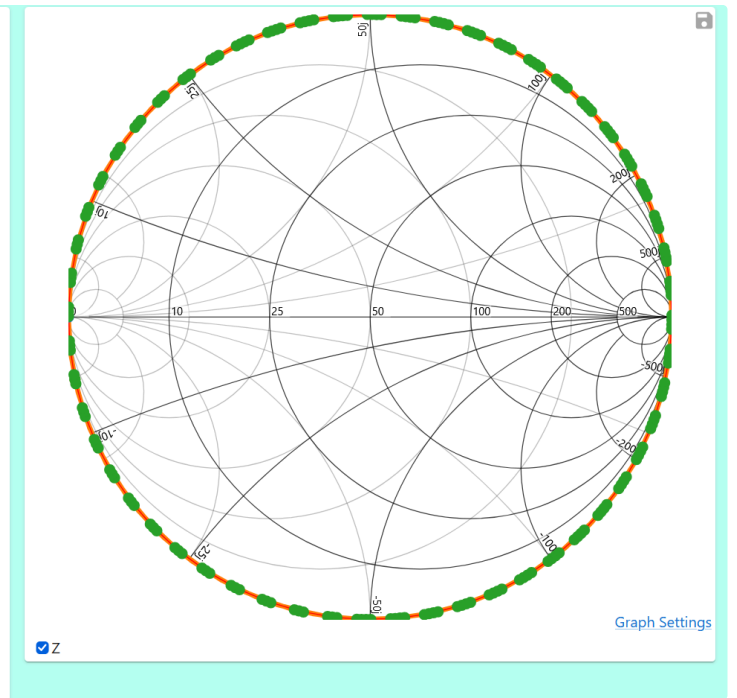
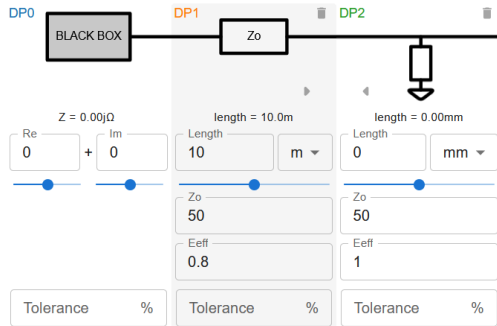
Frequency (MHz): -- █ $|S11|$ (dB): -- █ $\angle S11$ ($^\circ$): --



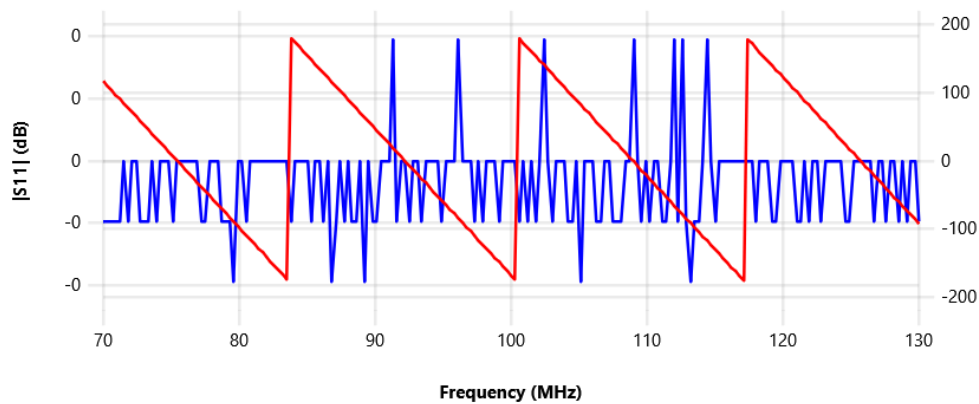
10 meter with shorted termination



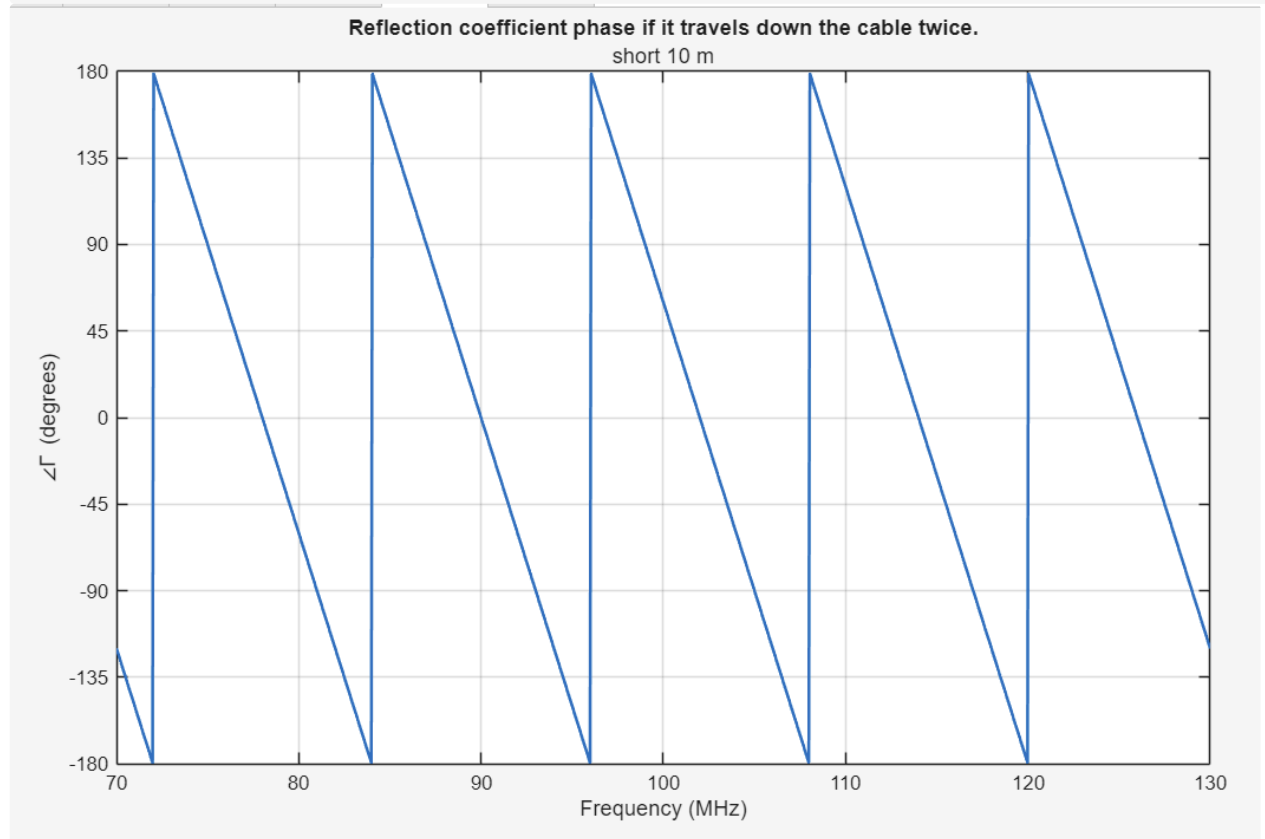
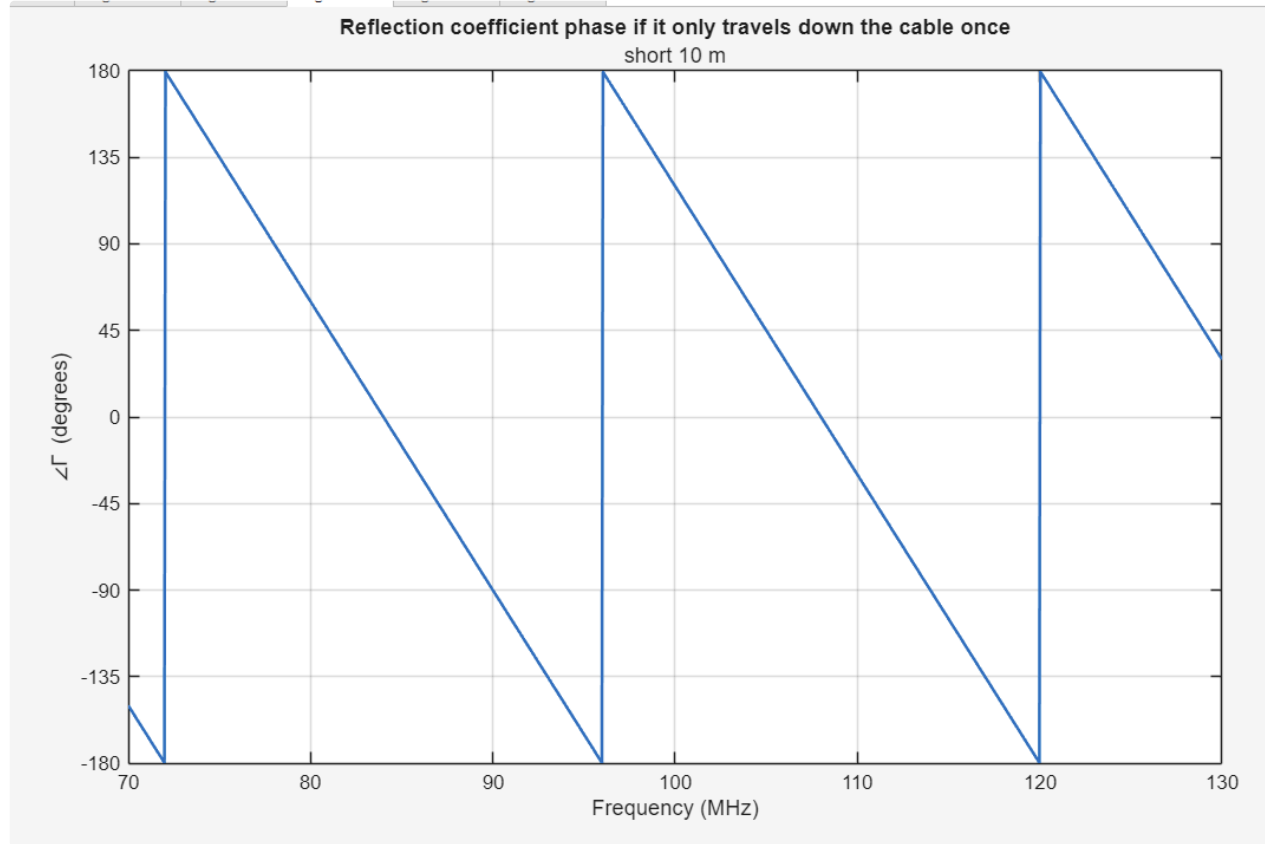
Click components above to add them to the circuit below. Impedance is looking towards the BLACK BOX



Frequency (MHz): -- | $|Z|$ (Ω): -- | $\angle Z$ ($^\circ$): --



My matlab is off by a negative sign for the phase, which I'm confused by.



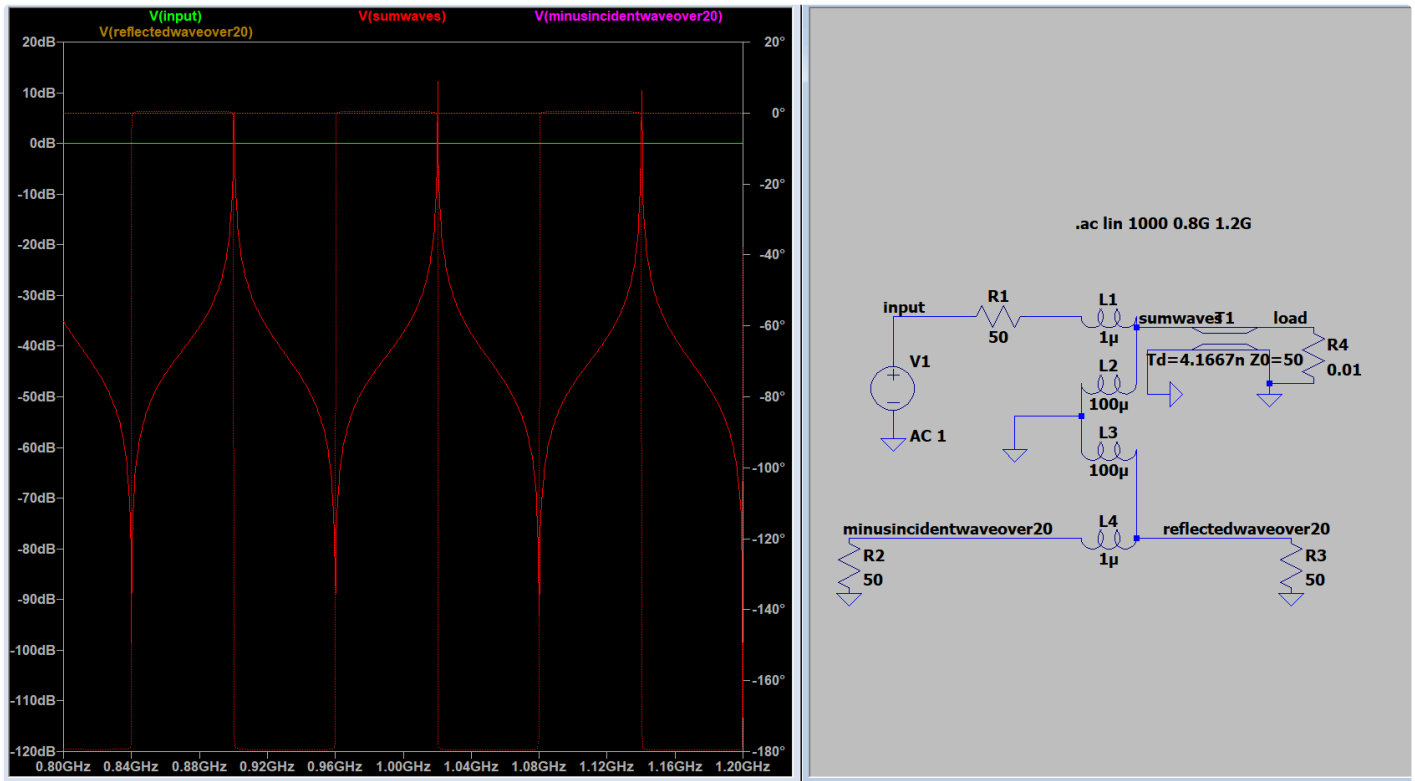
When you calibrate a VNA (more on how we do that later), you multiply your reflected wave by a complex number that cancels out the effect of the cable. The most prominent effect of the cable is that it adds a lot of phase to our measurements, which you can see by reflecting on the propagation equation:

$$V(x, t) = V(0, t) * e^{-\alpha x} * e^{-jkx}$$

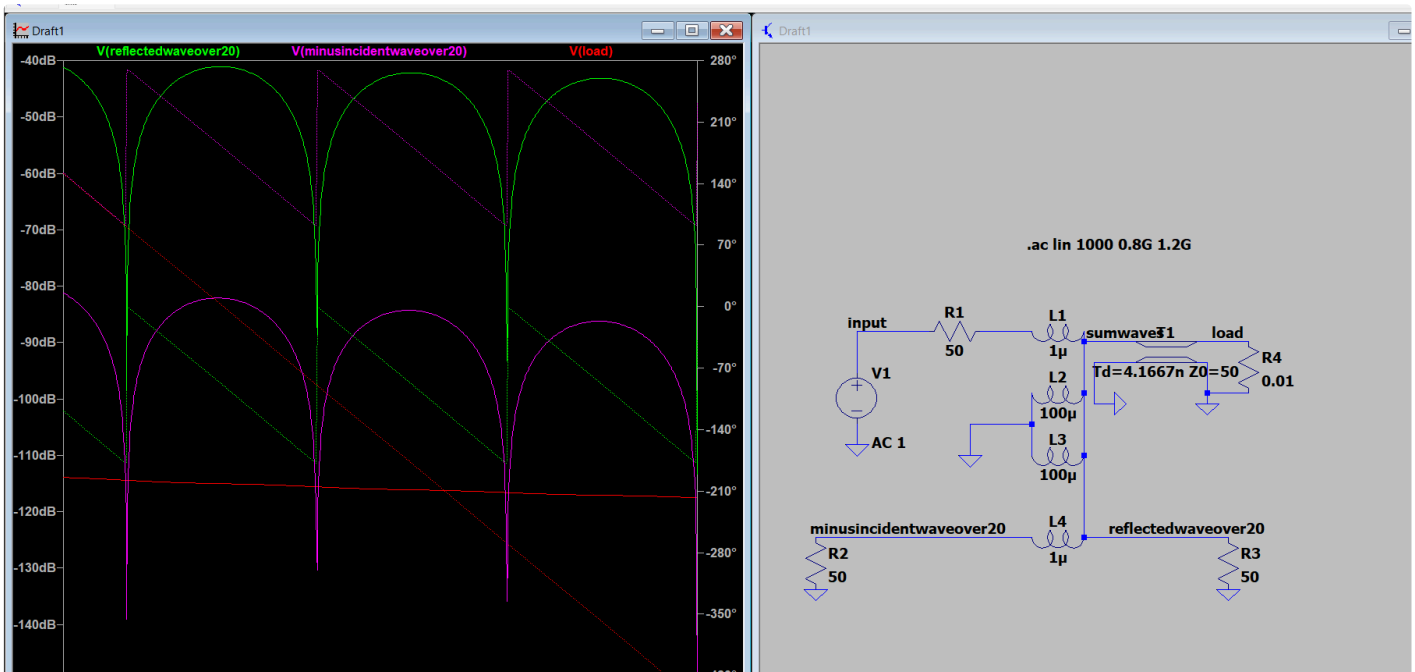
. We can capture the effect of

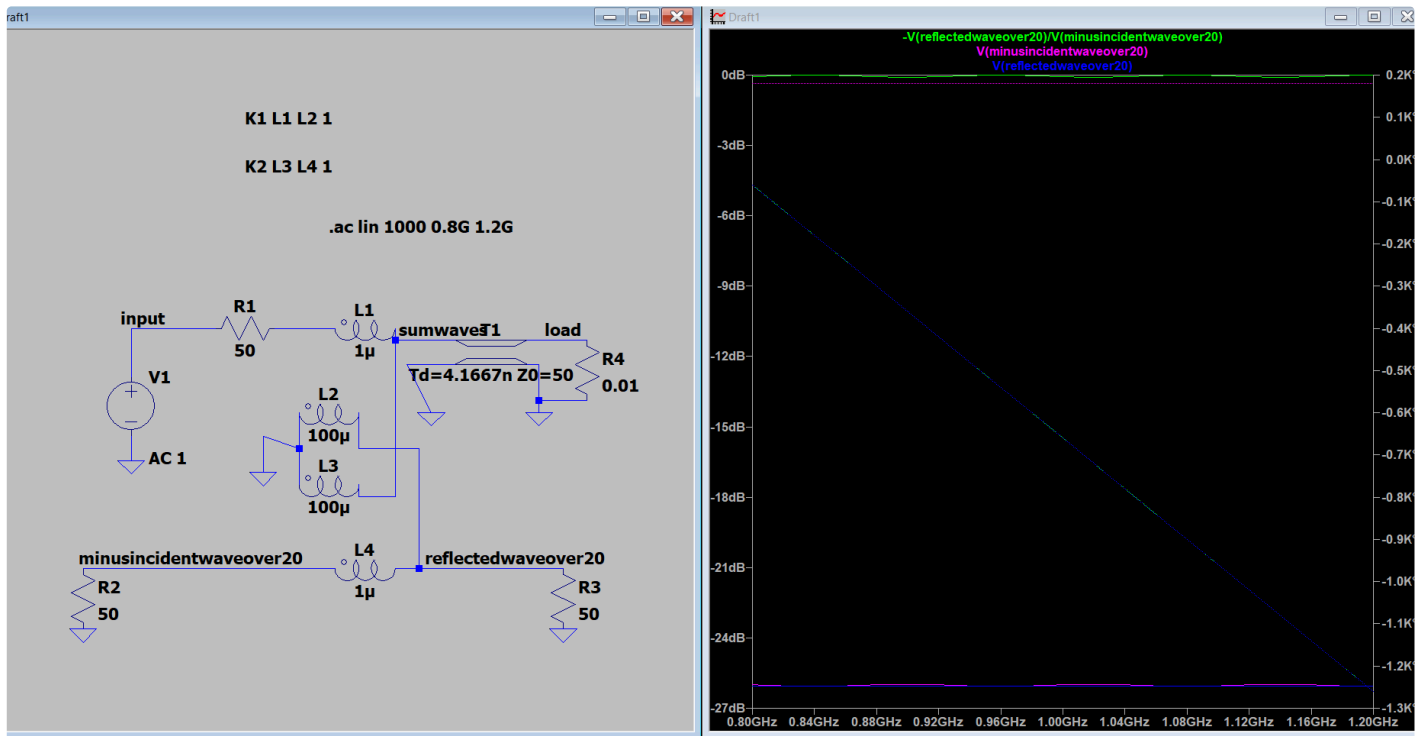
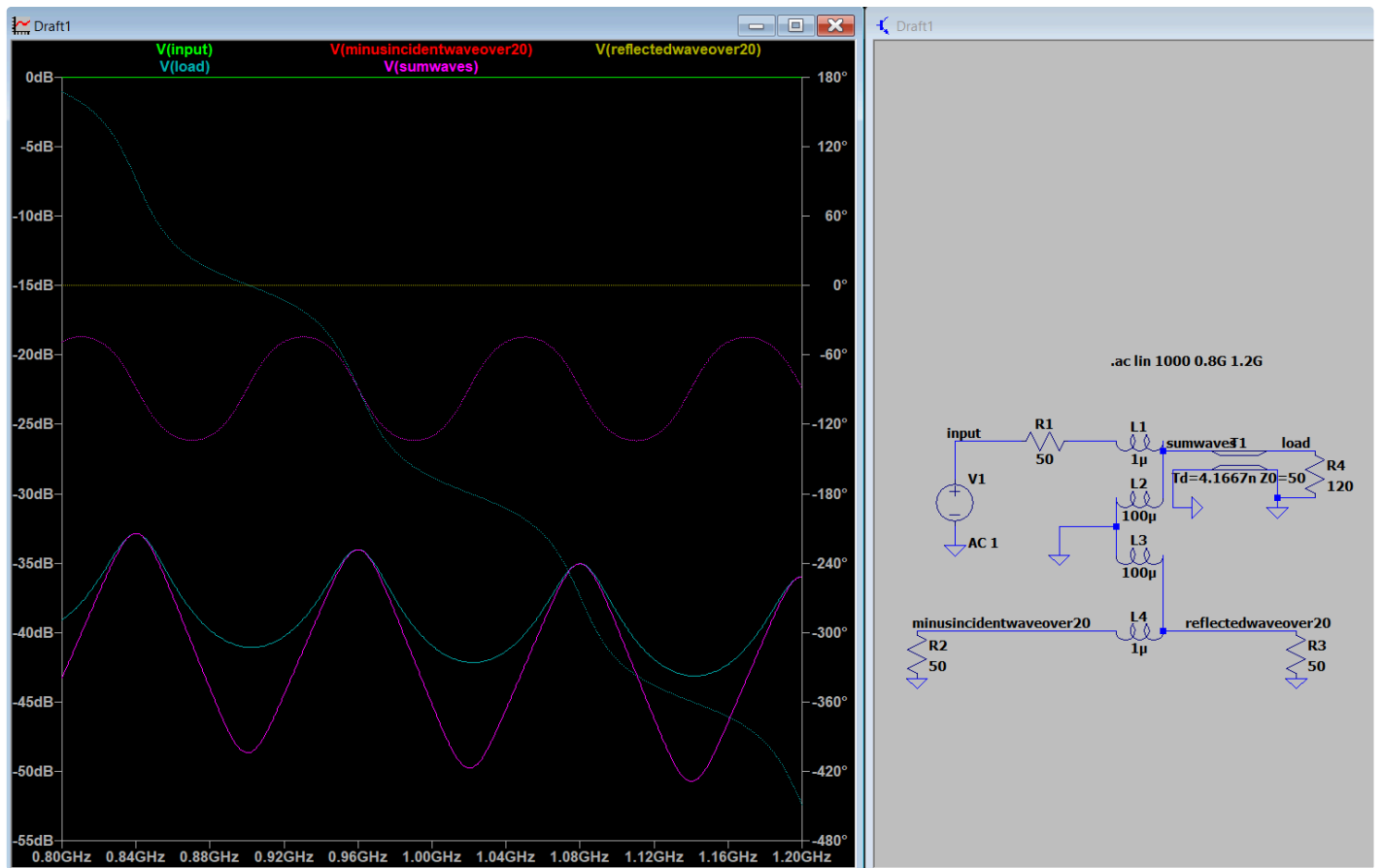
calibration in our analysis and simulation by multiplying our uncalibrated results by e^{+jkx} to cancel out the phase from the cables. In this question, we're going to practice doing that using a simulation of a VNA.

3. Figure 3 (below) is a starting point for simulating a VNA. The schematic uses ideal mutual inductance (the K statements) to represent the behavior of a directional coupler, a circuit element that separates forward and reverse waves. (We'll talk about these more in a few weeks.) The node labels indicate the wave voltages that will be observed on that node, so the reflectedwaveover20 node will display the reflected wave voltage scaled by a factor of 20. Build and simulate Figure 3.



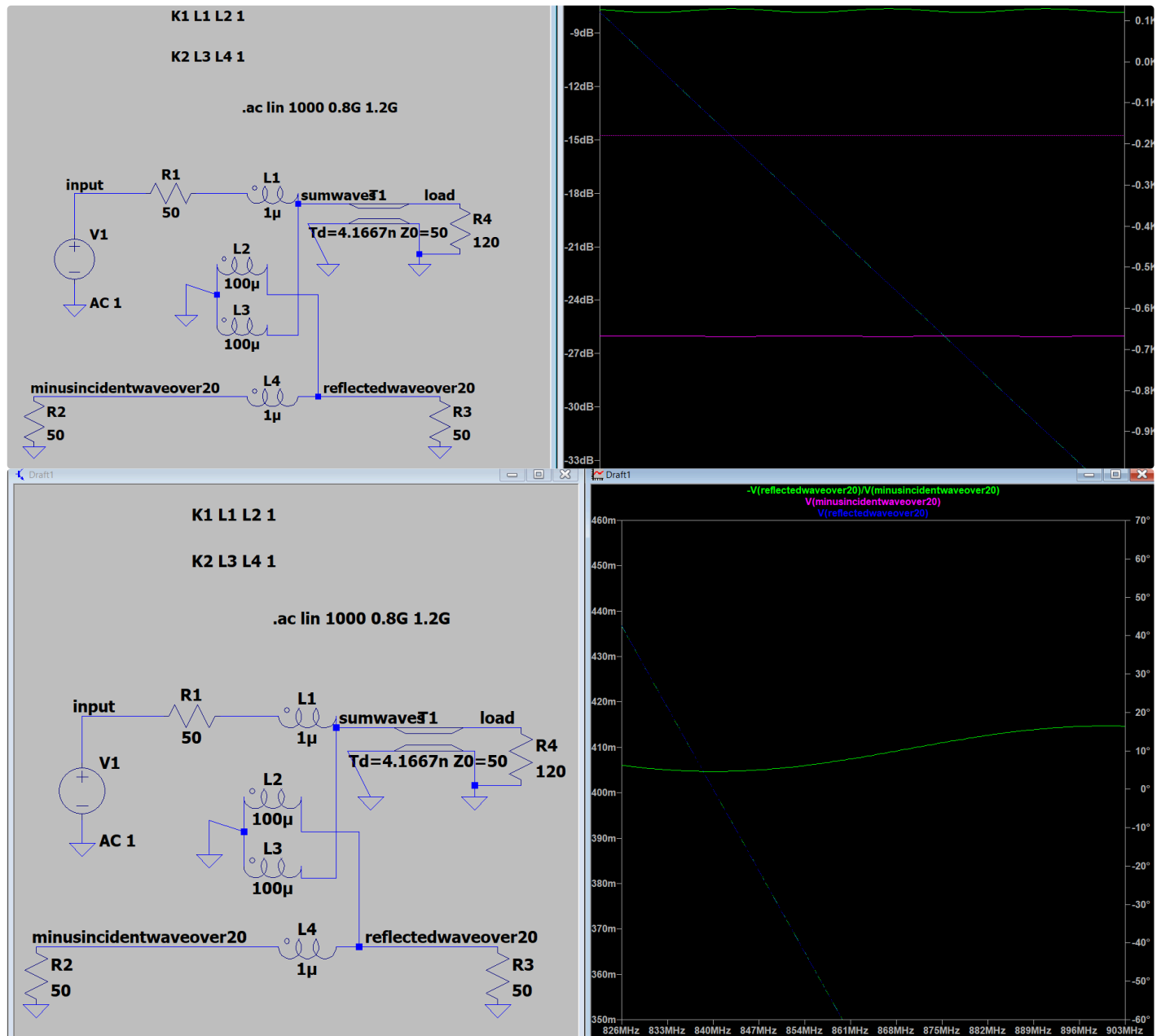
OR





3. Modify figure 3 to express each of the situations in theory question 1. Make sure simulations and analysis match.

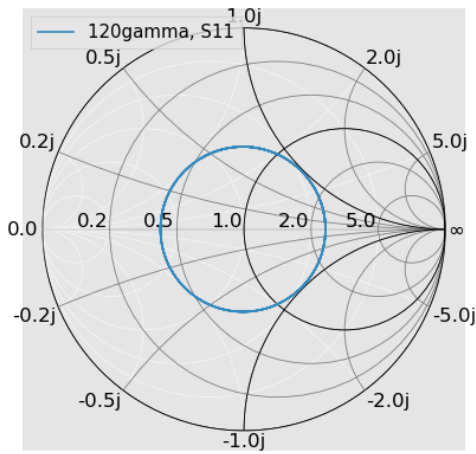
120 Ohm Resistor



zooming in, this is about 0.412 (it oscillates up and down) which is what we calculated.

Using scikitrf (which was a friggen pain).

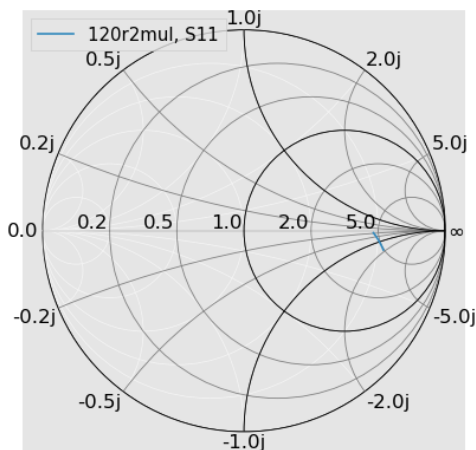
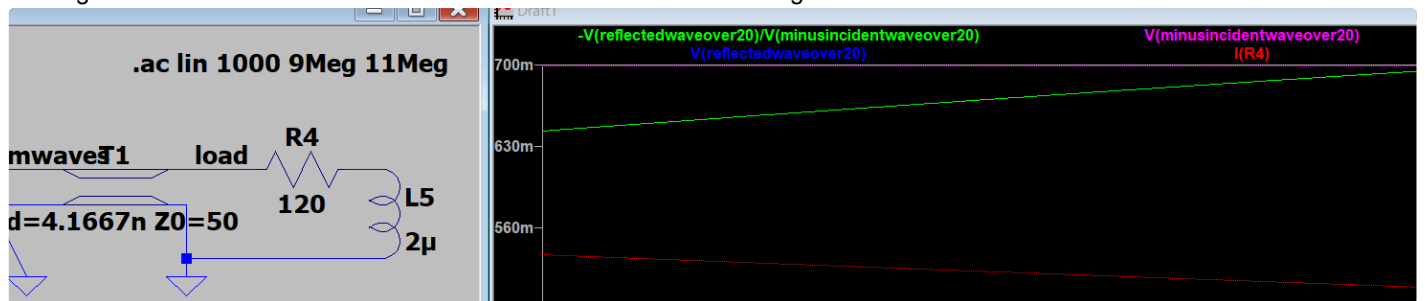
Note: I did the plots that look like this myself, and then later i have an "ai section" where I just had it modify my code to plot all of them on top of each other because I didn't want to waste time copying the new file paths. I didn't realize they had to be overlaid. I'm lazy. Pass fail.



This doesn't look right, but maybe it is because its reflection that we care about.

120 Ohm Resistor in series with a 2 uH inductor at 10 MHz

I changed the simulation to be 9 MHz to 11 MHz to show the 10 MHz region.



I had calculated $\gamma = 0.68 \angle 24.4^\circ$. We see roughly this - 0.64 ish V.

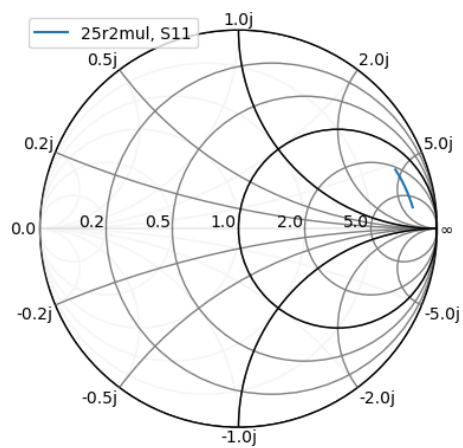
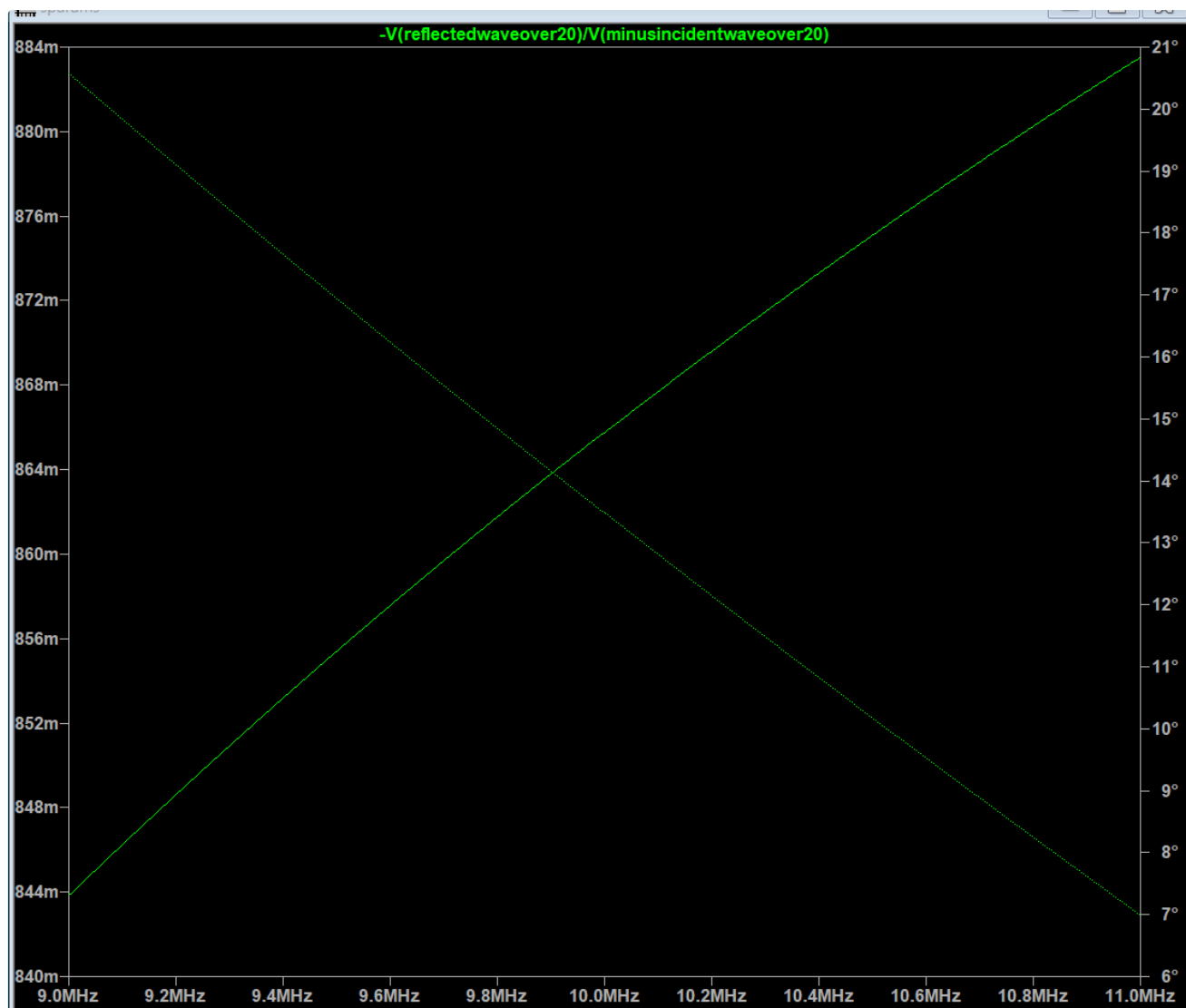
This has a negative angle near zero, which doesn't agree. However, this was the reflection coefficient at the load, so maybe we have to go back across the line? At 10 MHz we have the phase change

$$e^{-j\omega T_d}$$

where $T_d = 4.1667n$ and $\omega = 2\pi * 10^7 Hz$, so a phase shift -0.26 rad, = -14 degrees, so the angle would become 10.4° . This is still a positive angle, but closer to zero? I'm not sure why they disagree. I want to know this, but pass fail <3

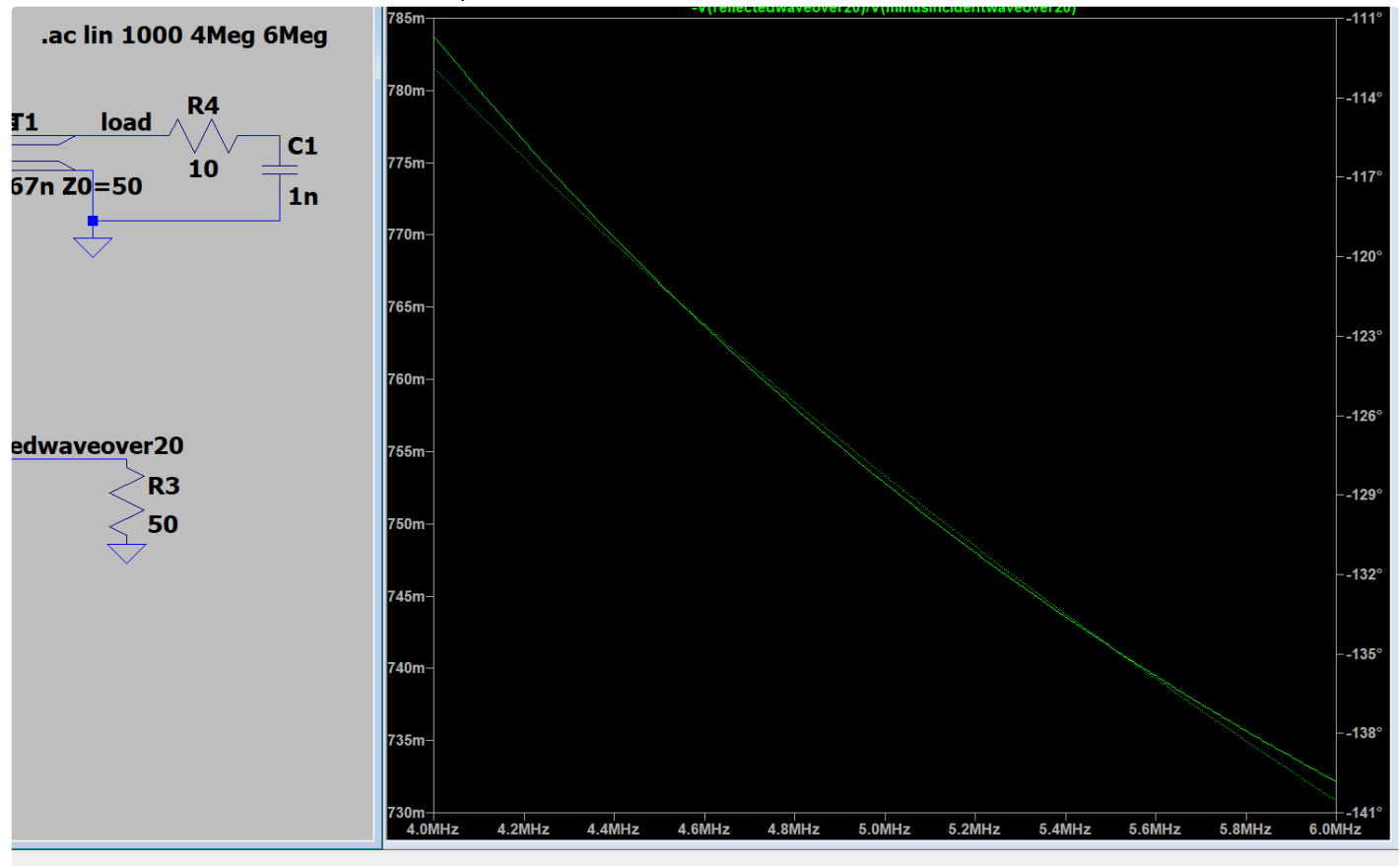
A 25-ohm resistor in series with a 2-uH inductor at 10 MHz

I had calculated $0.88 \angle 45^\circ$. We see roughly 0.86 at 10MHz, which is close enough.



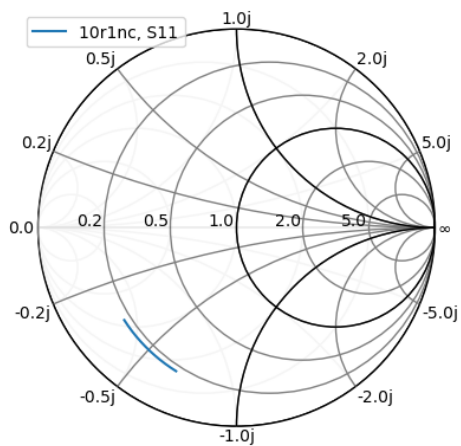
Positive angle, a bit less than 45. Same frequency, so we would maybe have the same phase shift - this looks to be about 20 degrees at the center, and the phase across the wire is about 10, so off by 10 degrees? Maybe its phase accumulated through the rest of the s parameter measuring circuit?

A 10-ohm resistor in series with a 1-nF capacitor at 5 MHz



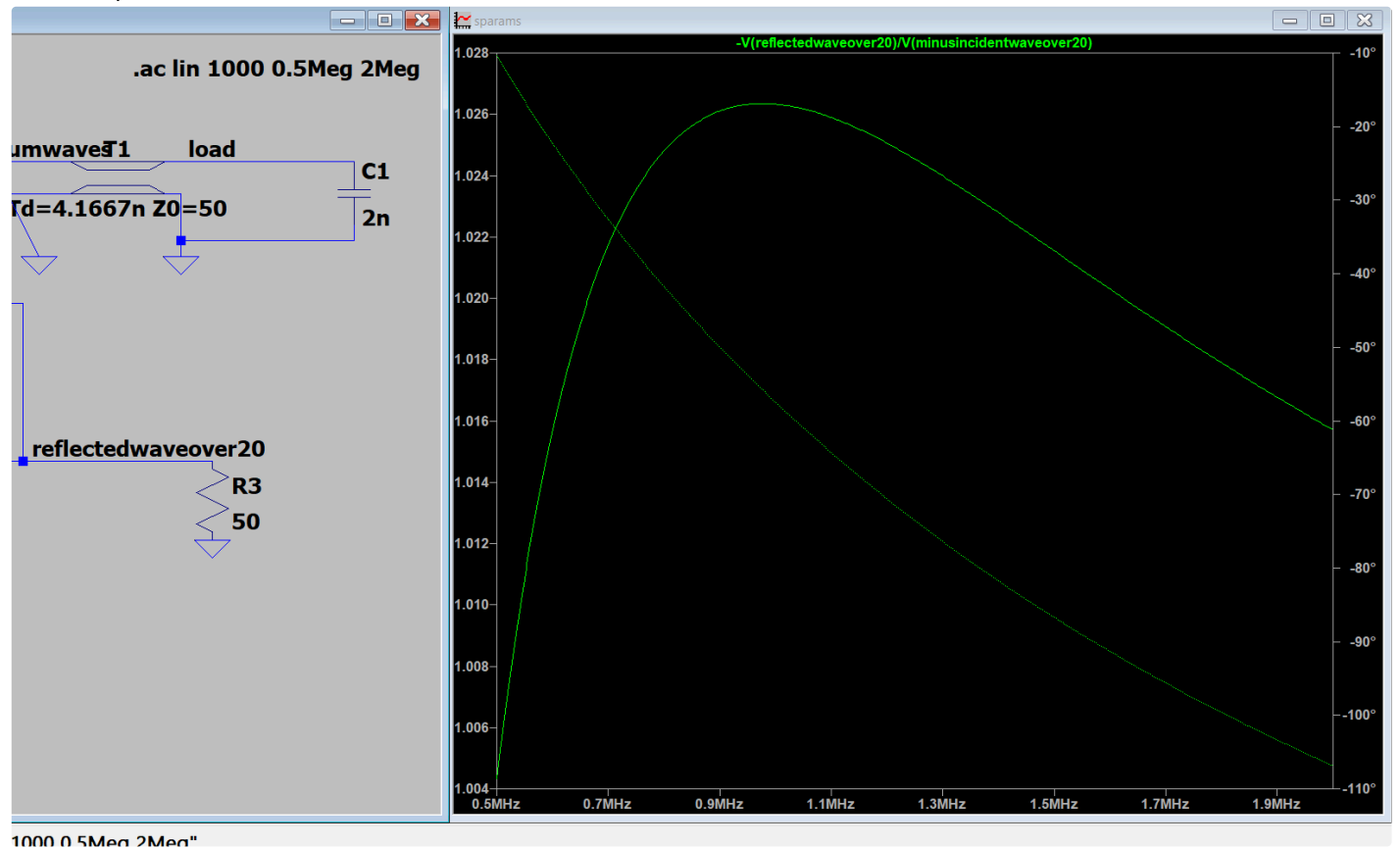
at 5 MHz, this is roughly 0.745. I calculated

$$\Gamma = 0.75 \angle -113.592$$

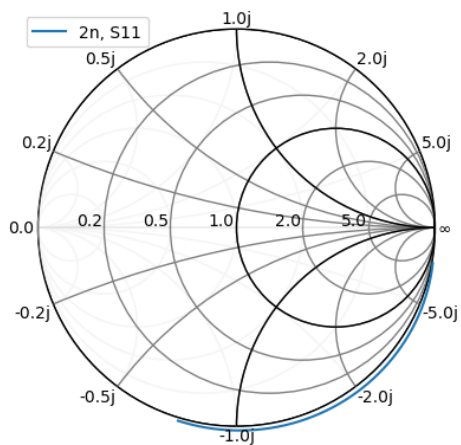


the angle looks a bit further than 23 past 90, but maybe it is. Close enough.

A 2-nF capacitor at 1 MHz

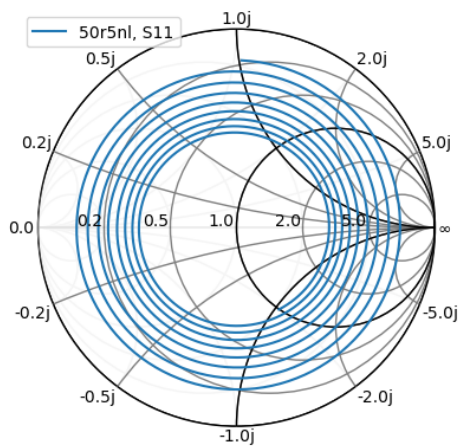
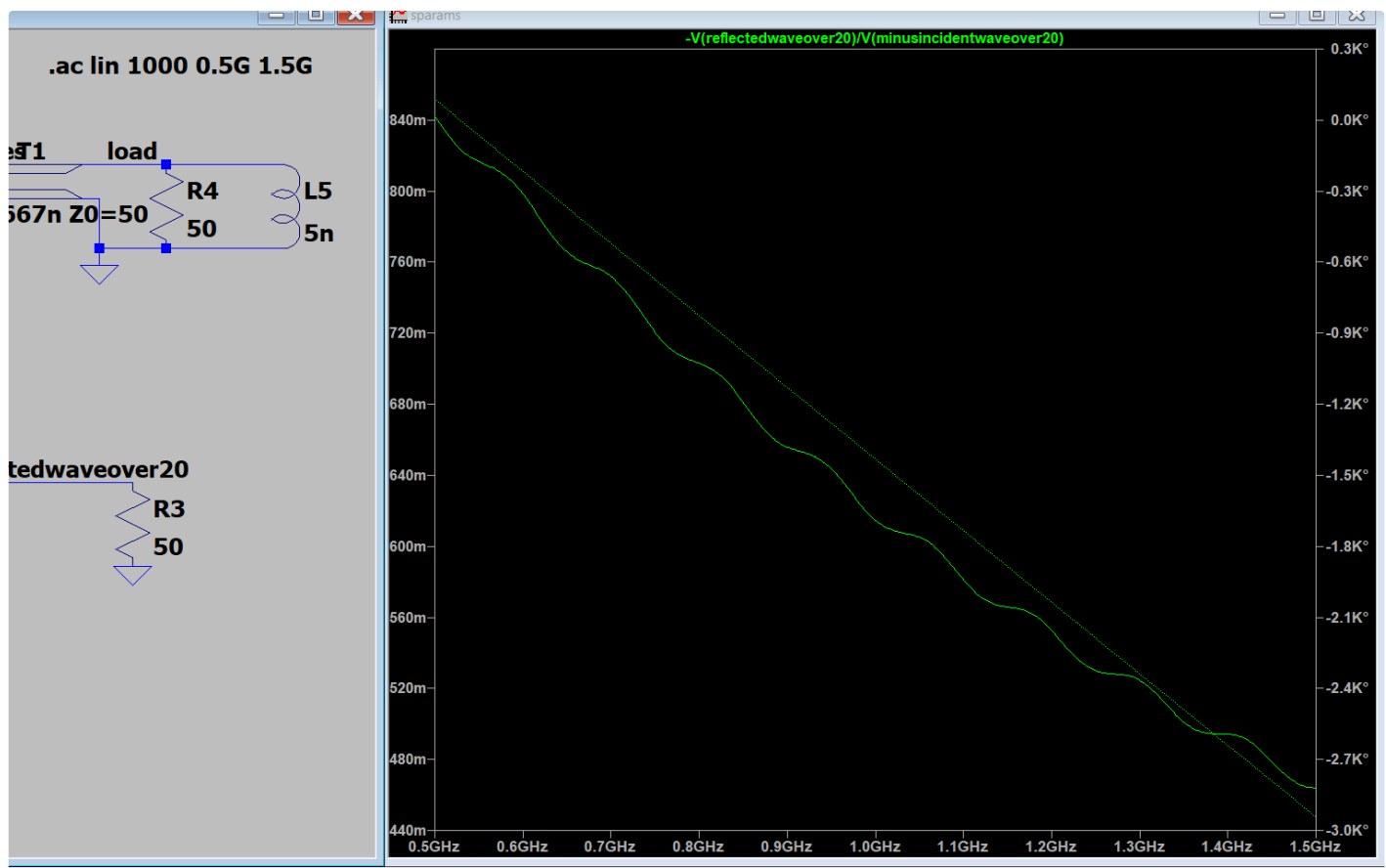


I calculated $\Gamma = 1 \angle -64^\circ$



one third of the way (bc i did 0.5MHz to 2 MHz) looks to be around there ish.

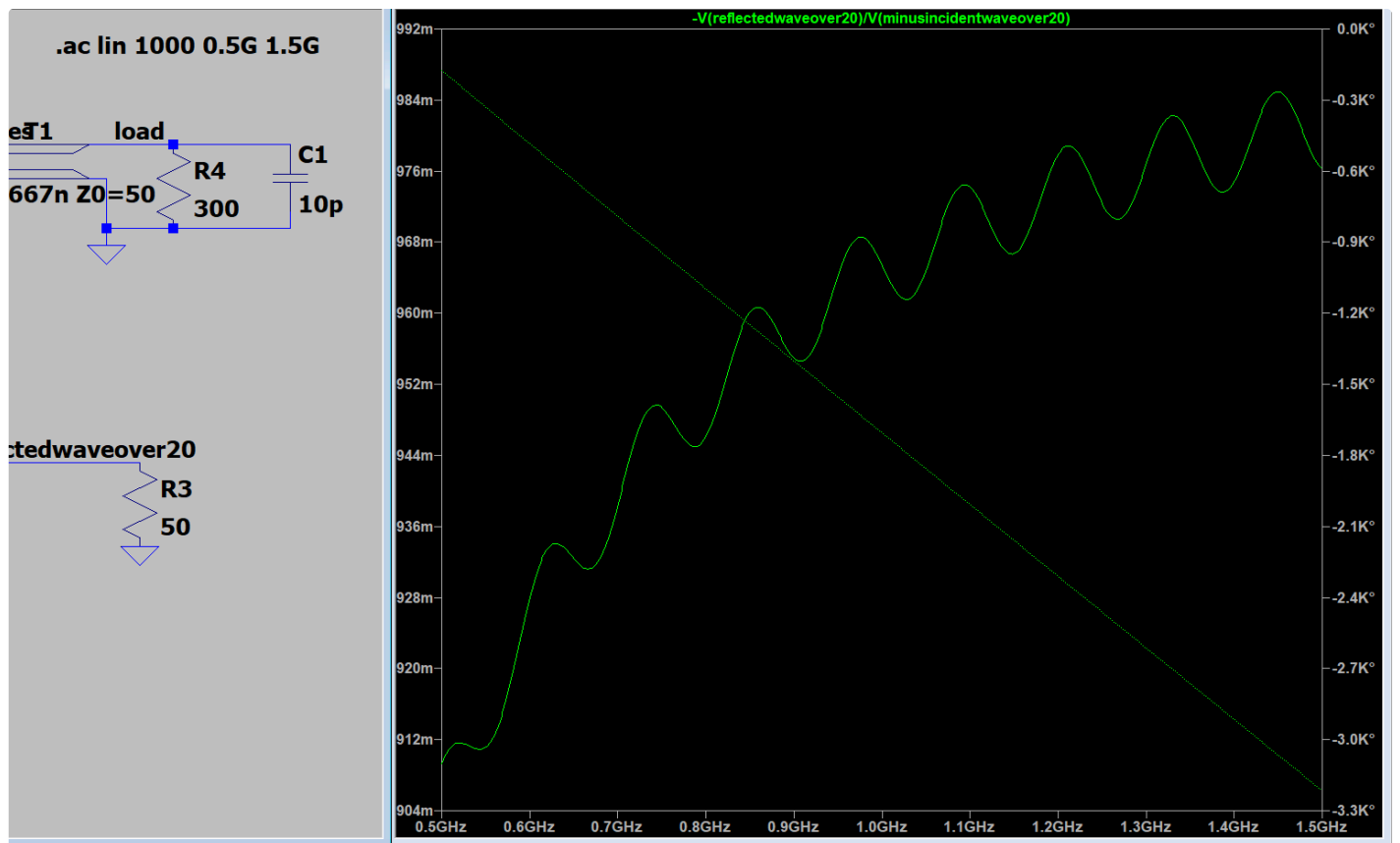
A 50-ohm resistor in parallel with a 5-nH inductor at 1 GHz



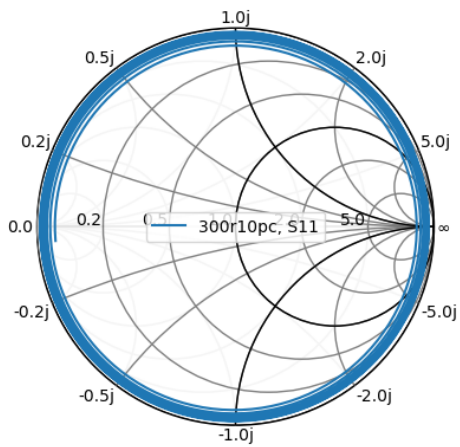
What?!?

That's funky, but makes some sense - we are changing the regime from $z = \frac{1}{\text{less than } 1}$ to $\frac{1}{\text{greater than } 1}$ because nF is 10^{-9} and Giga is 10^9 . We are on our way to becoming an open circuit, aka going to the outer circle. Cool.

A 300-ohm resistor in parallel with a 10-pF capacitor at 1 GHz

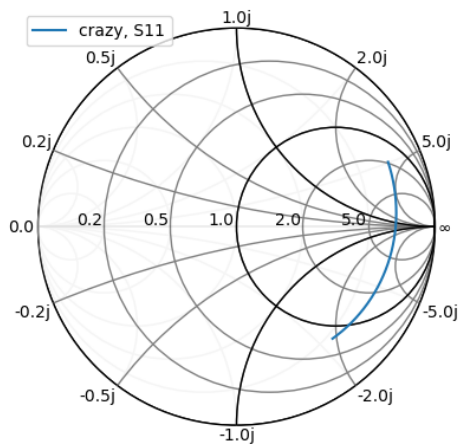


Same here!



Makes sense.

A 30-ohm resistor in parallel with a 50-nH inductor, in series with a 20-pF capacitor, at 100 MHz.



I calculated

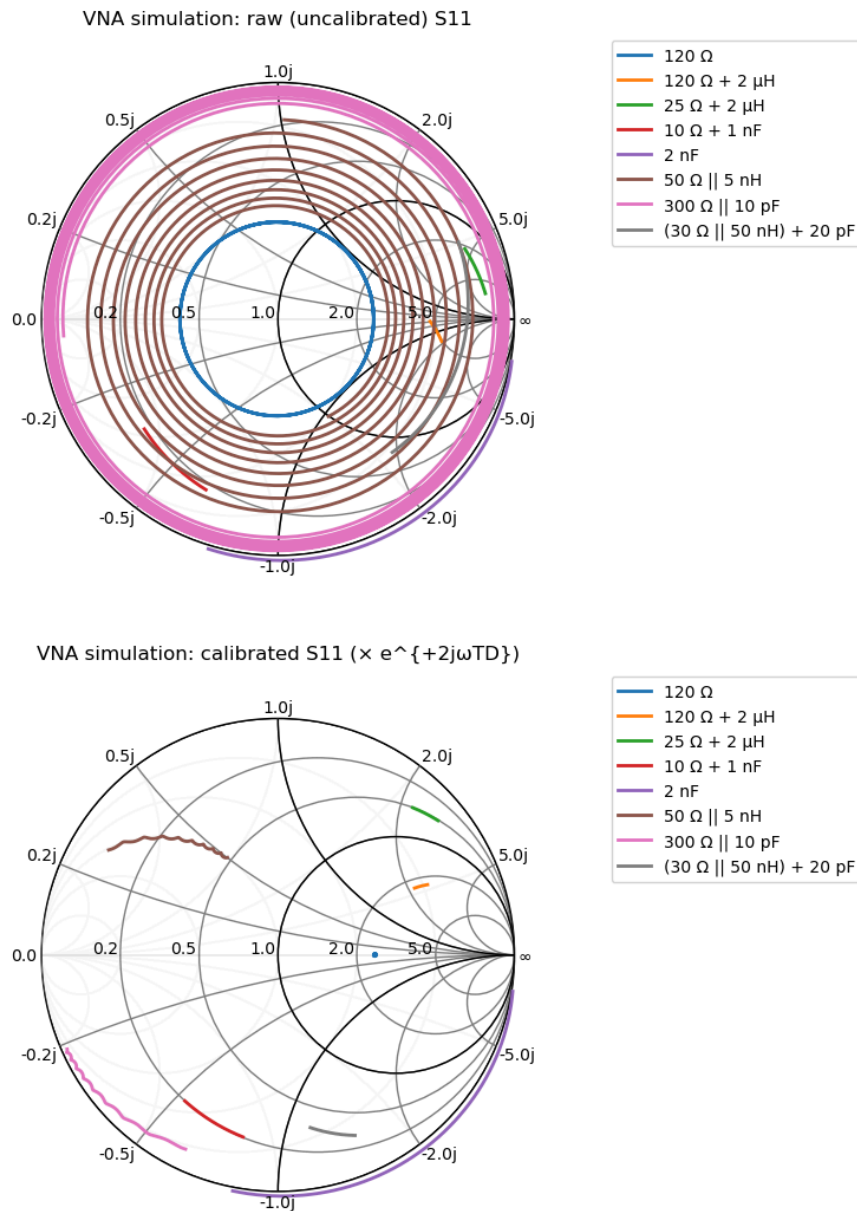
$$\Gamma = 0.96 \angle -144^\circ$$

Close to the edge (right $|\Gamma|$), but the phase is totally different than what I got.

- Plot $\angle \Gamma(f)$, $|\Gamma(f)|$ and a Smith Chart of $\Gamma(f)$ for each of your schematics. ItSpice won't plot a Smith Chart for you, so you'll need to export your data and use an external program capable of making polar plots. Be sure to overlay Smith Chart reactance and resistance curves on any plot you make.
- I find that `scikit-rf`, a Python package, simplifies this process a great deal. Your mileage may vary. <http://scikit-rf.org/> for more information.
- Alternatively, you can convert an ItSpice AC simulation output file to a Touchstone s1p file, which can be read by Matlab. Be sure to use the in the real, imaginary output format for your AC simulation file, not the magnitude, phase format. Then (1) save the AC simulation output with a filename that ends in ".s1p", (2) replace the first line with "# Hz S RI R 50", and (3) replace all commas with spaces. Note that S_{11} and Γ are synonyms (for this lab). (here is the s1p spec, FYI: https://ibis.org/connector/touchstone_spec11.pdf)

7. Multiply your exported, uncalibrated data by a correction factor that removes the phase of the cables and then replot magnitude, phase and Smith Charts for your newly calibrated Γ . Compare your points to analysis.

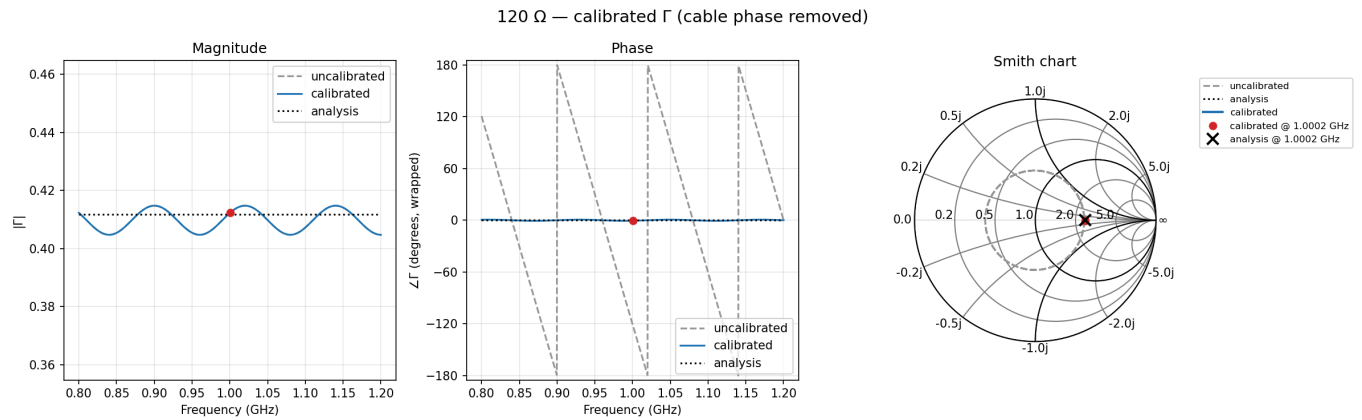
using ai and not doing the question manually. I went back and put these all on one plot using scikit-rf, shown first



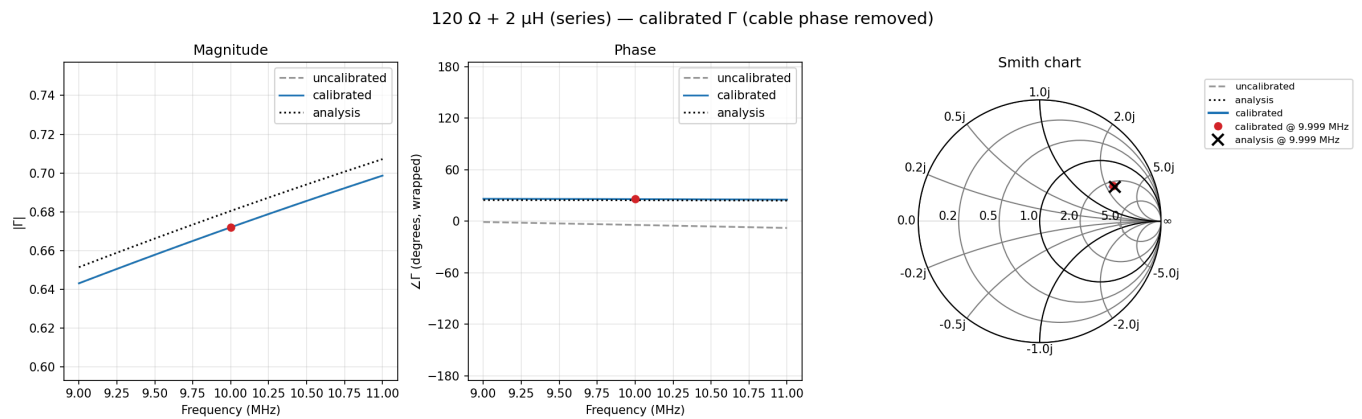
I uploaded the code I had to plot (it was super basic) and the filepaths for all my saved .s1p files, and each of the situations involved.

The only change was just " $g_{\text{cal}} = g_{\text{raw}} \cdot \exp(2j \omega * TD)$ ", for each ω and Γ that I calculated, with some random extra decoding stuff and plotting garbage.

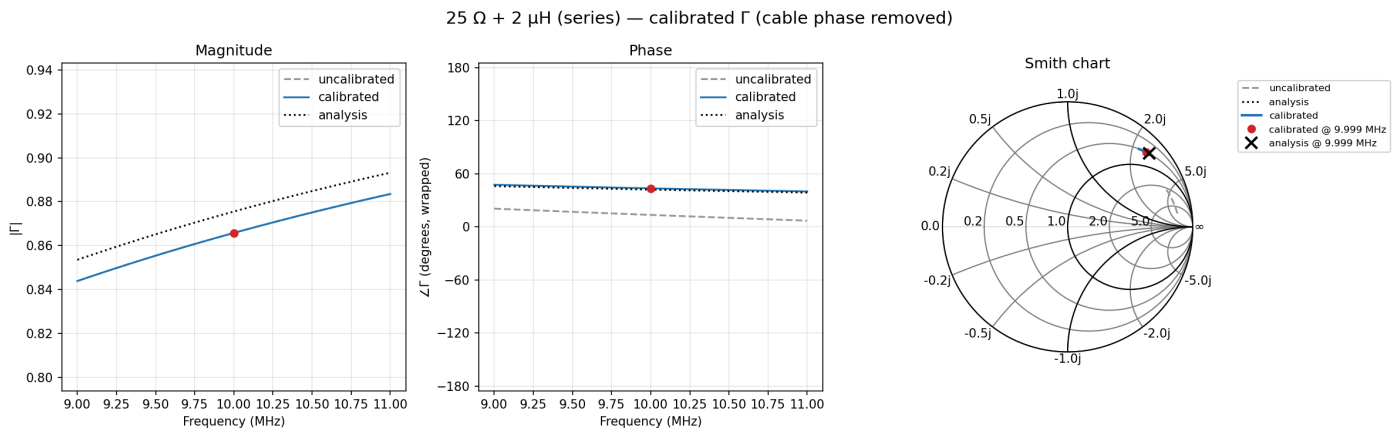
1. A 120-ohm resistor



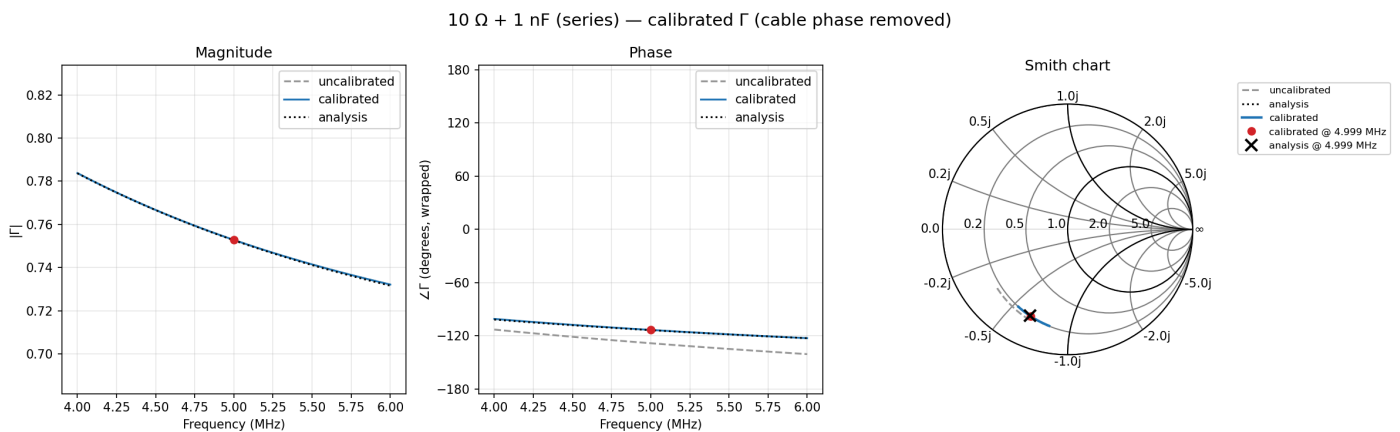
2. A 120-ohm resistor in series with a 2- μ H inductor at 10 MHz



3. A 25-ohm resistor in series with a 2- μ H inductor at 10 MHz

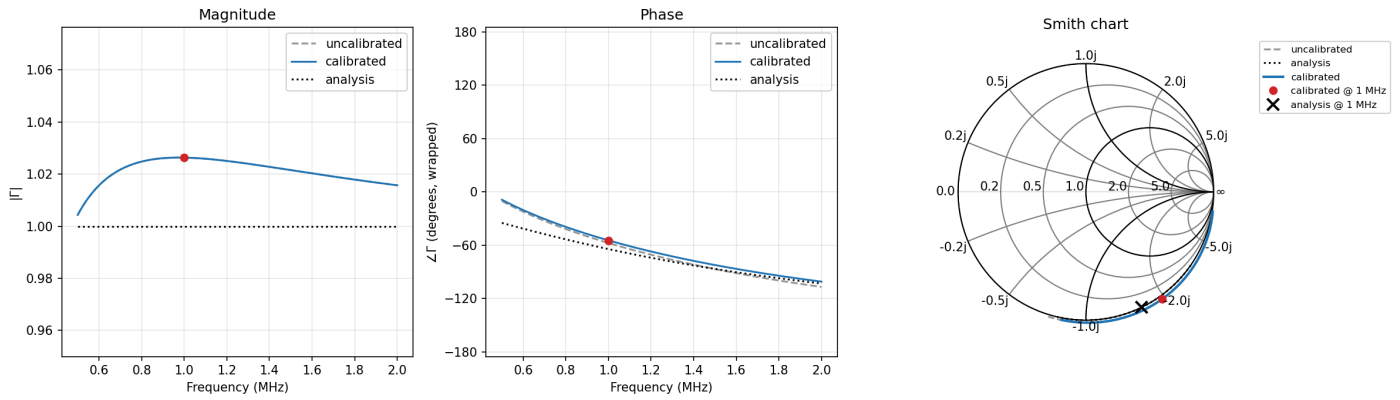


3. A 10-ohm resistor in series with a 1-nF capacitor at 5 MHz



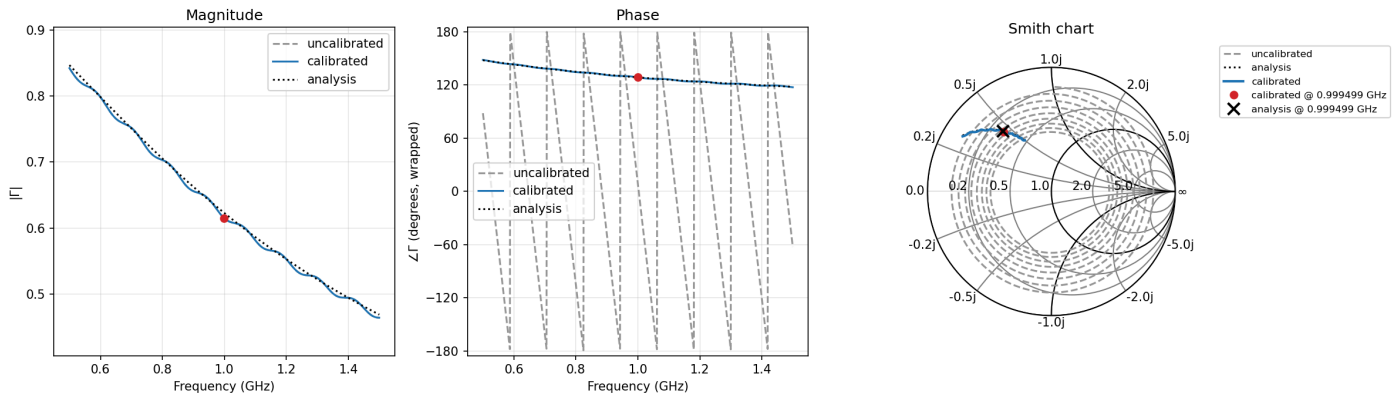
4. A 2-nF capacitor at 1 MHz

2 nF — calibrated Γ (cable phase removed)



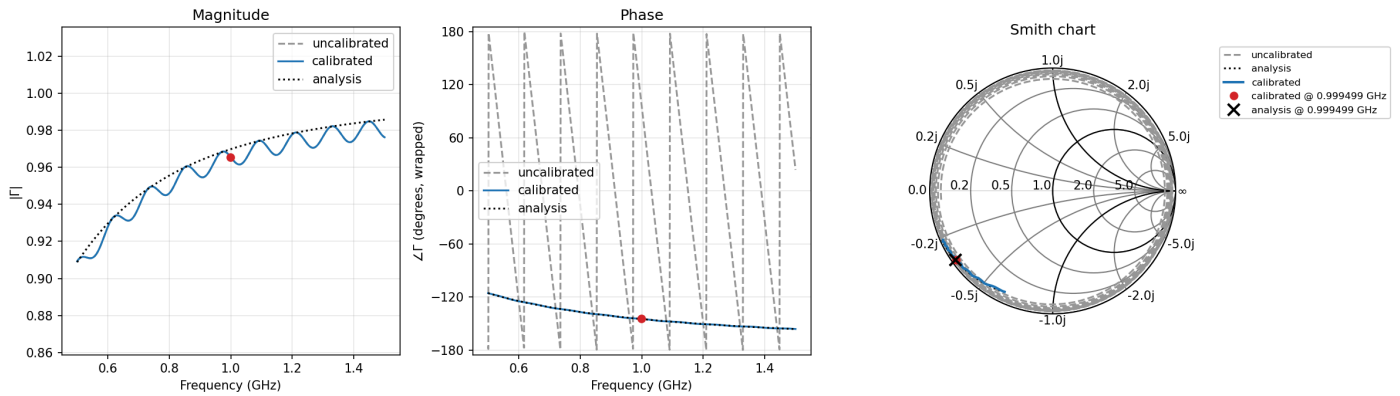
5. A 50-ohm resistor in parallel with a 5-nH inductor at 1 GHz

50 Ω || 5 nH — calibrated Γ (cable phase removed)



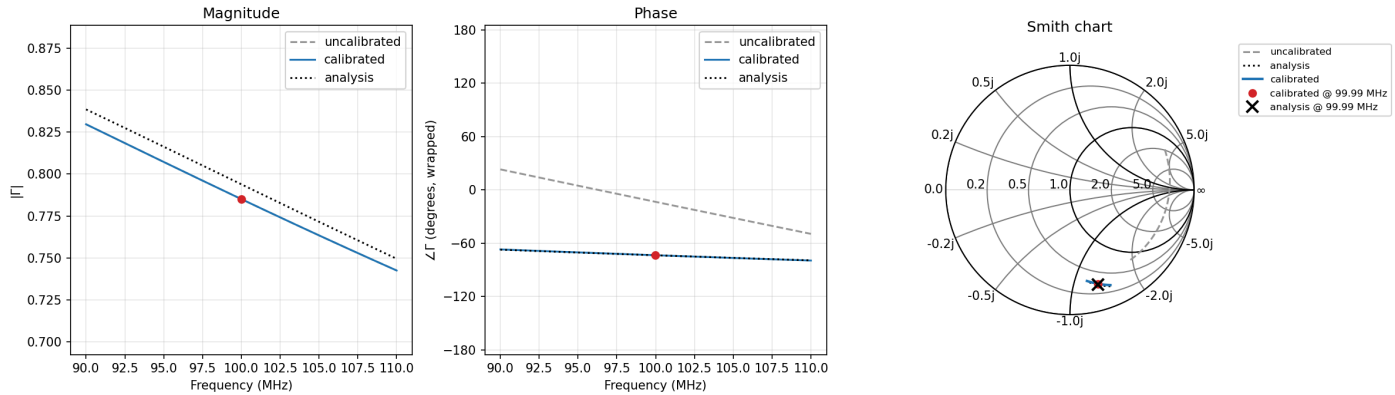
6. A 300-ohm resistor in parallel with a 10-pF capacitor at 1 GHz

300 Ω || 10 pF — calibrated Γ (cable phase removed)



7. A 30-ohm resistor in parallel with a 50-nH inductor, in series with a 20-pF capacitor, at 100 MHz. To clarify, you are finding the impedance looking into one end of a 20-pF capacitor, while the other end of the capacitor is attached to a 50-nH inductor and 30-ohm resistor in parallel with each other.

(30 Ω || 50 nH) + 20 pF — calibrated Γ (cable phase removed)



End of AI section that should be marked as a zero

8. Design an L-match from a 50 Ohm transmission line to a 270 Ohm load at 100 MHz using a series capacitor. What is the bandwidth of this matching network?

1. You can do this analytically with a series-to-parallel transformation, or graphically using a Smith chart. Only after you complete it one of these two ways, use a digital Smith chart tool to check your work.

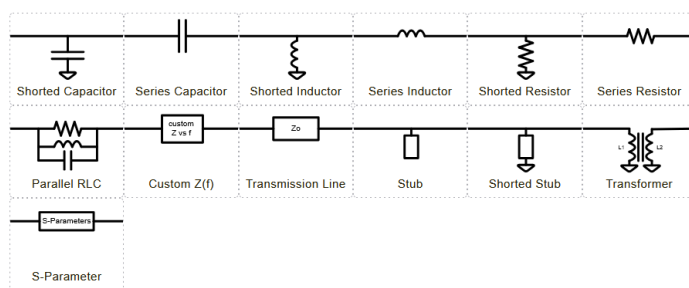
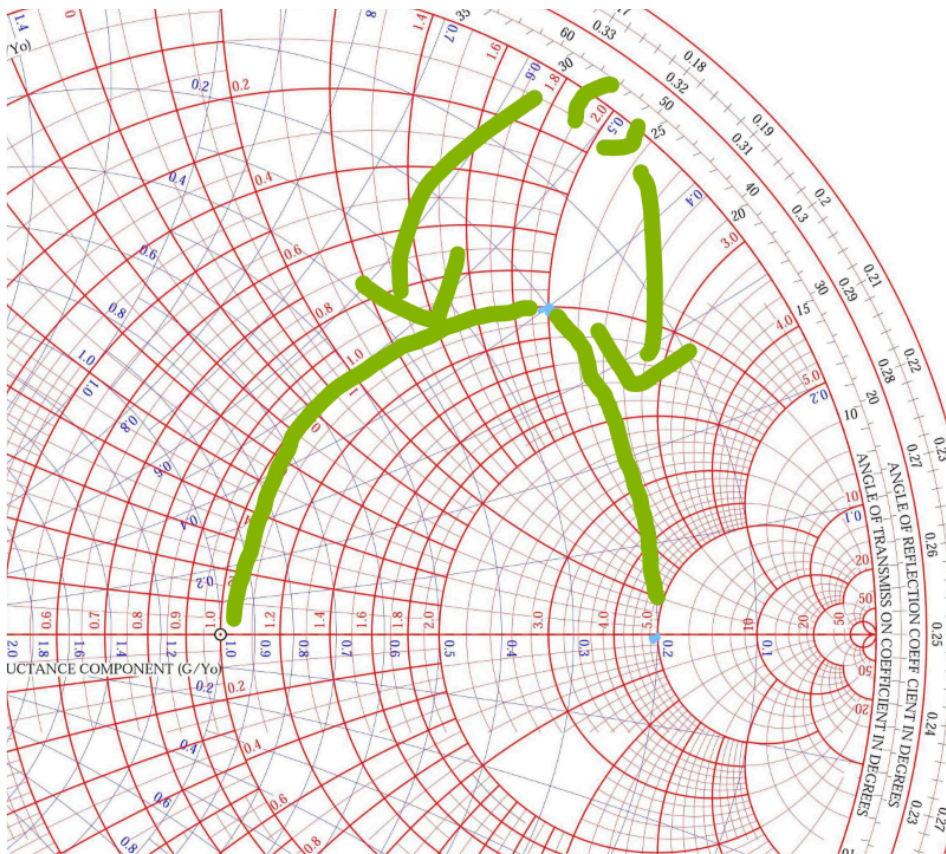
To do it analytically, we would try to find where

the real part of an inductor, capacitor, and two resistors (one in series with L and the other in parallel with C) would have a real part equal to -220Ω . I don't want to do this.

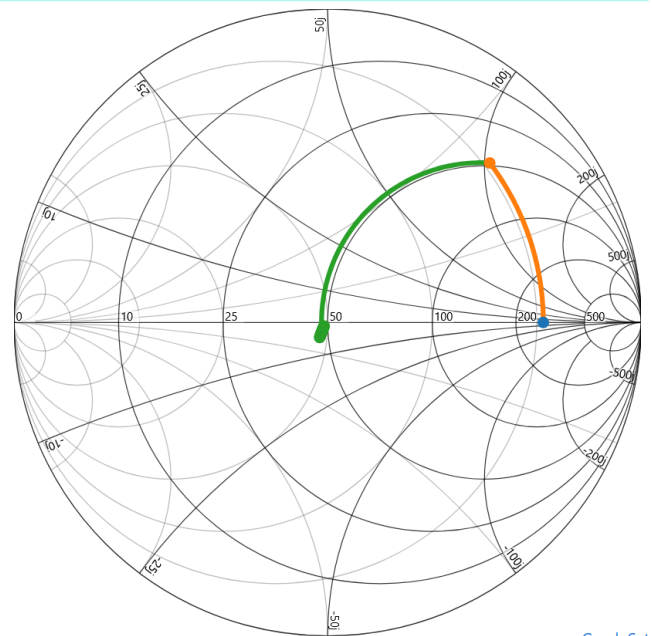
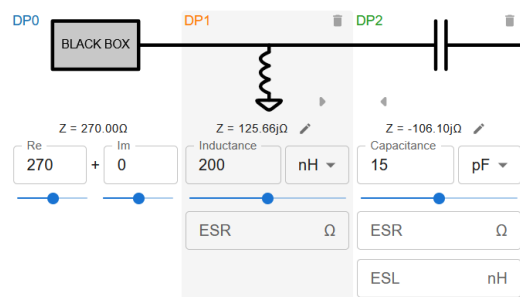
We can start with a shunt inductor to go to the line where $\frac{1}{\omega l} = \frac{0.4}{z_0}$, and then a capacitor to go down to the origin. We see that we are lying on the 2 line of constant inductance, so a capacitor to undo that would have

$$\begin{aligned}\frac{1}{\omega c} &= 2z_0 \\ c &= \frac{1}{4\pi f z_0} \\ \frac{1}{\omega l} &= \frac{0.4}{z_0} \\ l &= \frac{z_0}{0.8\pi f}\end{aligned}$$

$$\begin{aligned}l &= 200\text{ nH} \\ c &= 15\text{ pF}\end{aligned}$$



Click components above to add them to the circuit below. Impedance is looking towards the BLACK BOX



Graph Set

WOOHOOO RAGH. That was harder than I thought it would be. I initially had forgotten to swap the z_0 for l , checked online, and saw that I was terribly wrong. I played around for a lil bit, realized the mistake, and fixed it.

Old work

Graphically: $\frac{270}{50} = 5.4$, so we start on the real axis there. We can go down with a shunted capacitor to $\frac{x}{z_0} = 0.4$ on the constant reactance curve. We then want a series inductor to move clockwise until we hit the x axis. We are starting at the spot the $\omega l = 2z_0$, which would move us back to the origin.

This gives

$$l = \frac{2}{\omega z_0} = \frac{1}{\pi f z_0}$$

and

$$\omega c = 0.4 z_0$$

$$c = \frac{1.25 * z_0}{2\pi f}$$

$f = 100\text{MHz}$, so

$$c = 800\text{pf}$$

10. Use a series-to-parallel transformation to find the quality factor of the network in Figure 4 at frequency ω_0 , the natural frequency of the resonant network.

We had this in notes, but i forgot it and found it online at [link](#)

$$\Rightarrow R_p = R_s (1 + Q^2) \quad \& \quad X_p = X_s \left(1 + \frac{1}{Q^2} \right)$$

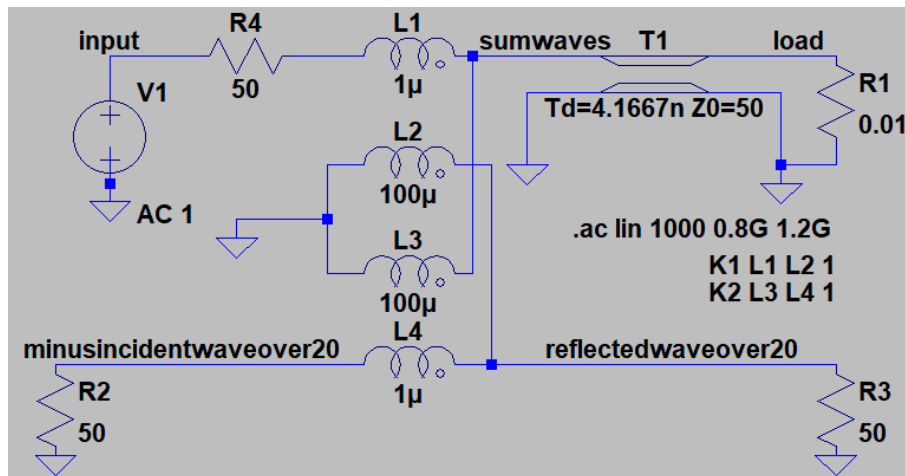


Figure 3: Simulated one-port vector network analyzer with directional coupler.

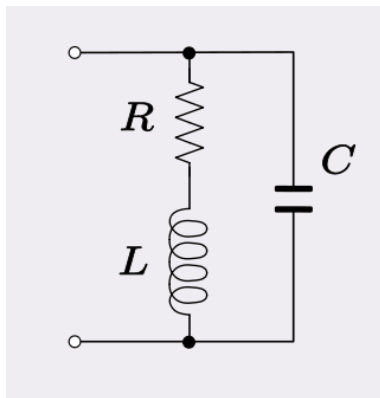


Figure 4

Lab Instructions

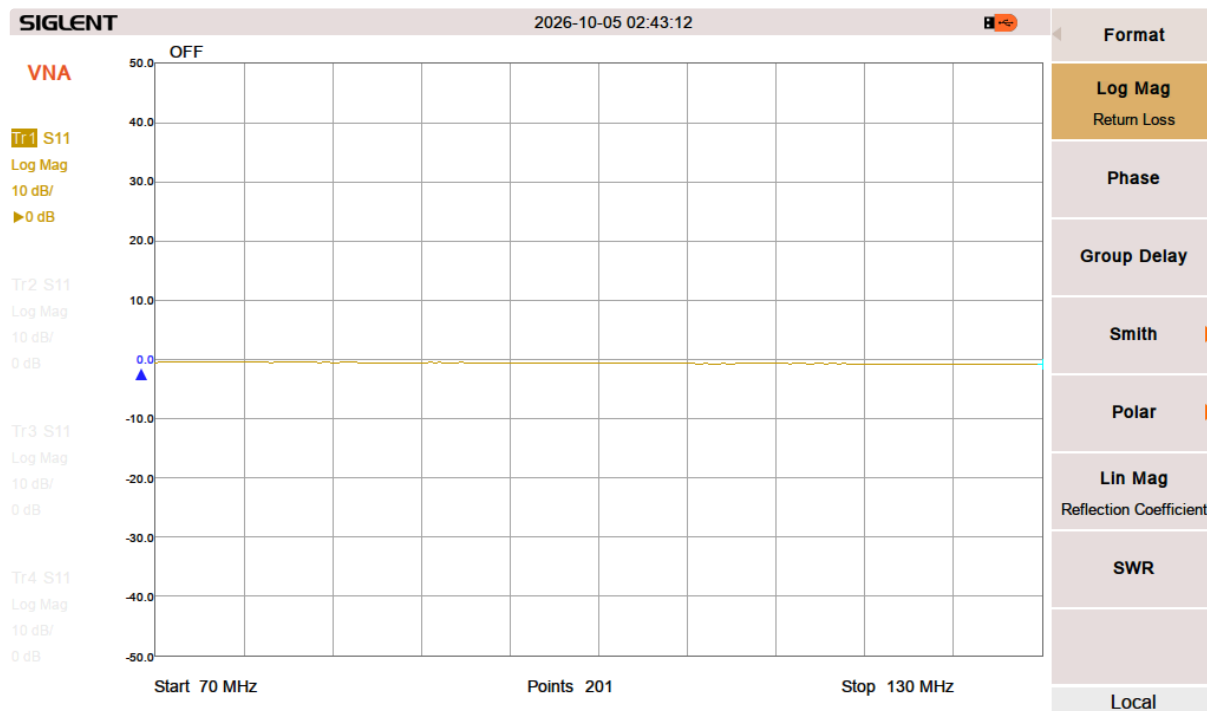
These lab instructions will walk you through the basics of VNA operation, VNA calibration and surface mount construction of radio frequency modules. We will use the Siglent SVA1000X VNA that sits on the bench in the North of the RF lab. This is technically not a full vector network analyzer, it's a spectrum analyzer with a tracking generator, but it is sufficient for this lab. Watch the videos and consult the manuals below to learn some details of operation:

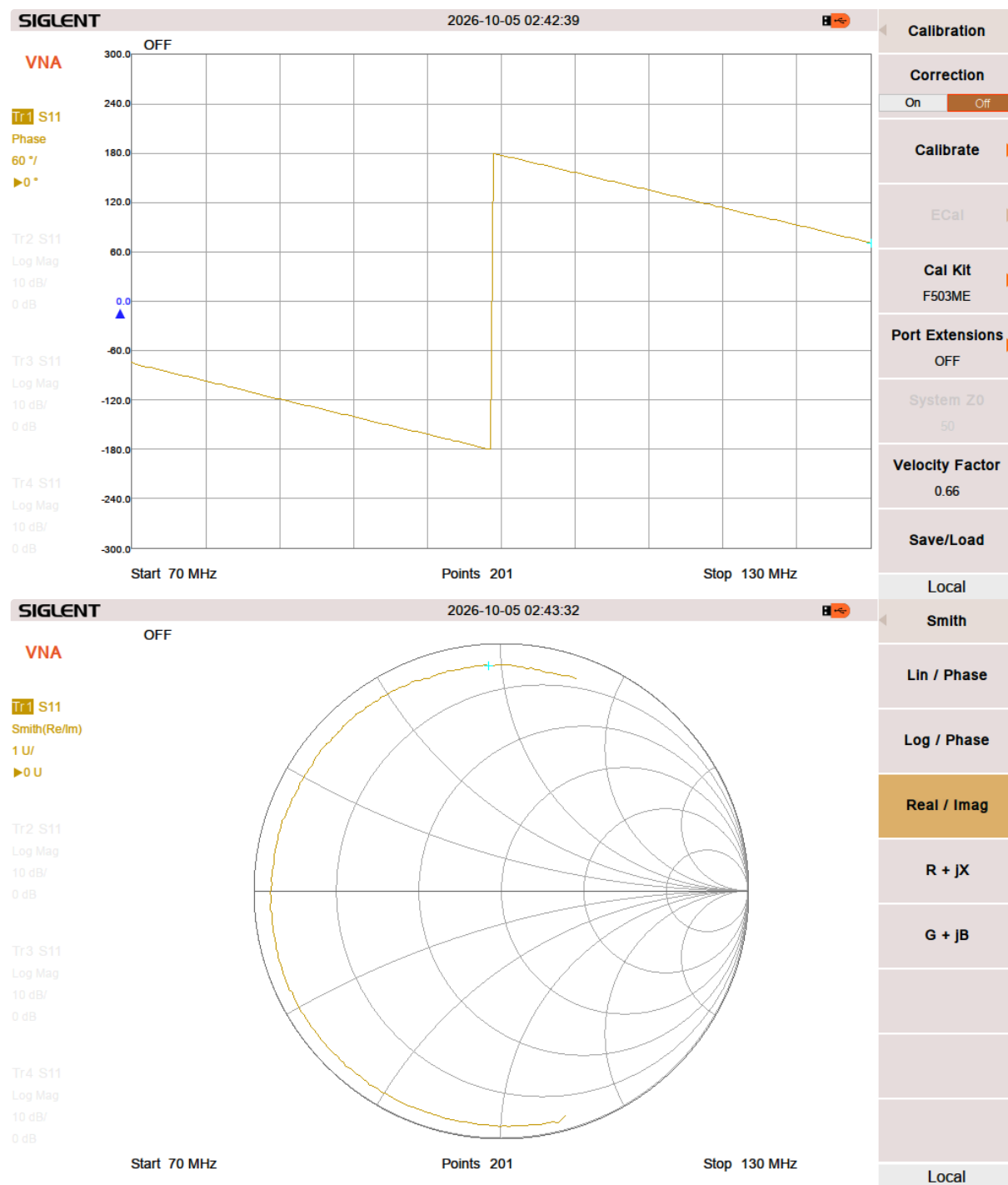
- <https://www.youtube.com/watch?v=AfKowzuxVtI&list=PLmTIWEcpkWvi8XD-j4zHAEUylsEBEZTe1&index=5>
- <https://siglentna.com/spectrum-analyzers/sva1000x-spectrum-vector-analyzer/>
- https://siglentna.com/wp-content/uploads/2020/07/UserManual_UG0703P_E02A.pdf

Note that we only use the TG port of the VNA during this lab. We will not use the RF input port. Also note that S11 is a synonym for reflection coefficient in this lab.

1. Observe operation of a VNA before calibration.
2. Connect an SMA cable to the TG port of the VNA. Terminate the cable with a short. (Don't use the calibration standard for this because it is fragile, there are plenty of other SMA shorts lying around the lab.)
3. Set your frequency to span to be 70MHz to 130MHz and then configure the instrument in VNA mode. (Side note: we're not messing with your tracking power level because it could be basically anything for this lab without jeopardizing your results, but I advise staying between -20dBm and 0dBm.)
4. Manipulate the display to plot S11 magnitude, S11 phase and a S11 Smith Chart, and record these data for your lab notebook. Screenshots may be OK for your lab notebook here if you make judicious use of markers. However, If you export a screenshot, always also export raw data in case you need to do analysis later. Cell phone photos of data are never OK for a lab manual. (That said, as some of you have discovered, digitization of photos or screenshots can sometimes work pretty well. If you can get accurate data that way, more power to you. I don't think that would be appropriate for a research paper or a datasheet, but it works for these labs.)
5. Use your analysis from theory question 1 to justify your observations. Also create a simulation that represents this scenario and compare it to your measurements and analysis.

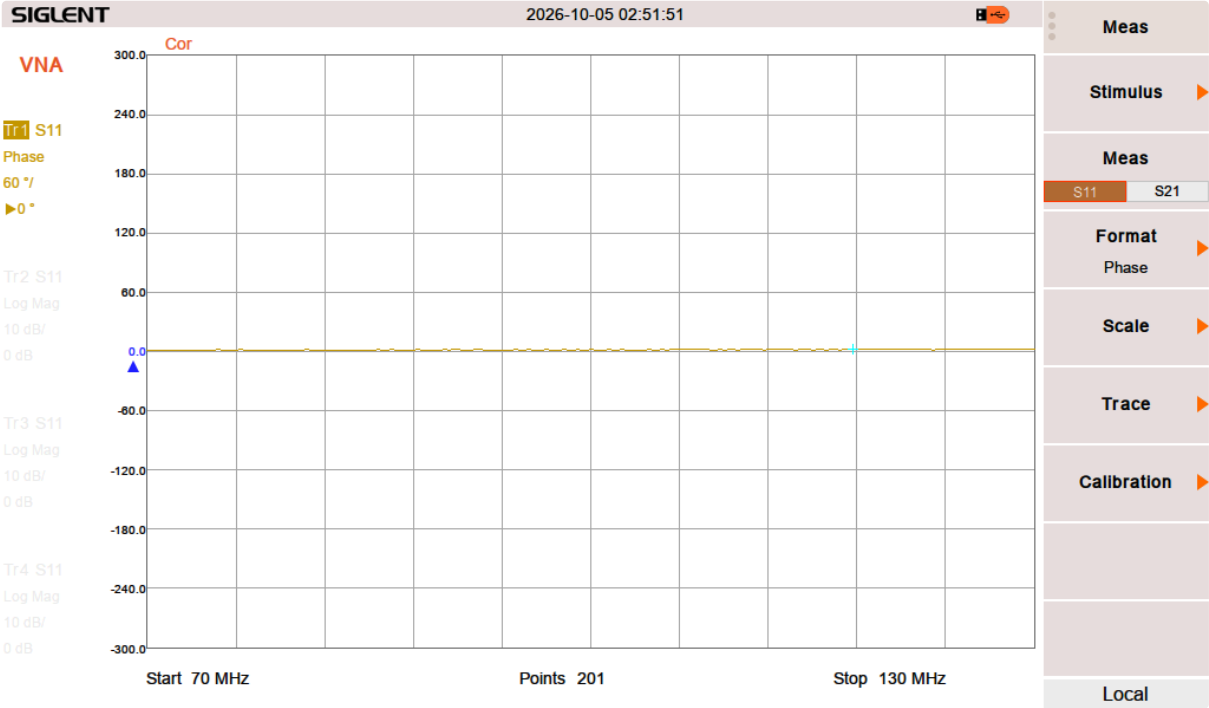
Short circuit Not Calibrated:





Short Circuit Calibrated

Phase



Cor

300.0

240.0

180.0

120.0

60.0

0.0

-60.0

-120.0

-180.0

-240.0

-300.0

Start 70 MHz

Points 201

Stop 130 MHz

Meas

Stimulus

Meas

S11 S21

Format

Phase

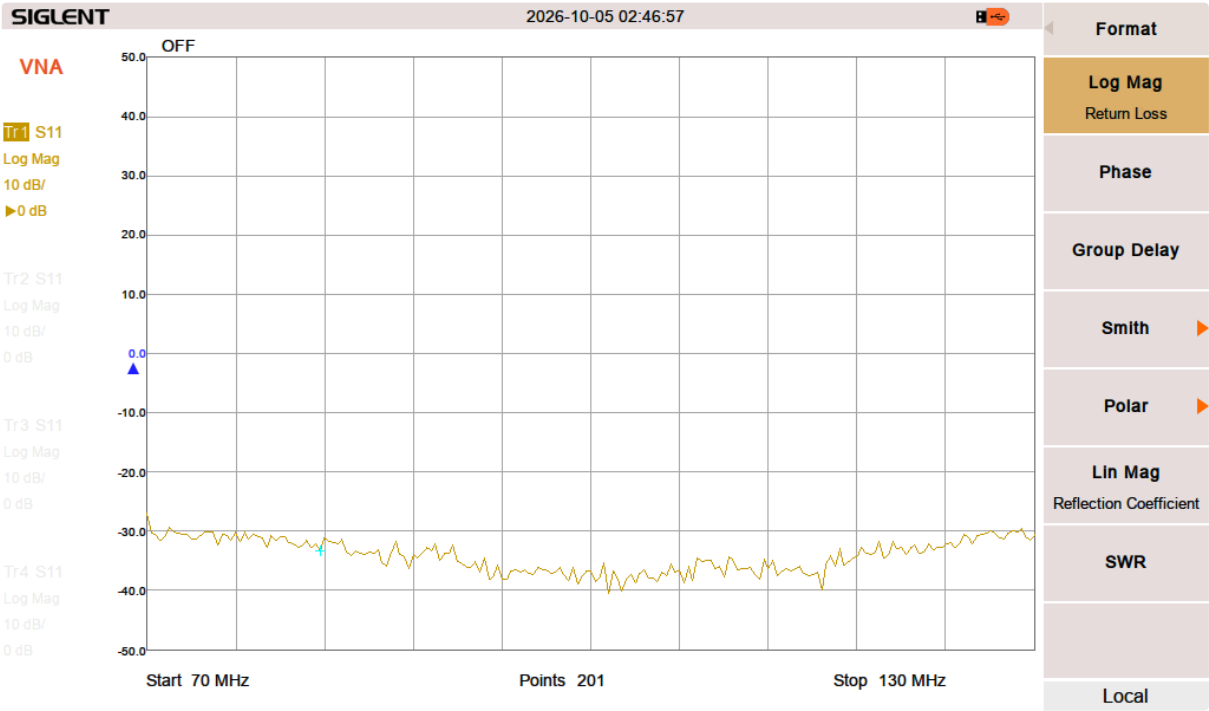
Scale

Trace

Calibration

Local

50 Ohm uncalibrated



OFF

50.0

40.0

30.0

20.0

10.0

0.0

-10.0

-20.0

-30.0

-40.0

-50.0

Start 70 MHz

Points 201

Stop 130 MHz

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

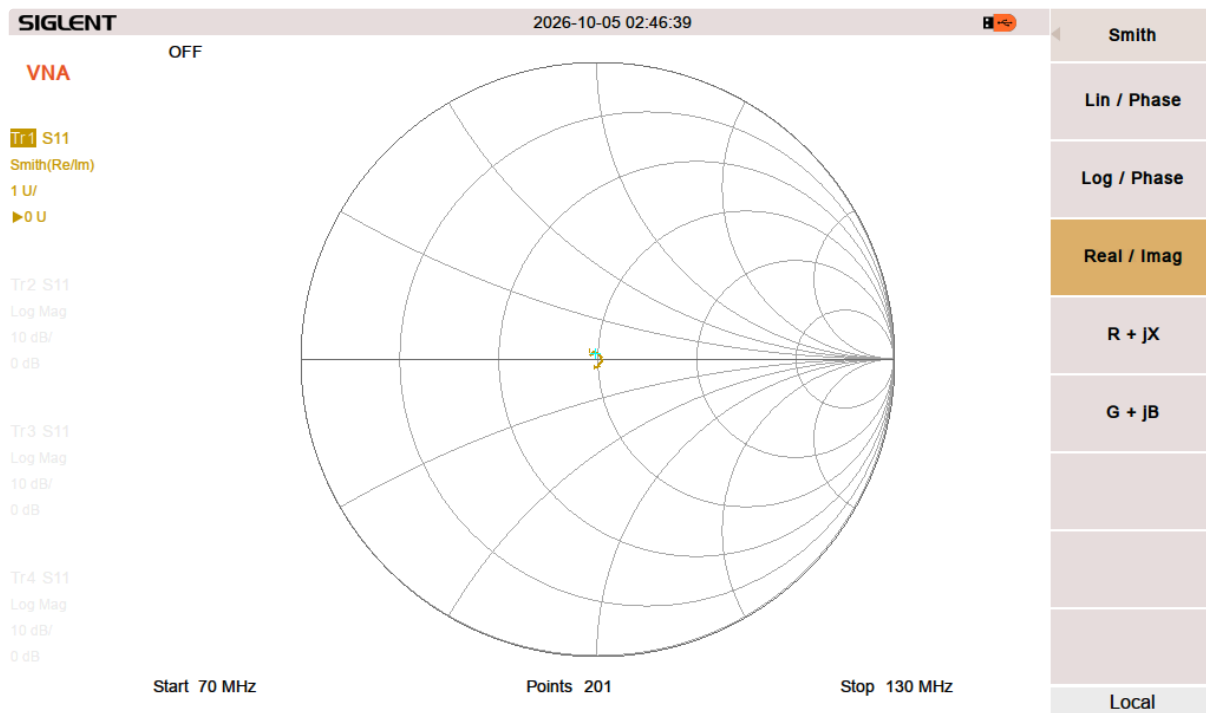
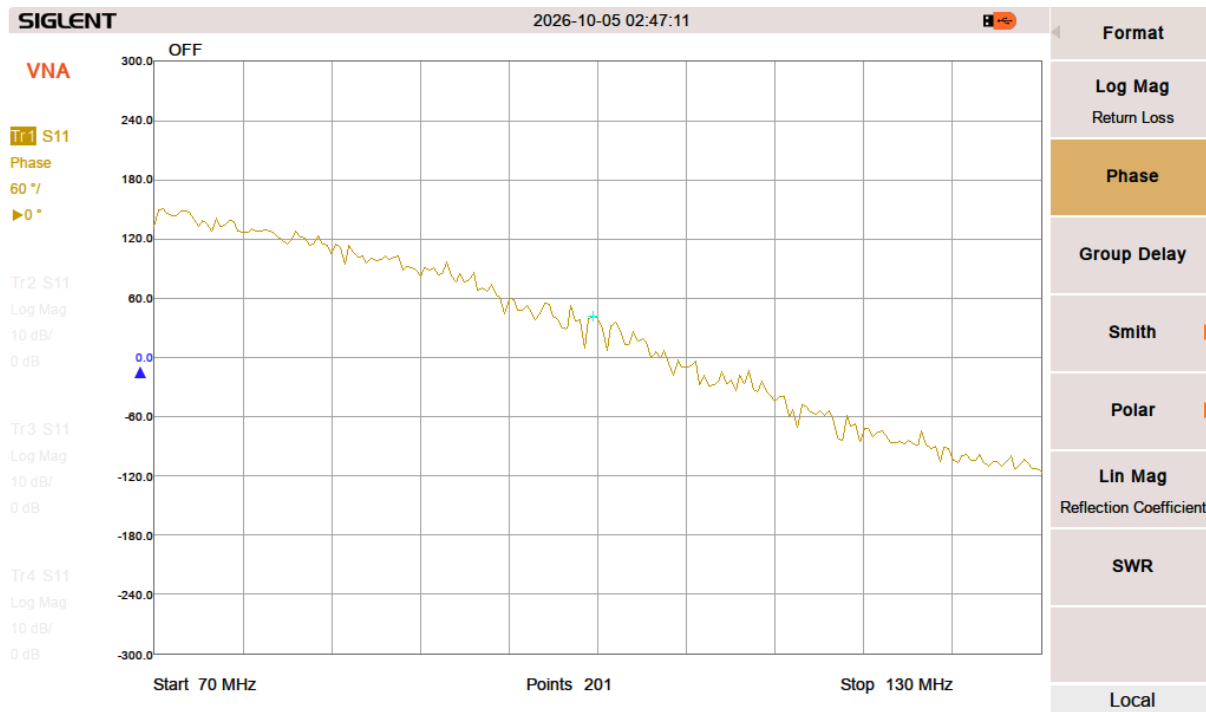
Polar

Lin Mag

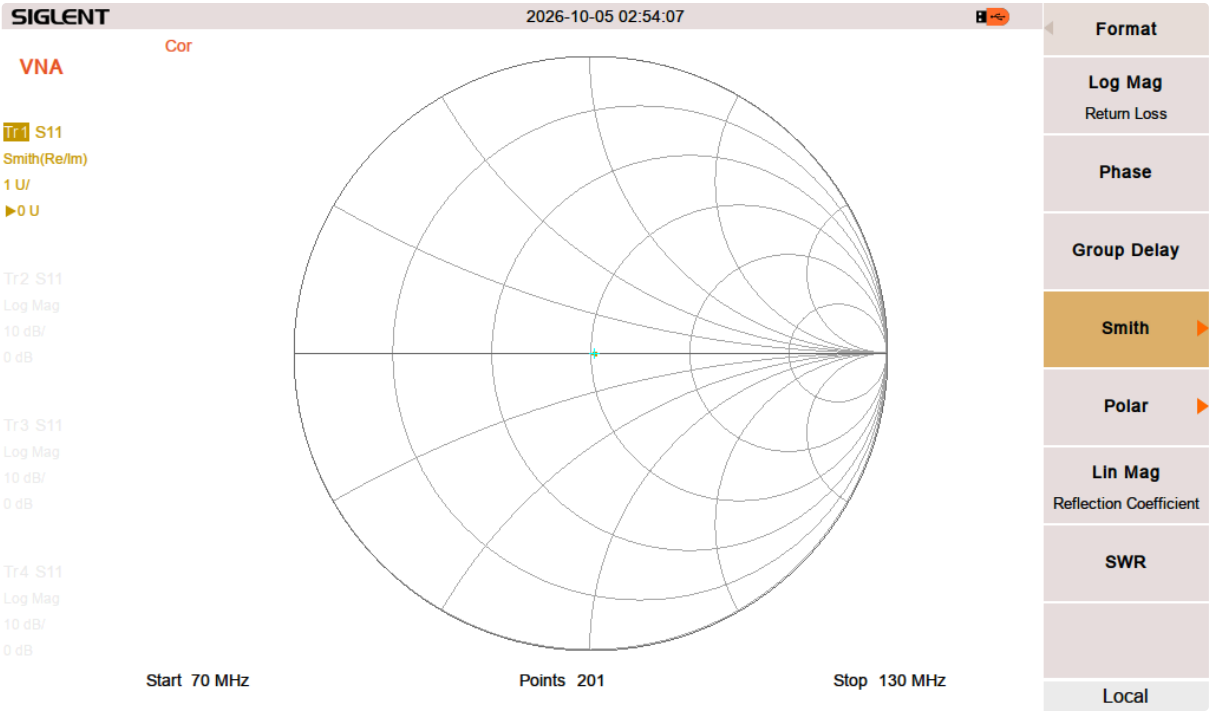
Reflection Coefficient

SWR

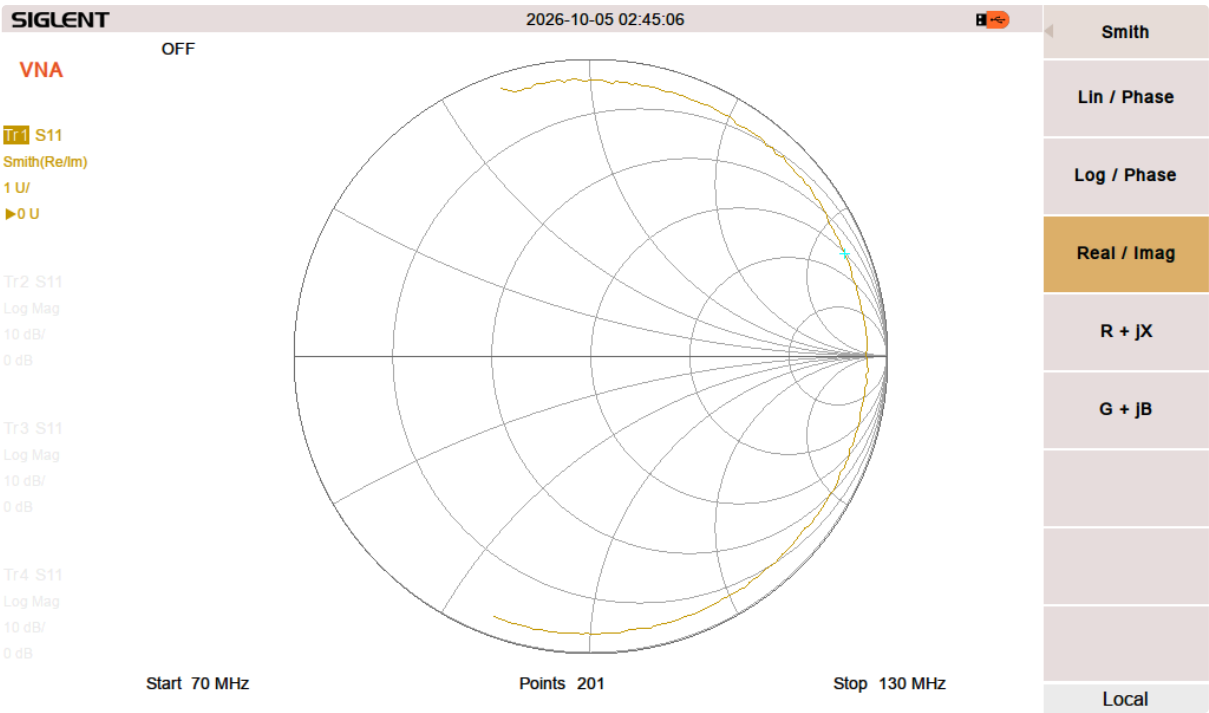
Local



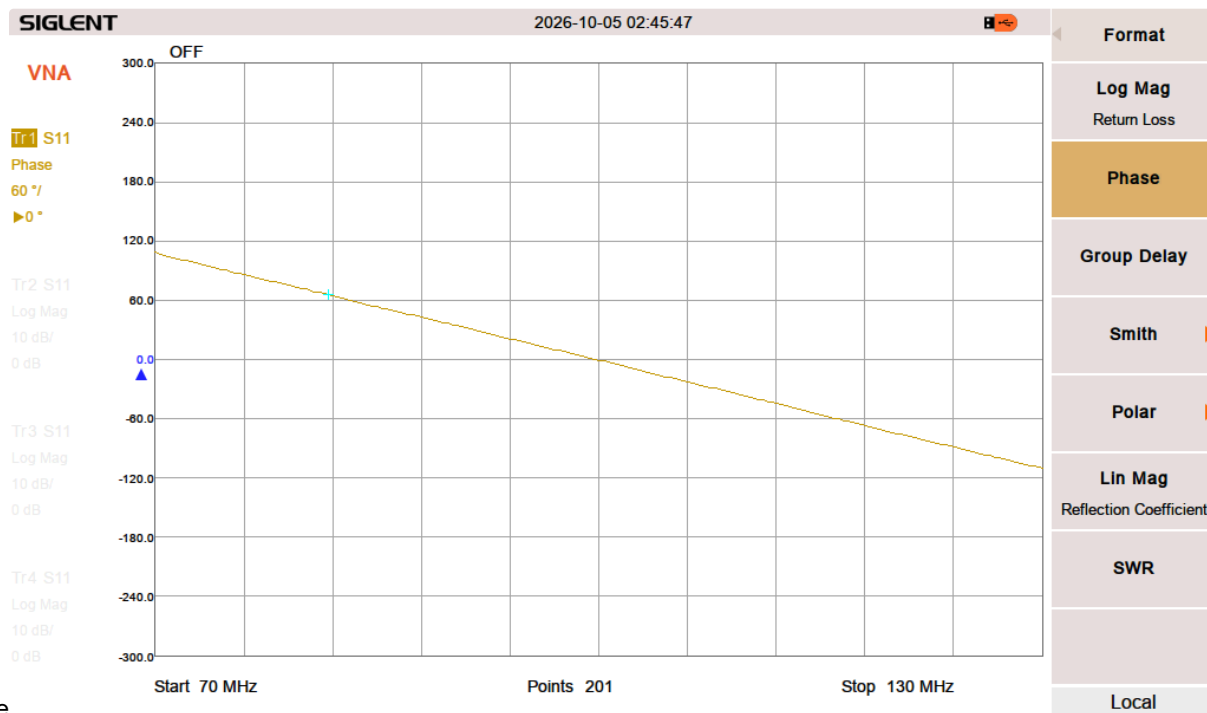
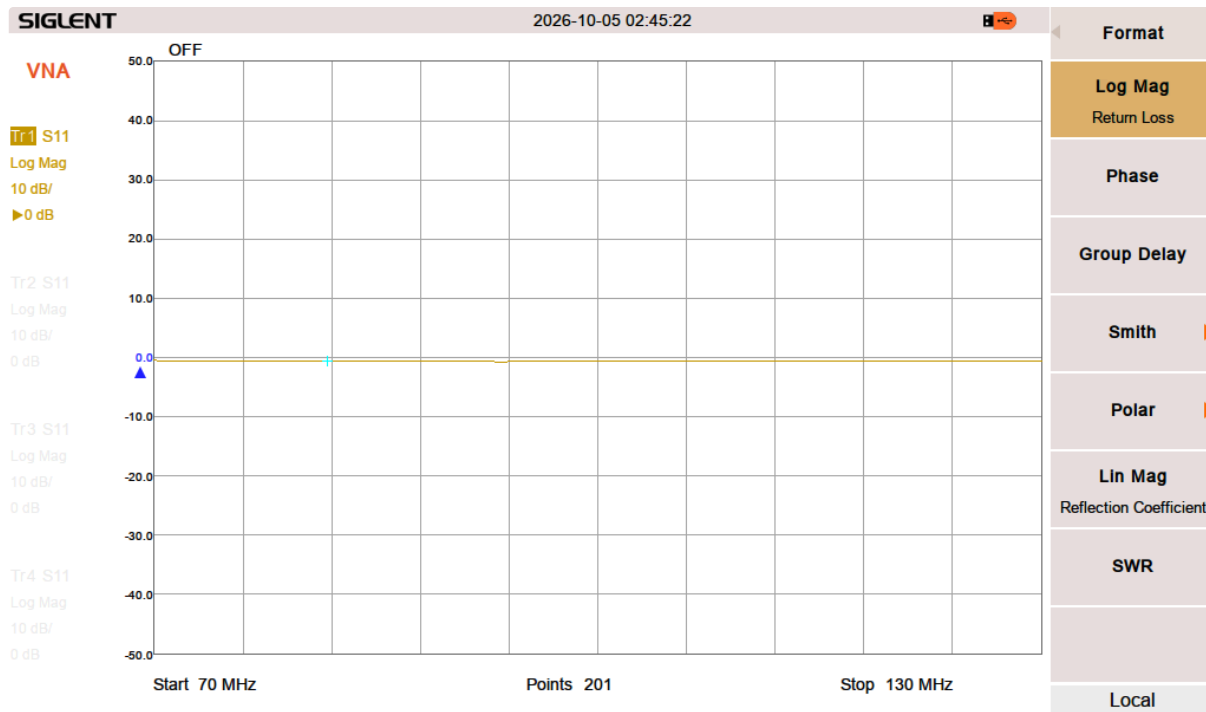
50 Ohm Calibrated



Open circuit uncalibrated

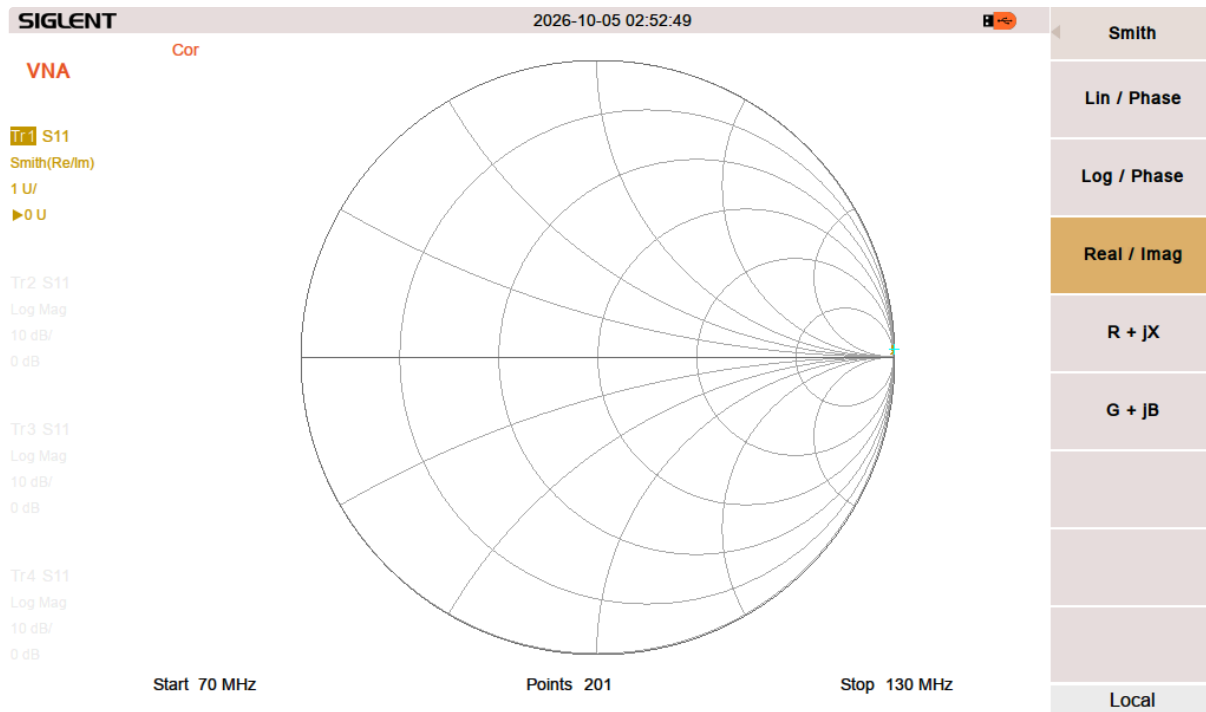


magnitude

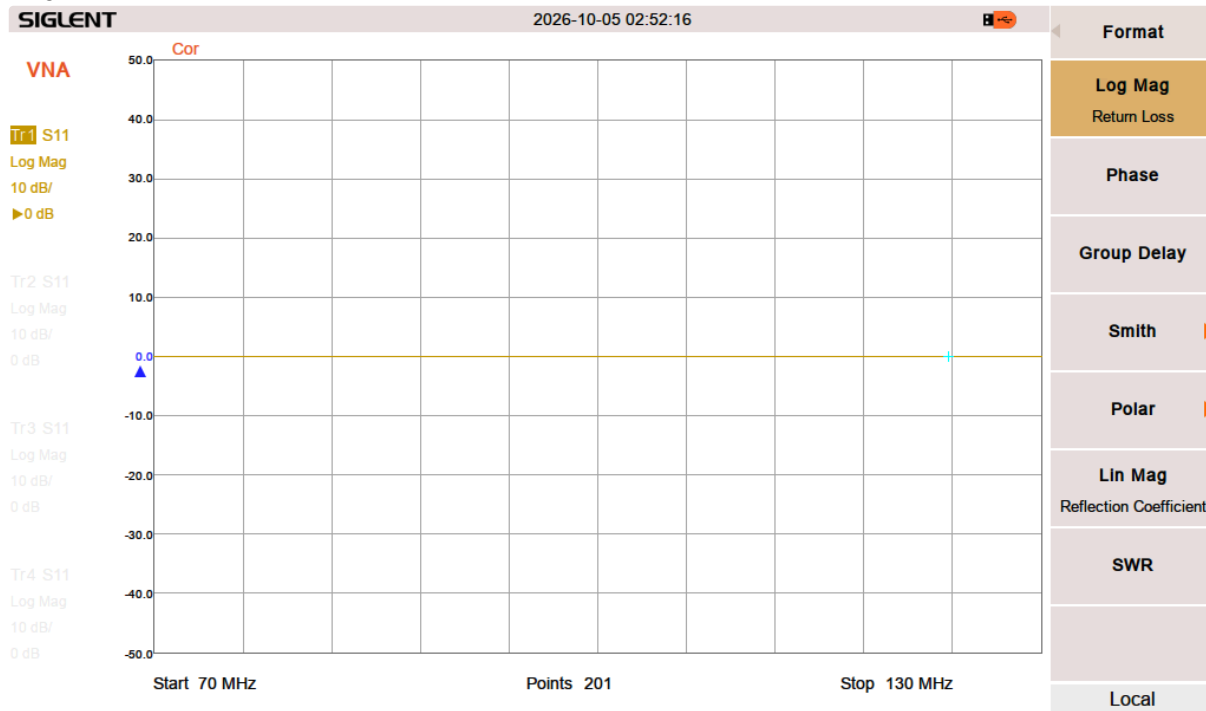


phase

Open circuit calibrated



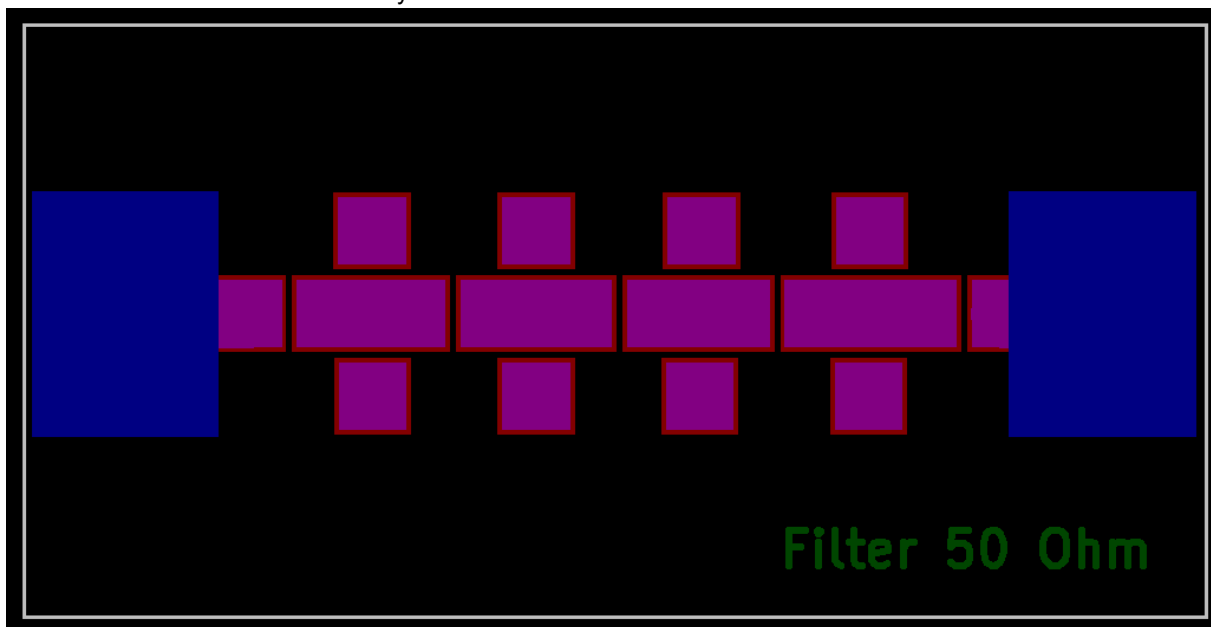
magnitude



- Repeat step c for the following two terminations: an open and a 50 Ohm load. Compare your results to appropriate analysis and simulation.
- Calibrate the VNA and observe how your measurements change.
- Follow the calibration process demonstrated in the video linked above.
- For each of our three terminations (open, short, 50 Ohm load), measure S11 magnitude, S11 phase and an S11 Smith Chart. Compare these results to your observations in lab instruction 1 and theory question 2. What has changed? Does it match theory?

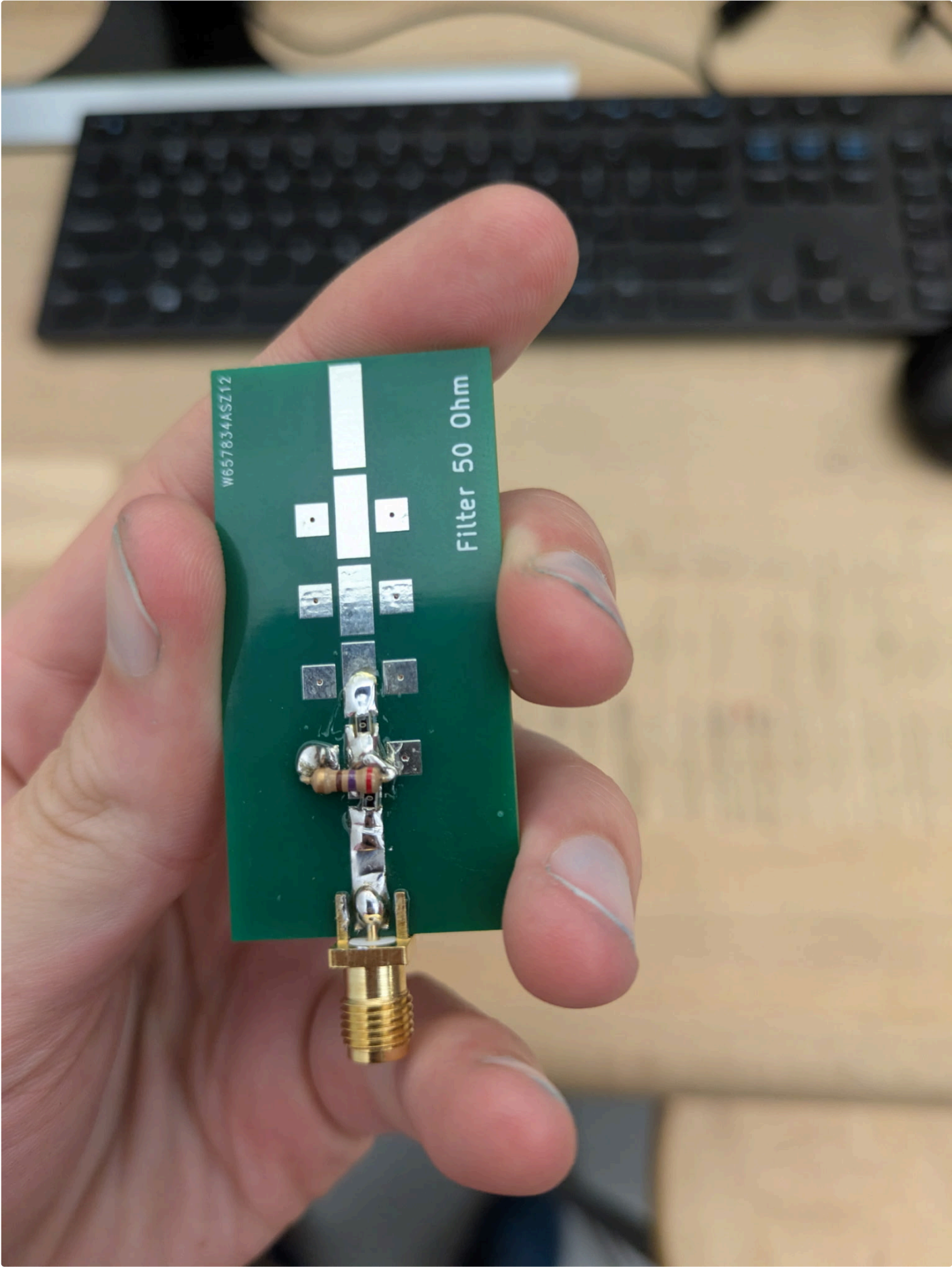
The phase accumulation across the wire without calibration looks like it does in the plots that I calculated. The calibration removes this phase accumulation due to the wire, and leaves only the results from the load / termination / parts that we care about.

11. Change your frequency span to 90MHz to 110MHz, and see if your results still look calibrated. Calibrations are specific to the frequency range, so you need to recalibrate when you change your span. This exercise is to warn you about that!
12. Set your span back to 70MHz to 130MHz and recalibrate.
13. Get used to soldering onto our boards
14. You will need to find many of these parts in our lab. They should all be somewhere in the red tool chest in between the two wood lab tables, except for the 0603 resistors, which are in a cardboard box next to the tool chest. It's all a little messy, but not completely disorganized (capacitors are together, inductors are together, 0-ohm resistors are together, etc.), so make sure you leave everything at least as organized and clean as you found it.
15. As an aside, the RF lab reorganization is a currently underway project. If you have ideas, requests, or contributions you would like to make, let me know!
16. Watch this video for a refresher on the basics of soldering:
<https://www.youtube.com/watch?v=oqV2xU1fee8>
17. Watch this video for a visual reference on how to solder surface-mount components, the kind we use in our lab, to a printed circuit board: <https://www.youtube.com/watch?v=PU7wLcuqc-I>. I've briefly mentioned the "double-soldering-iron" method to many of you as well, and that is how I have had the most luck with soldering surface-mount parts personally.
18. Find the filter boards in our lab. They look like this:

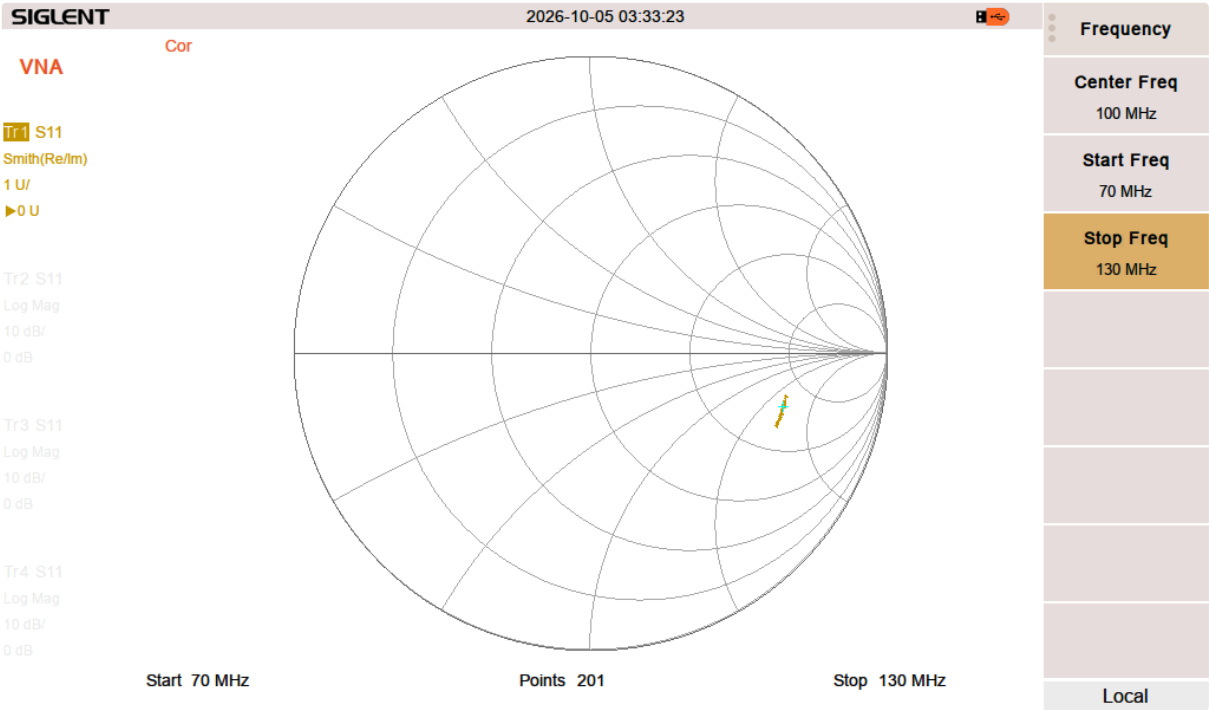


19. Solder a side launch SMA connector to one side of your filter board. Pay very close attention to your ground connections on the back side (they need to be rock solid) and don't over-solder your signal connection on the front side.
20. Starting from your connector, solder two series 0 Ohm resistors and a shunt 270 Ohm resistor.
21. Measure the (calibrated) S11 magnitude, phase and Smith Chart of this combination of zero Ohm series plus 270 Ohm load, and compare to analysis and simulation.
22. Build the L match that you designed in "VNA Theory and Simulation" question 3.

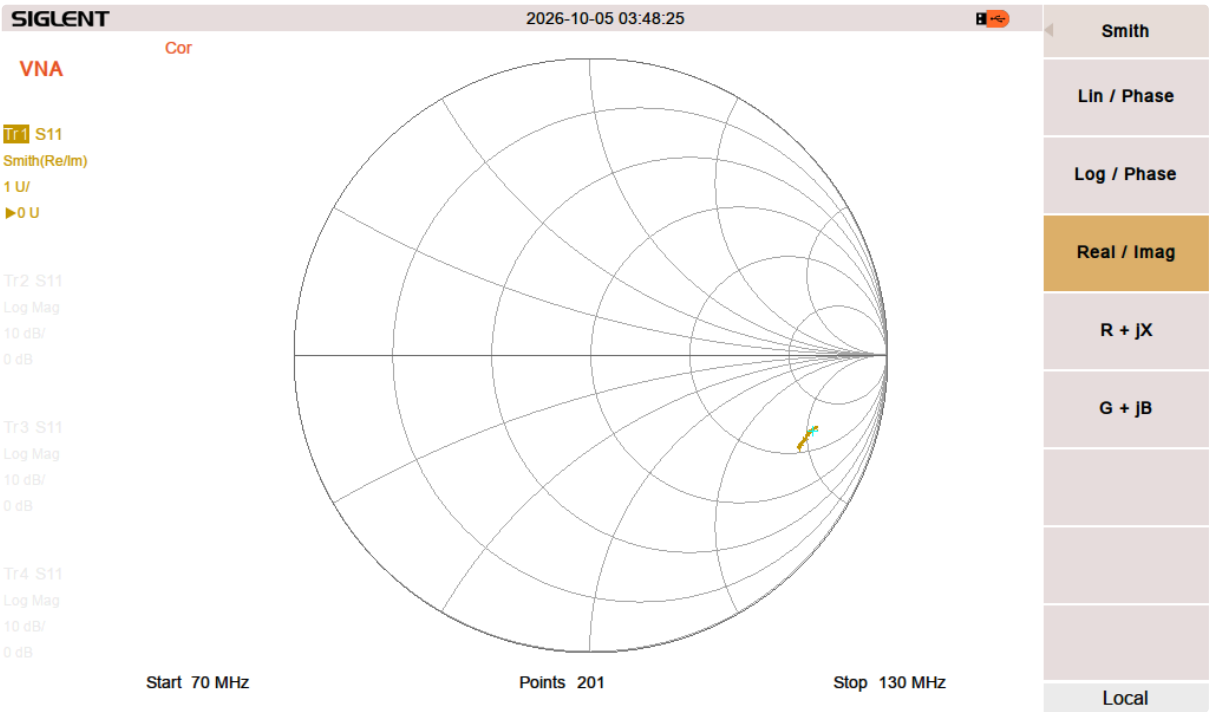
Completed circuit (we didn't take photos of it in stages)



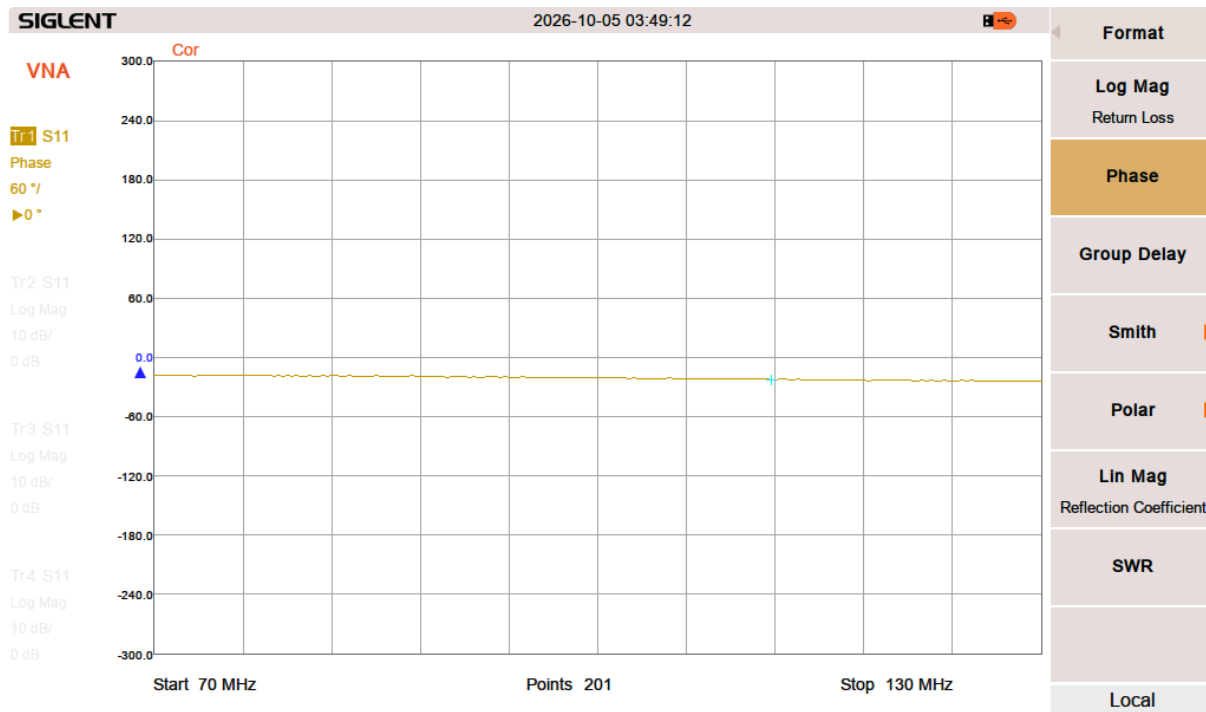
resistor only



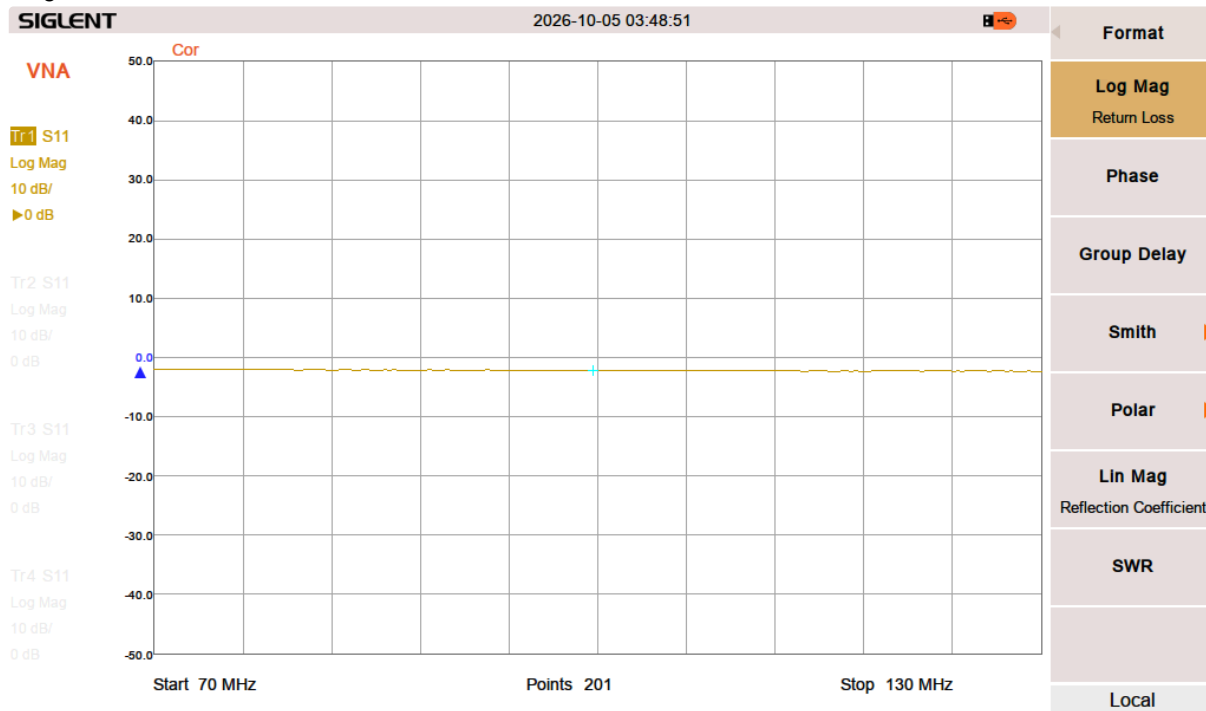
270 ohm resistor and only the 16 pf capacitor



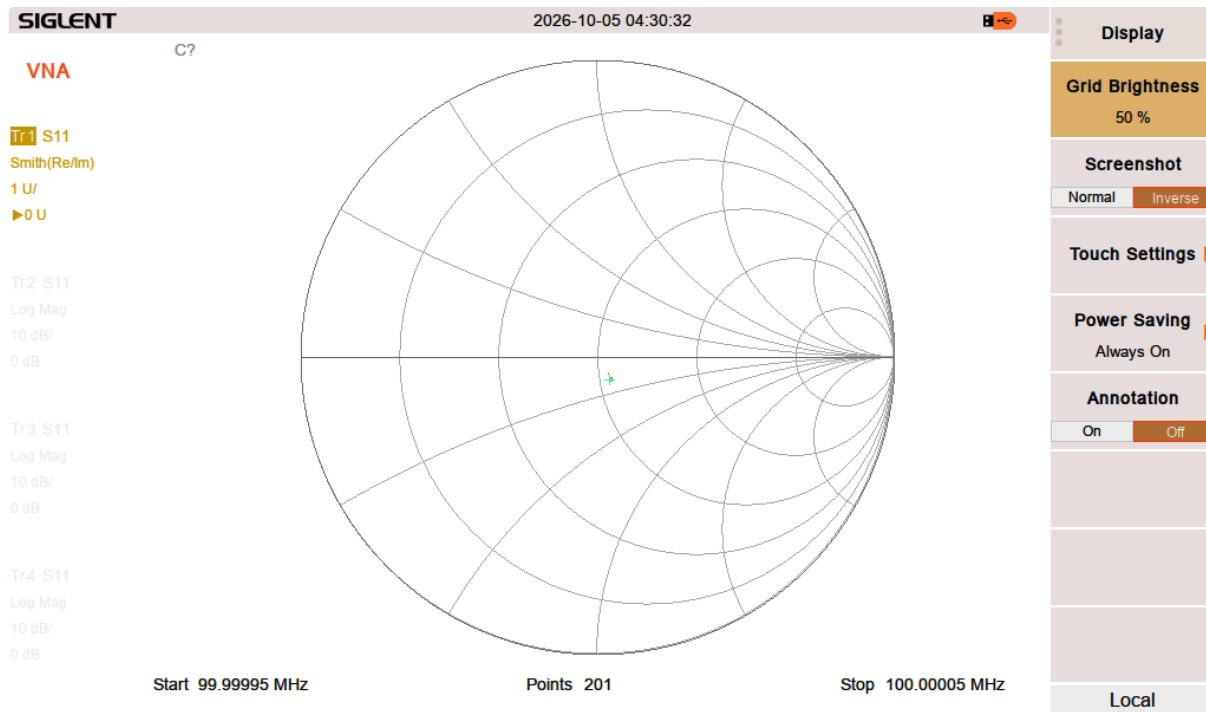
phsae



magnitude



180 nH inductor and 15 pf capacitor at only 100 MHz (its pretty close ish)



22. DON'T START WITH A FRESH BOARD! This section asks that you start by manipulating the board you've already populated in section 3.
23. Instead, add the component of your L match that is closest to your load to your board. If the component that's closest to your load is a series component, replace one of the zero Ohm resistors with this component. If it's shunt, you can just solder it next to your load.
24. Look at (and record) the S11 Smith Chart for the combination of this first component and your load. You can account for errors in your manufacturing by looking at the 100MHz point on this Smith Chart. If it isn't where you designed it to be, you can just compute the appropriate inductance to make the best of your situation, and use that instead of your original design. Designing matching networks component by component, looking at the Smith Chart between each soldering job, is an important trick for achieving good results in bespoke, hand-manufacturing of RF boards.
25. Solder in your second component, then look at the S11 Smith Chart for your completed L match.
26. Finally, when the network is complete, extract the Q value of the network from your results. Compare it to theory.

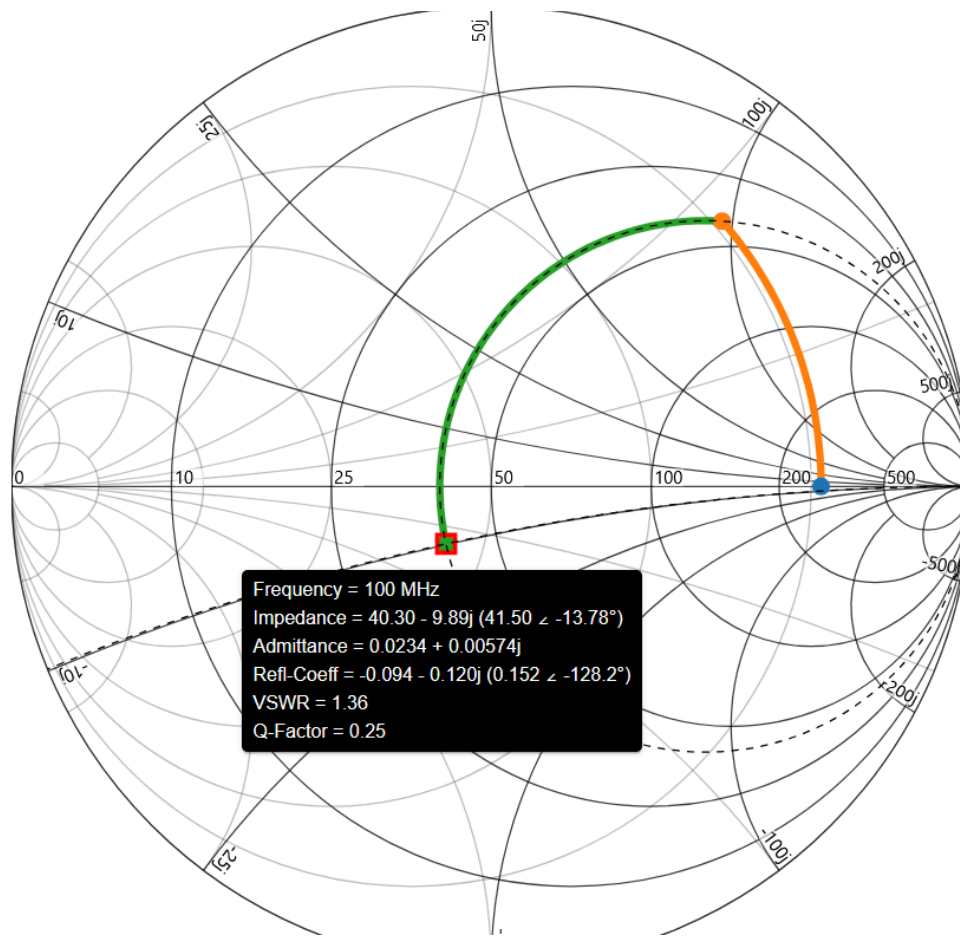
We can read the reactance and the resistance from the smith chart and use

$$Q = \frac{X}{R}$$

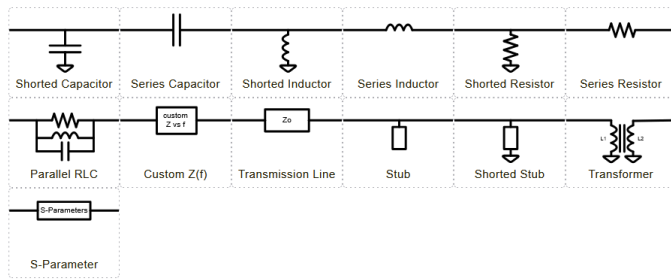
The impedance is roughly $60 - 7.5j$ by reading the rough location off of a smith chart, which gives

$$Q = 0.13$$

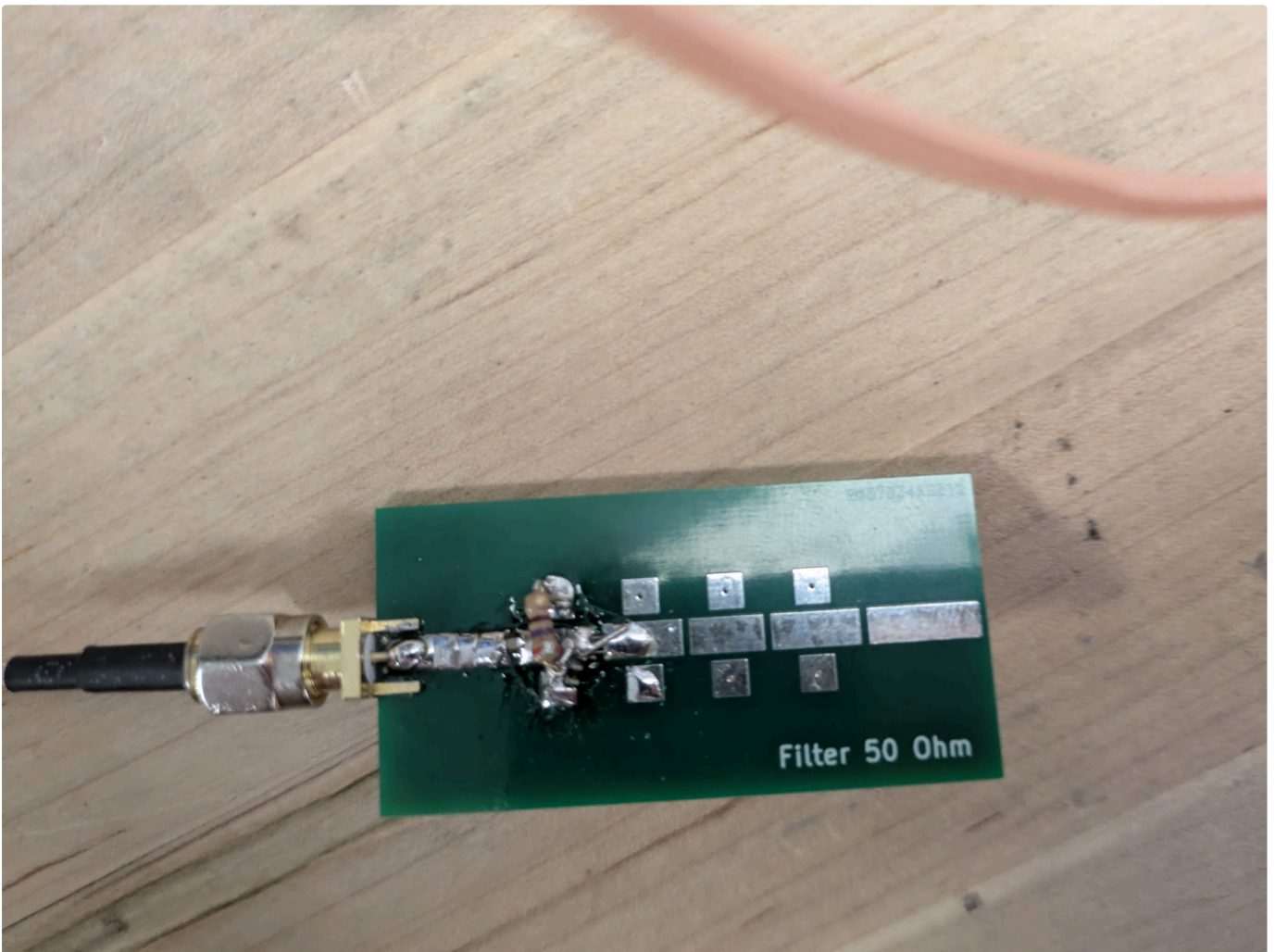
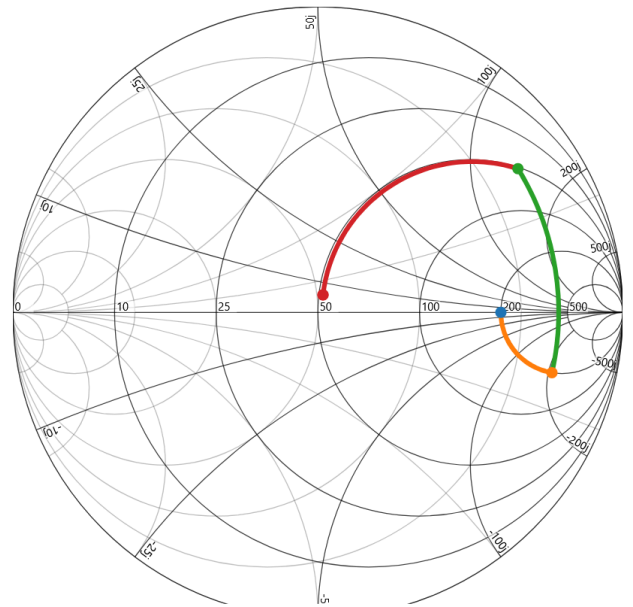
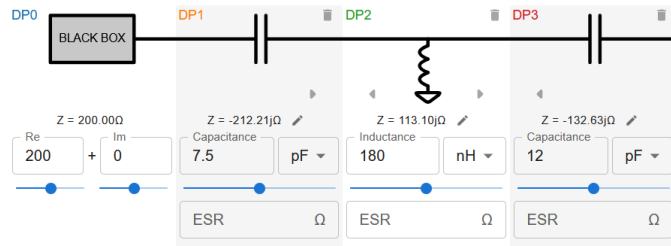
In theory, with the component values we used we would get a Q around 0.25. Potentially parasitic components helped, or the tolerances went in the right direction.



27. Optional Extra Credit: Design a T match from a 200 Ohm load, simulate it, assemble it and measure it. We designed this to use real world component values that we could find in the lab.



Click components above to add them to the circuit below. Impedance is looking towards the BLACK BOX

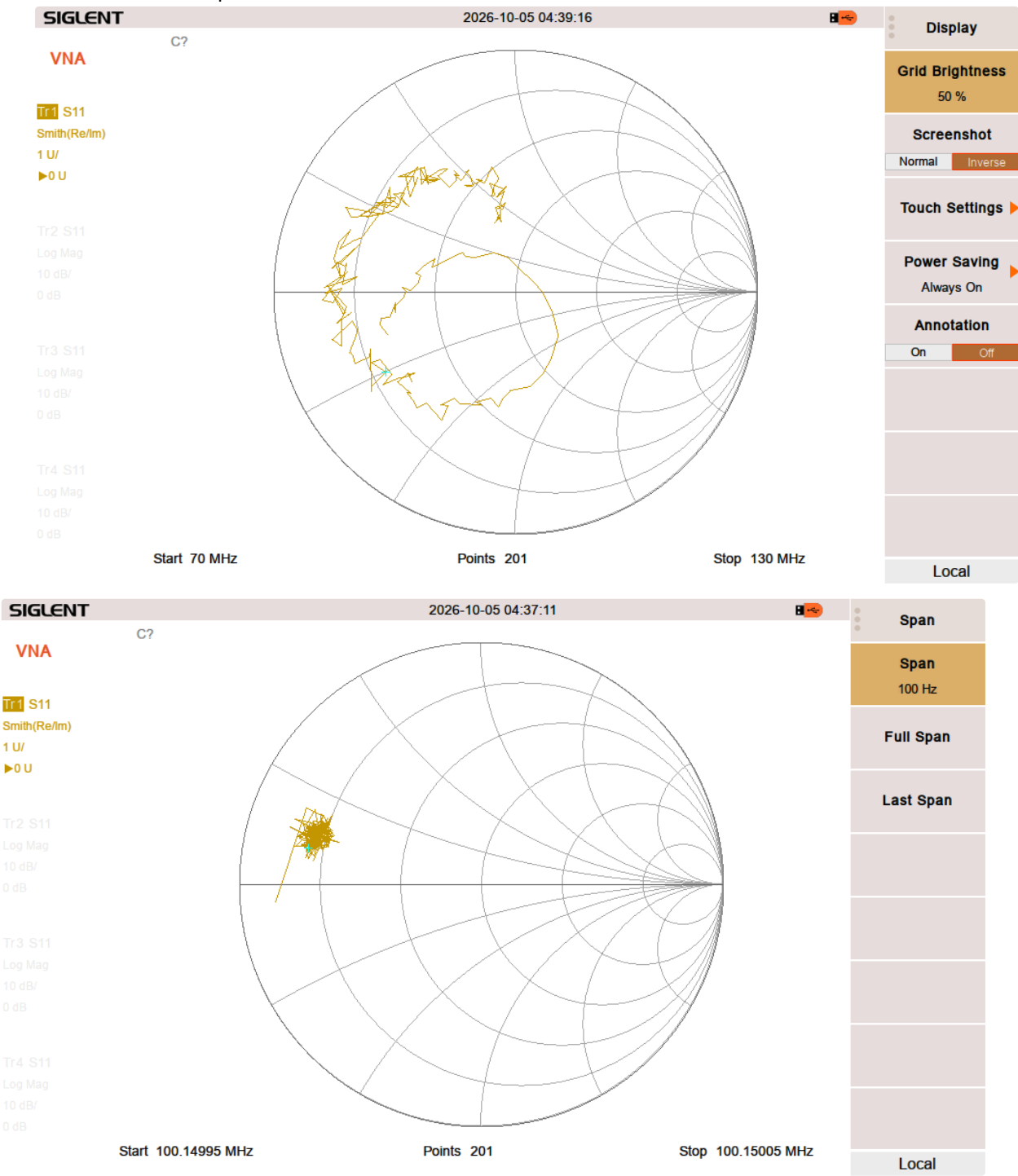


TODO PASTE SMITH CHART PHOTO

28. Optional Extra Credit: Parasitic components are very important in RF design, and small parasitics in your board may be affecting your matching network designs. For extra credit, you can use zero Ohm resistors to extract your parasitic pad capacitance and parasitic via inductance as follows.
29. Solder a zero Ohm resistor in series from the SMA connector to a pad, then solder a large capacitor in shunt from the pad to ground. It's important to pick a capacitor with modest ESL, but it needs to be large enough to swamp out other sources of

parasitic capacitance. Observe resonance in S11, then back out what inductance causes it. That inductance is most likely a byproduct of your via.

30. Remove the shunt capacitor and replace the series zero Ohm with a series inductor. Observe resonance again. The inductance causing the resonance is your series inductor plus your via inductor. The capacitance causing the resonance comes from the parasitic capacitance of the pad.
31. Optional Extra Credit: Measure that one “mystery load” using the VNA! You’ll need to get creative with cables and some kind of SMA-to-wires adapter.



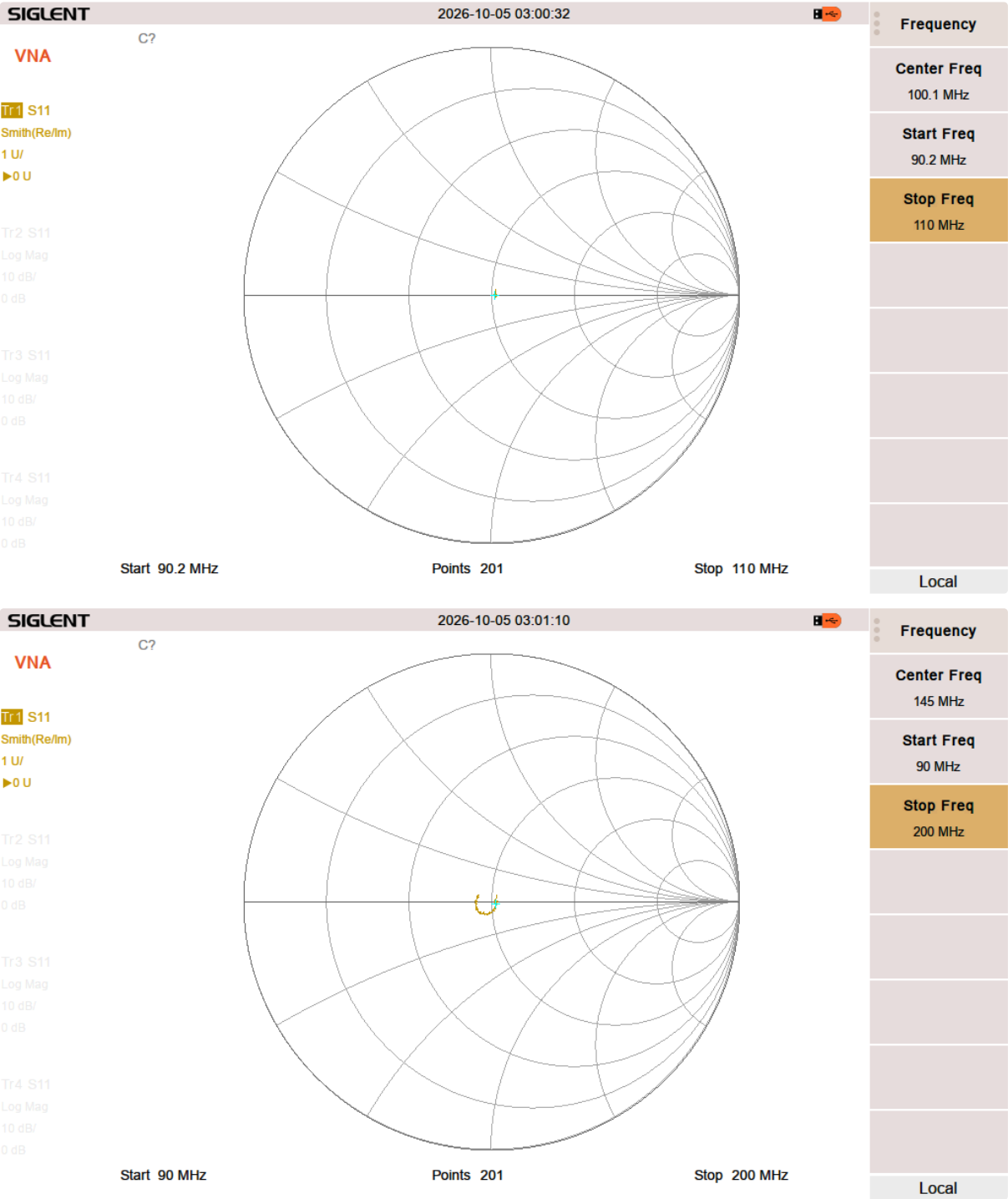
Submission Instructions

These are the things you will need to include in your submission. You do not need to include more than this, or any justification, reasoning, or calculations except what is requested here.

- One Smith chart showing all of the loads from the “Smith Chart Practice Theory” section.

- Your three uncalibrated measurements (one for each termination) in instruction 1. For all future VNA measurements, only provide calibrated results. However, the uncalibrated results are interesting in this specific part of this lab.
- Your three post-calibration measurements in instruction 2. You may omit the measurement when you change to 90MHz to 110MHz.
- The measurement of your zero Ohm resistors + 270 Ohm termination at the start of your soldering practice.
- The measurement you took after each step of assembling your L match. Make sure to include a marker on your Smith Chart showing where 100MHz is.
- Also, be sure to include photographs of your filter board as you construct it. Pair these with the measurements of each step.
- Finally, provide measurements showing the Q of your matching network, and show that the Q matches your designed Q value. If you provide screenshots (instead of post-processing your data), then be sure to include markers at the points you measured to calculate Q.

Photo dump in case I forgot a photo but have one (its so many files)



VNA

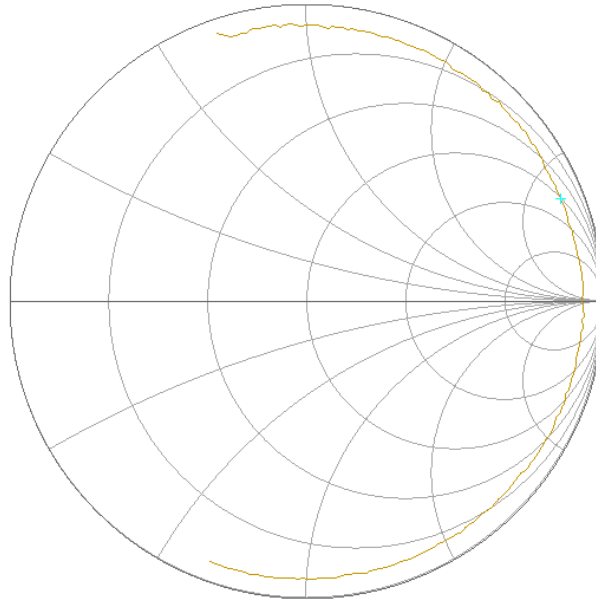
OFF

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Smith

Lin / Phase

Log / Phase

Real / Imag

R + jX

G + jB

Local

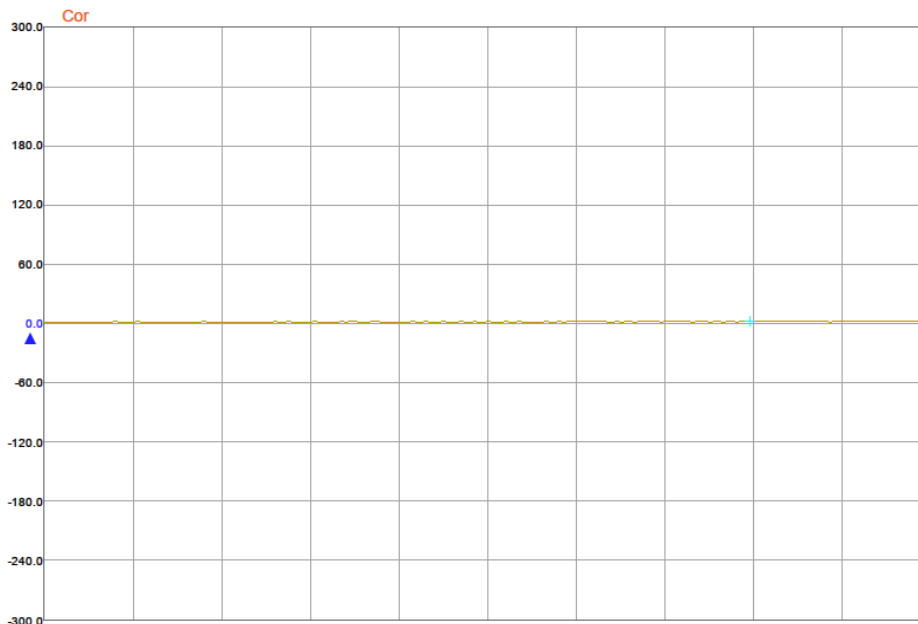
VNA

Tr1 S11
Phase
60 °/
► 0 °

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Meas

Stimulus

Meas

S11 S21

Format

Phase

Scale

Trace

Calibration

Local

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Tr1 S11

Log Mag

10 dB/

0 dB

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

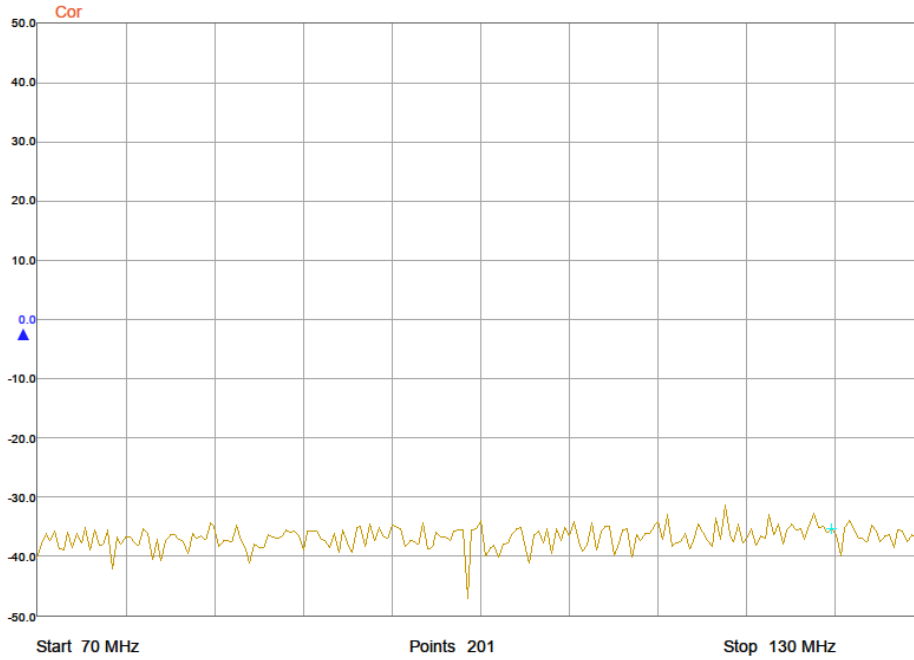
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Tr1 S11

Phase

60 °/

0 °

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

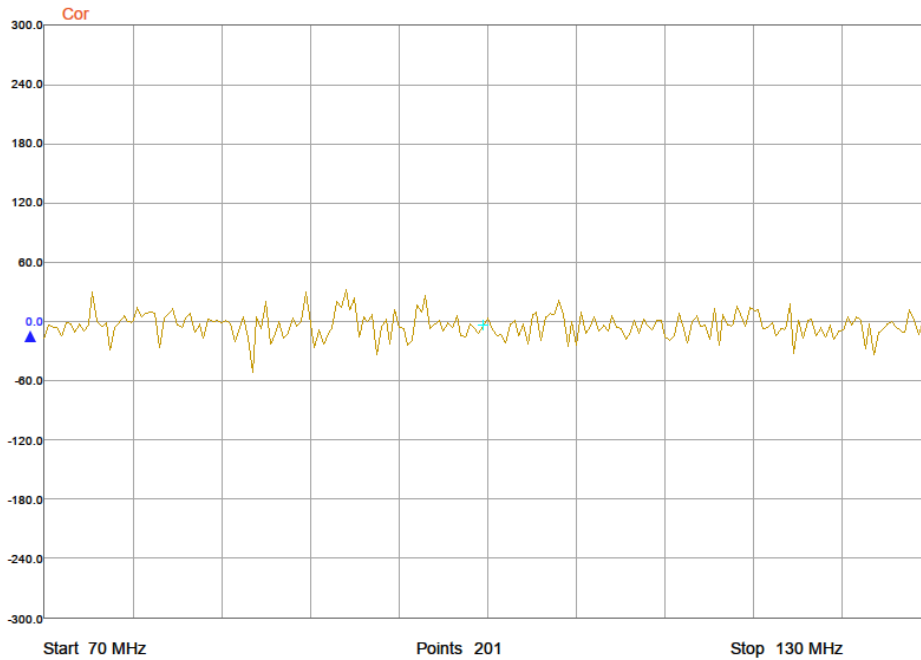
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

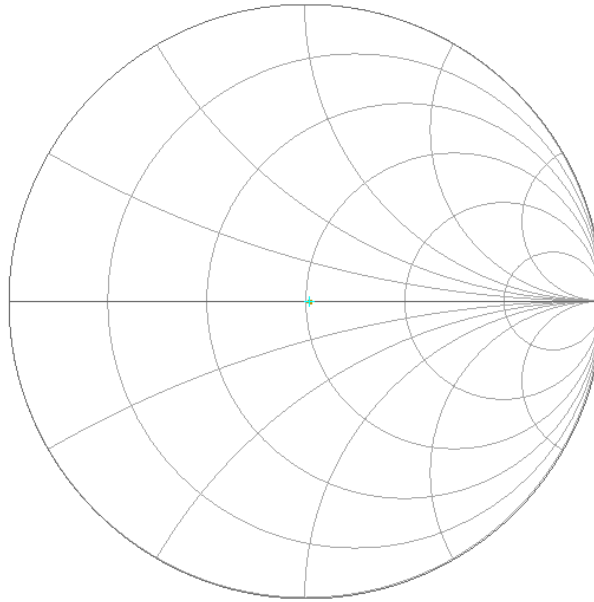
Cor

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

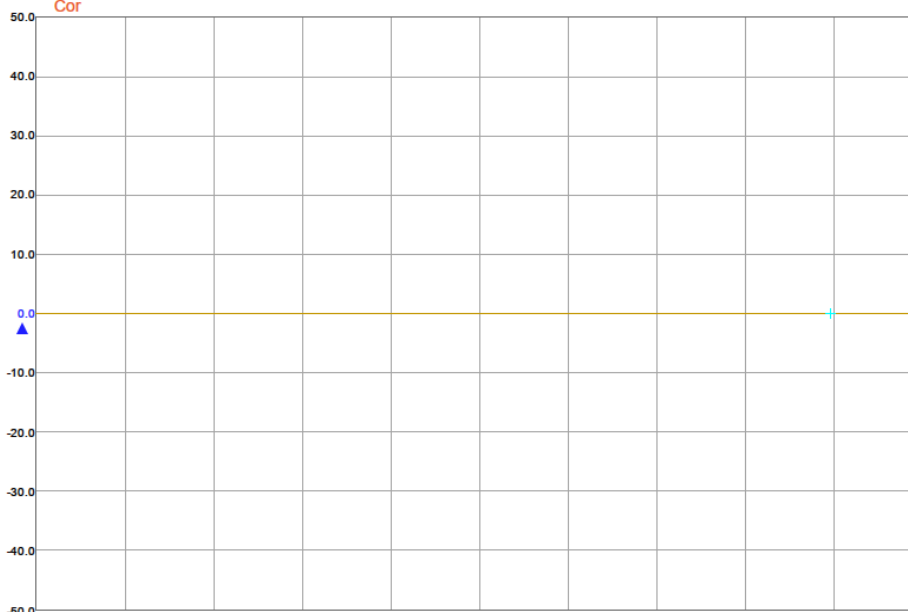
Cor

Tr1 S11
Log Mag
10 dB/
► 0 dB

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

VNA

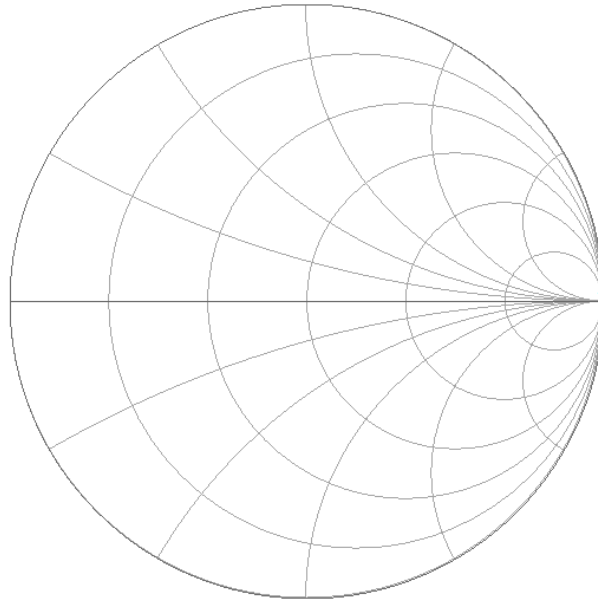
Cor

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Smith

Lin / Phase

Log / Phase

Real / Imag

R + jX

G + jB

Local

VNA

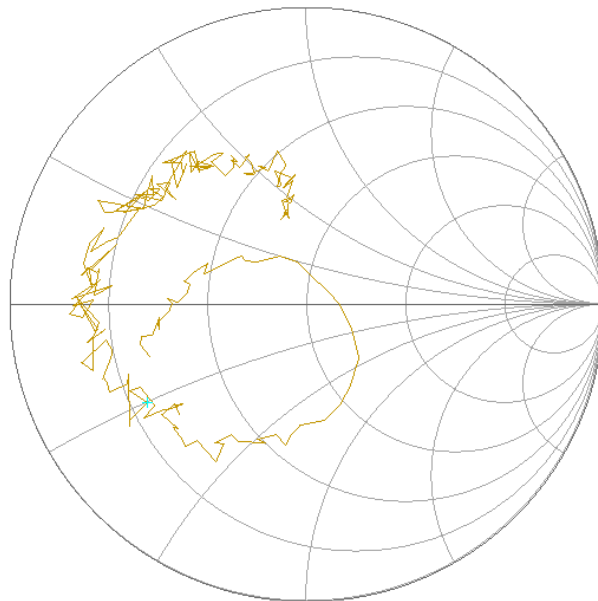
C?

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Display

Grid Brightness
50 %

Screenshot

Normal

Inverse

Touch Settings ►

Power Saving ►

Always On

Annotation

On

Off

Local

VNA

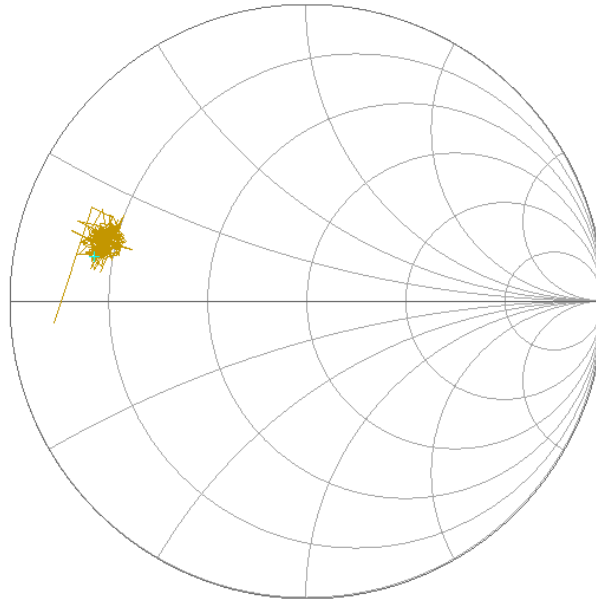
C?

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 100.14995 MHz

Points 201

Stop 100.15005 MHz

Span

Span

100 Hz

Full Span

Last Span

Local

VNA

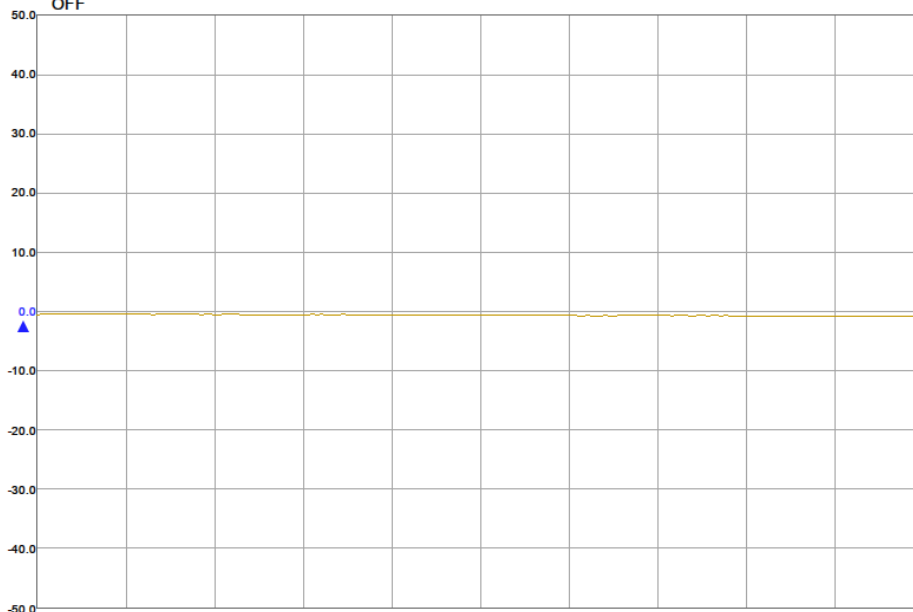
OFF

Tr1 S11
Log Mag
10 dB/
► 0 dB

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Tr1 S11

Phase

60 °

▶ 0 °

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

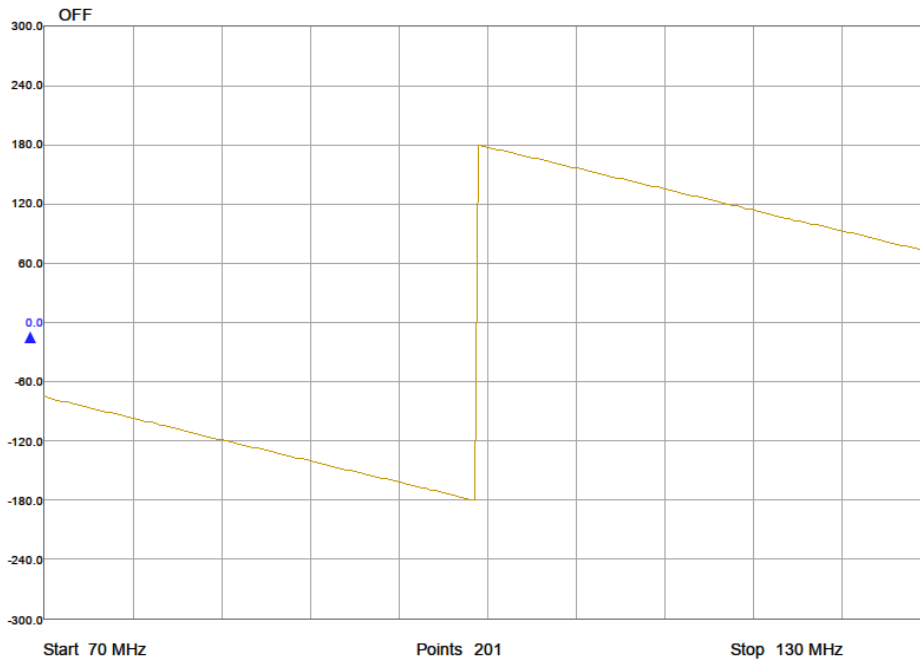
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Calibration

Correction

On

Off

Calibrate

ECal

Cal Kit

F503ME

Port Extensions

OFF

System Z0

50

Velocity Factor

0.66

Save/Load

Local

VNA

OFF

Tr1 S11

Smith(Re/Im)

1 U/

▶ 0 U

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

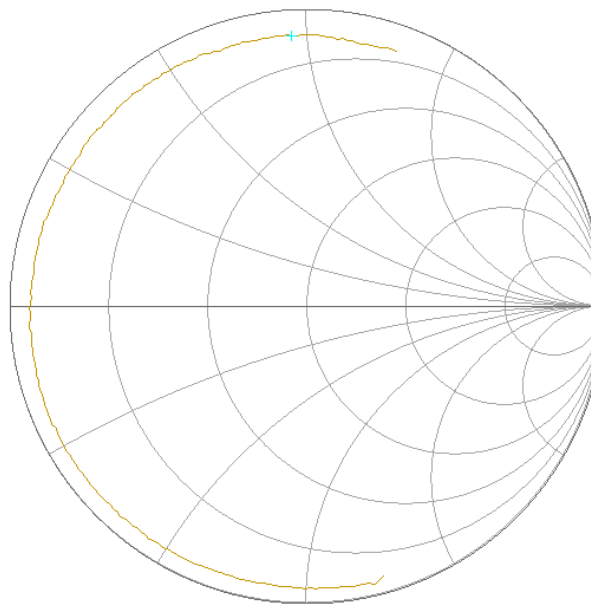
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Smith

Lin / Phase

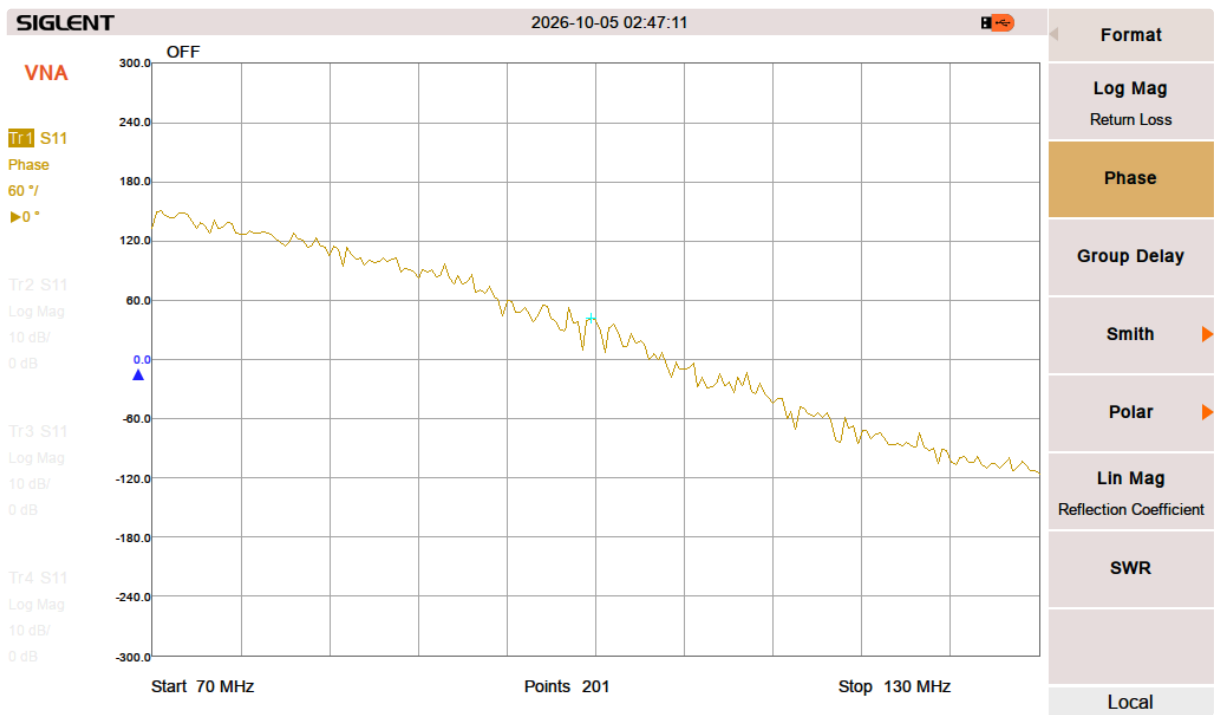
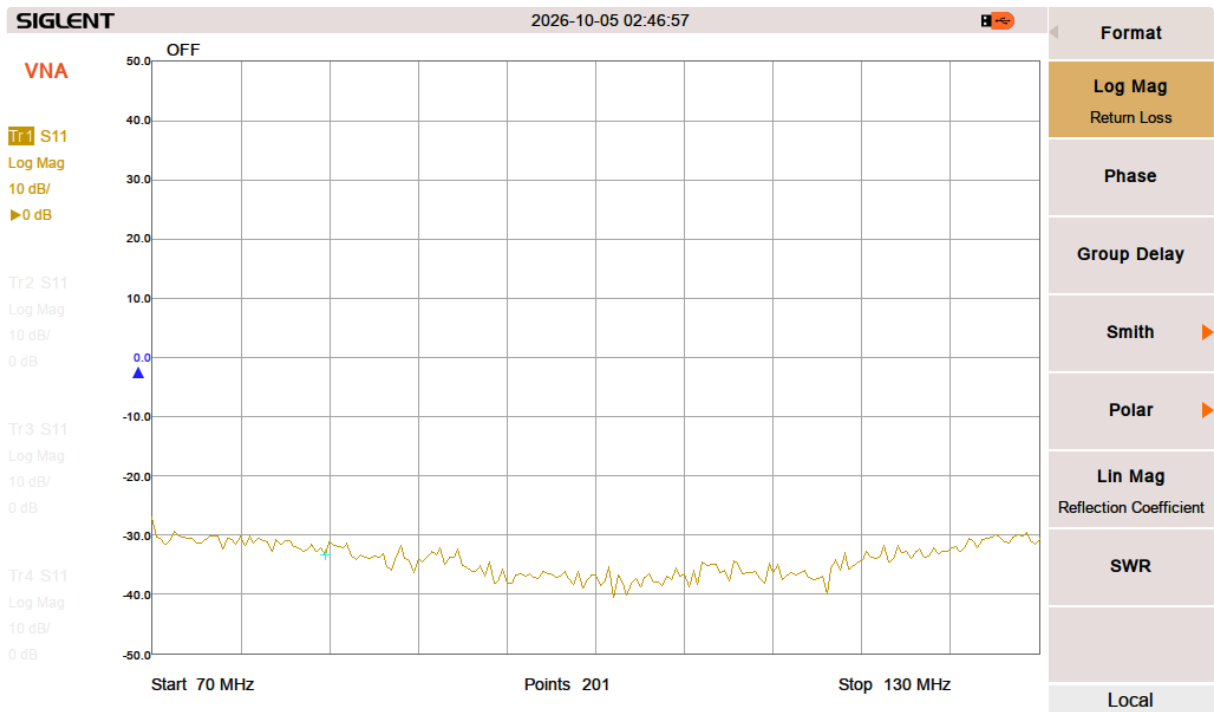
Log / Phase

Real / Imag

R + jX

G + jB

Local



SIGLENT

2026-10-05 02:46:39



VNA

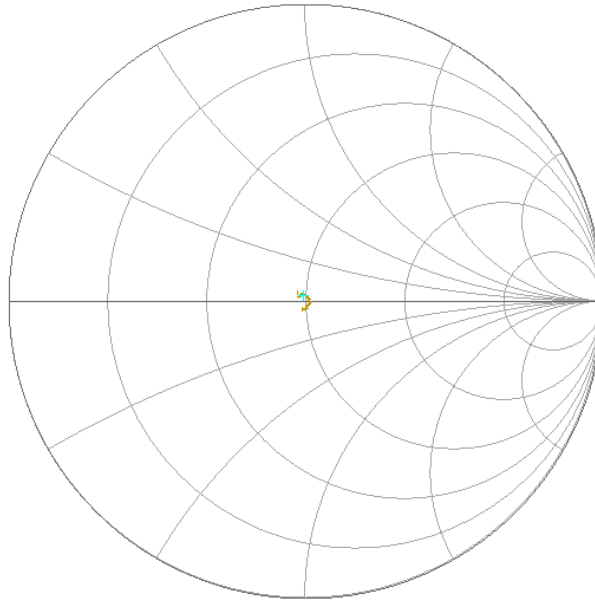
OFF

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Smith

Lin / Phase

Log / Phase

Real / Imag

R + jX

G + jB

Local

SIGLENT

2026-10-05 04:30:32



VNA

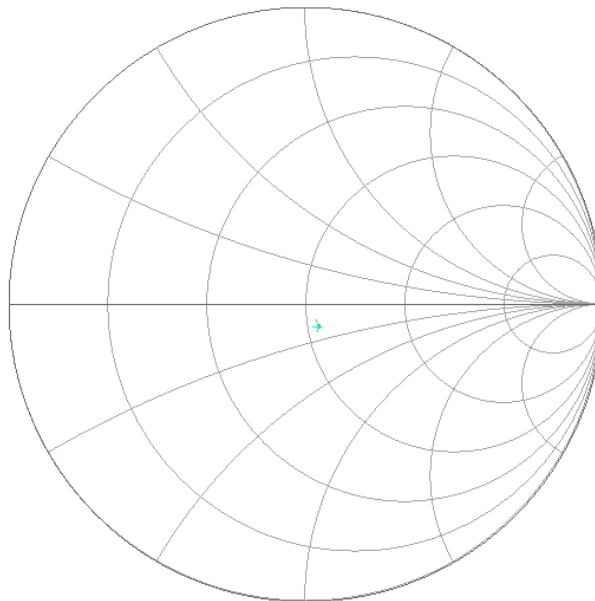
C?

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 99.99995 MHz

Points 201

Stop 100.00005 MHz

Display

Grid Brightness
50 %

Screenshot

Normal

Inverse

Touch Settings ►

Power Saving ►

Always On

Annotation

On

Off

Local

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Tr1 S11

Log Mag

10 dB/

0 dB

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

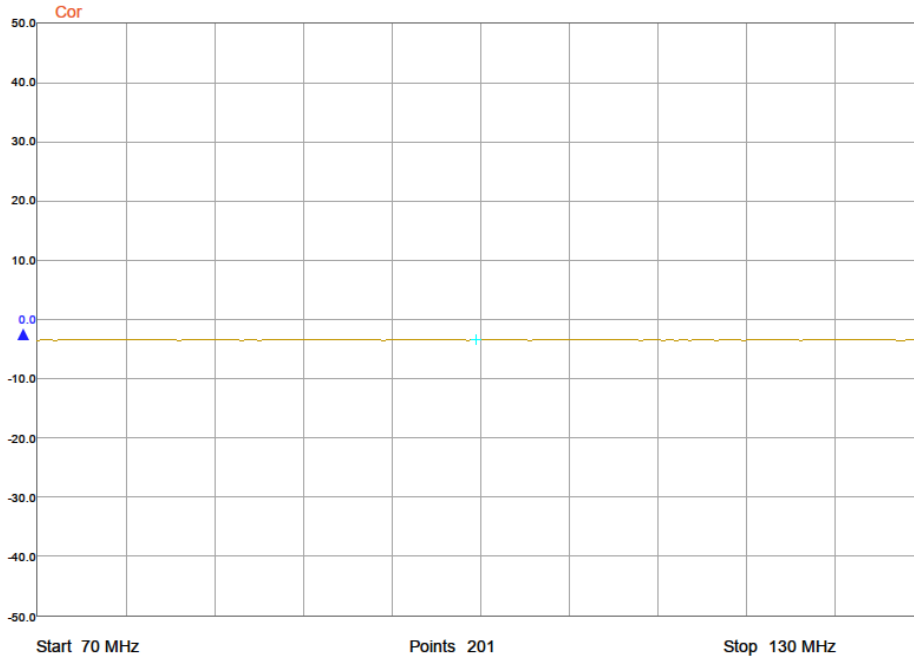
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Tr1 S11

Phase

60 °/

0 °

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

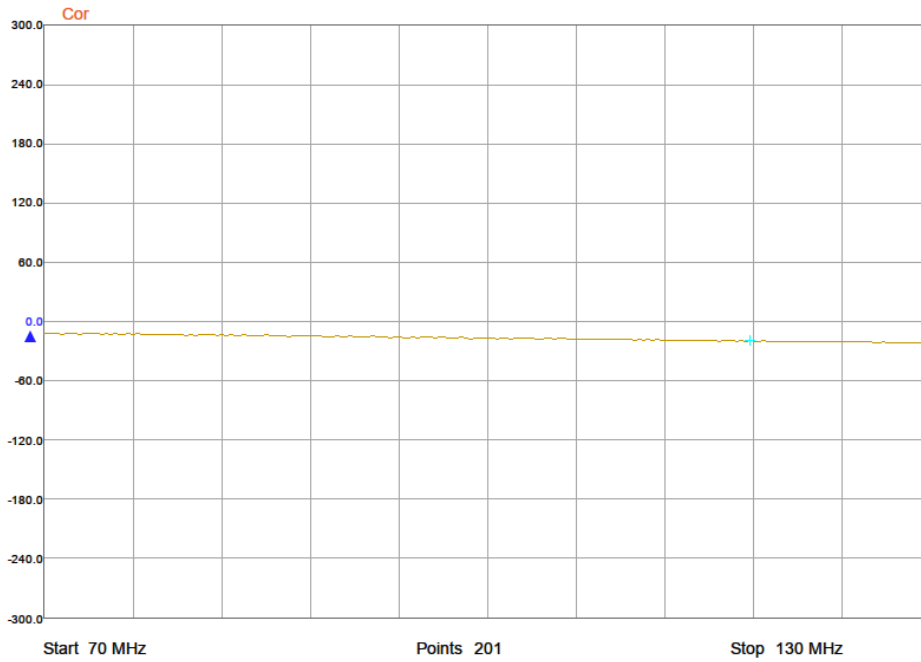
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Frequency

Center Freq

100 MHz

Start Freq

70 MHz

Stop Freq

130 MHz

Local

VNA

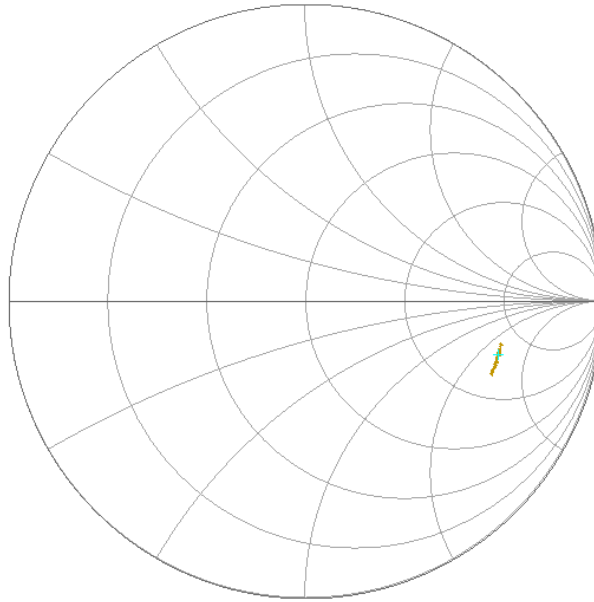
Cor

Tr1 S11
Smith(Re/Im)
1 U/
► 0 U

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Cor

Tr1 S11
Log Mag
10 dB/
► 0 dB

Tr2 S11
Log Mag
10 dB/
0 dB

Tr3 S11
Log Mag
10 dB/
0 dB

Tr4 S11
Log Mag
10 dB/
0 dB



Start 70 MHz

Points 201

Stop 130 MHz

Format

Log Mag

Return Loss

Phase

Group Delay

Smith

Polar

Lin Mag

Reflection Coefficient

SWR

Local

VNA

Tr1 S11

Phase

60 °

► 0 °

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

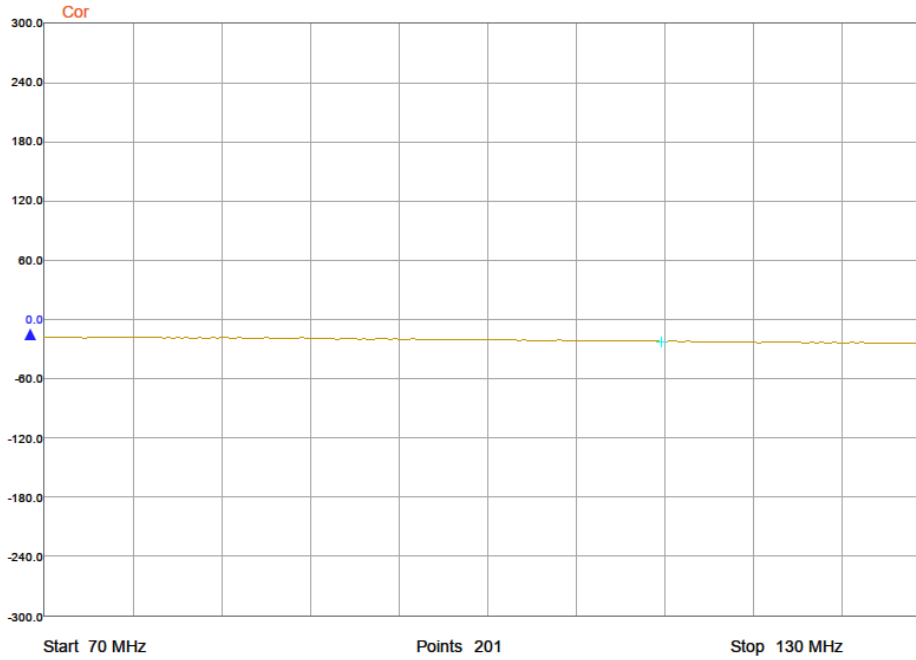
0 dB

Tr4 S11

Log Mag

10 dB/

0 dB



Smith

Lin / Phase

Log / Phase

Real / Imag

R + jX

G + jB

Local

VNA

Tr1 S11

Smith(Re/Im)

1 U/

► 0 U

Tr2 S11

Log Mag

10 dB/

0 dB

Tr3 S11

Log Mag

10 dB/

0 dB

Tr4 S11

Log Mag

10 dB/

0 dB

